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Dynamics of Stochastic Gradient Descent

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Abstract

Stochastic dynamics provides a rich framework for modelling and understanding optimization processes in neural networks. It has been empirically observed that optimization with stochastic gradient descent drives the system towards flatter loss minima than pure gradient descent. Flatter minimal are hypothesized to correlate with better generalization performance. This work reproduces relevant empirical findings, investigates modelling approaches and discusses theoretical explanations for these training dynamics. The aim of this work is to give a broad introduction into this research direction.

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Chapter 1

Introduction

> I propose to consider the question:
> 'Can machines think?'
>
> *Alan Turing, [1]*

It's been 75 years since Alan Turing posed the question 'Can machines think?' in his article 'Computing machinery and intelligence' [1] in which he introduces the Turing Test and engages with earlier remarks, e.g. by Ada Lovelace in her notes on Babbage's Analytical Engine [2]. In 1956 McCarthy, Minsky, Rochester and Shannon coined the term artificial intelligence in their overly ambitious Dartmouth Conference, see [3]. There, the idea of working on artificial neural nets was already outlined, building on work by McCulloch and Pitts, see [4]. Research on artificial intelligence (AI) went through booms and winters and only in two last decades did the focus switch back from expert systems to artificial neural networks, but this time much more compute was available, see [5]. In 2012, Alex Krizhevsky, Ilya Sutskever and Geoffrey Hinton demonstrated with AlexNet, see [6], the dominance of artificial neural networks trained on GPUs in computer vision tasks with the backpropagation algorithm [7], which is an efficient implementation of the chain rule. In particular, deep learning [8], where neural networks with many layers are trained on large amounts of data, has proven very successful in recent years. Milestones and important use cases include the application of reinforcement learning for Atari Games [9], progress on the protein folding problem [10], medical diagnostics [11], [12] and modern large language models, like GPT4 [13], that are based on the transformer architecture [14]. But there are also significant risks associated with advanced AI systems. These can range from over reliance by individuals in critical situations like medical professionals, potentially faster spreading of misinformation through human speech mimicking by large language models, the possible misuse by groups or governments for increased surveillance capacity, cyber-attacks, autonomous weapon systems and the possibility of loss of control to sufficiently advanced AI systems. For extensive discussions of the risks associated with modern and future AI systems see for example [15], [16], [17], [18]. Again, Alan Turing comes to mind. The following is a short excerpt from a lecture titled 'Intelligent machinery, a heretical theory' given by him in 1951.

> [...] for it seems probable that once the machine thinking method had started, it would not take long to outstrip our feeble powers. There would be no question of the machines dying, and they would be able to converse with each other to sharpen their wits. At some stage therefore we should have to expect the machines to take control, in the way that is mentioned in Samuel Butler's Erewhon. [19]

How can these concerns be addressed? The field of mechanistic interpretability tries to infer learned emergent capabilities from the weight structure of neural networks. Specifically, understanding the training dynamics of neural networks can potentially provide the tools to observe emergent capabilities of an AI system before it is deployed. Of particular interest is the capability to generalize to data outside the training distribution. Training a neural network typically means applying optimization algorithms to minimize a loss function that evaluates the discrepancy between the output of the model and the desired behavior. An important class of optimization algorithms are gradient based optimization methods like gradient descent (GD) and stochastic gradient descent (SGD), see [5], [20]. While, GD is deterministic and stops as soon as a local minimum is found, SGD introduces stochastic noise that leads to more interesting training dynamics, see [21]. Reference [21] hypothesizes that stochastic gradient descent drives the loss function of neural networks towards flatter minima than gradient descent and that flatter minima lead to better generalization. The link between generalization and flat minima can be traced back to research by Hochreiter and Schmidhuber in 1997, see [22]. In this work we primarily focus on the connection between SGD and flat minima, while additionally providing some empirical evidence for the claim regarding generalization. This hypothesis is visualized in Fig. 1.1. We investigate this hypothesis following a structure of first making empirical observations, then constructing a data independent model based on these observations and in a last step derive theoretical results from the model. We will recognize that statistical physics and in particular stochastic dynamics is an ideal framework to model the training dynamics. Explicitly, we start by introducing relevant mathematical concepts and definitions from measure theory, probability theory and stochastic processes. These concepts build the foundations of this text and will be referred to frequently. We then proceed with explaining the standard framework from the machine learning literature. Building on this we implement and extend on the experiments performed by Ref. [21]. In this way we derive the relevant empirical results that underlie our theoretical assumptions. Relying on these assumptions we carefully reconstruct the model developed by Ref. [21] in great detail and derive some of their simulation results. In a last step we thoroughly explain the theoretical observations made by Ref. [21], while providing the necessary connections to the theory of stochastic dynamics.

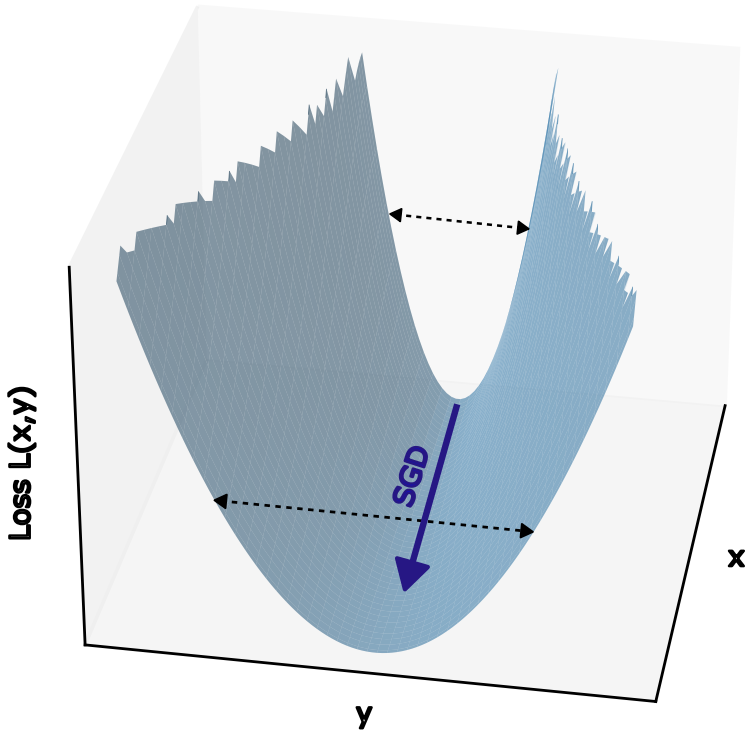


Figure 1.1: A simplified illustration of the hypothesis in a two-dimensional loss landscape. The loss $L(x, y)$ is plotted against the variables x and y. The hypothesis is that with respect to pure gradient descent (GD), stochastic gradient descent (SGD) drives the loss towards a flatter minimum, i.e. a point with less curvature. A similar figure is found in Ref. [21].

Chapter 2

Mathematical foundations

This chapter briefly introduces relevant mathematical definitions that are either directly referred to or build relevant foundations for discussions in this work. For a detailed introduction to measure theory see [23] and [24]. For a detailed introduction to probability theory and stochastic processes see [23], [25], [26] and [27].

2.1 Measure theory

Let Ω be a set and let $A, B \subseteq \Omega$ be subsets of Ω.
Then the **power set** $\mathcal{P}(A) := \{D | D \subseteq A\}$ is the set containing all subsets of A.
The **complement** of A is $A^c := \Omega \backslash A := \{x | x \in \Omega \wedge x \notin A\}$. So $A \backslash B = A \cap B^c$.

Definition 2.1.1 (σ-field)**.** A **σ-field** $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ on Ω is a subset of the power set of Ω with these properties:

(i) $\Omega \in \mathcal{A}$

(ii) $A \in \mathcal{A} \implies A^c \in \mathcal{A}$

(iii) If $A_n \in \mathcal{A}$ for all $n \in \mathbb{N}$ then $\bigcup_{n \in \mathbb{N}} A \in \mathcal{A}$

An element $A \in \mathcal{A}$ is a **measurable set** and $(\Omega, \mathcal{A})$ is a **measurable space**.

Definition 2.1.2 (Borel σ-field)**.** Let $C \in \Omega$ be a subset of Ω and define $\sigma(C) := \bigcap_{\mathcal{A} \in F} \mathcal{A}$ with $F = \{\mathcal{A} : \mathcal{A}\ is\ \sigma - field\ on\ \Omega\ and\ C \in \mathcal{A}\}$. Then $\sigma(C)$ is the intersection of all σ-fields on Ω that contain C. Let $\mathcal{A}$ be a σ-field on Ω and let $\mathcal{O}$ be the set of all open subsets $O \in \mathcal{A}$. The **Borel σ-field** on Ω is defined as $\mathcal{B}(\Omega) := \sigma(\mathcal{O})$.

Definition 2.1.3 (Positive measure)**.** A **positive measure** $\mu : \mathcal{A} \to [0, \infty[$ on a measurable space $(\Omega, \mathcal{A})$ is a function with the properties:

(i) $\mu(\emptyset) = 0$

(ii) (σ-additivity): Let $A_n \in \mathcal{A}$ for all $n \in \mathbb{N}$ be disjoint subsets of Ω, i.e. $\forall m, n \in \mathbb{N}$ with $m \neq n : A_m \cap A_n = \emptyset$. Then $\mu(\bigcup_{n \in \mathbb{N}} A_n) = \sum_{n \in \mathbb{N}} \mu(A_n)$.

If $\mu(\Omega) = 1$, then μ is a **probability meassure** and $(\Omega, \mathcal{A}, \mu)$ is a **probability space**.

Definition 2.1.4 (Measureable Functions)**.** Let $(\Omega, \mathcal{A})$ and $(\Psi, \mathcal{C})$ be measurable spaces. A function $f : \Omega \to \Psi$ is **measurable** if for all $C \in \mathcal{C} : f^{-1}(C) \in \mathcal{A}$. In the case that $\mathcal{A}$ and $\mathcal{C}$ are the Borel σ-fields $\mathcal{B}(\Omega)$ and $\mathcal{B}(\Psi)$, we call f **Borel measurable**.

2.2 Probability theory

Let $(\Omega, \mathcal{A}, P)$ be a probability space.

Definition 2.2.1 (Event)**.** A measurable subset of Ω, i.e. an element $A \in \mathcal{A}$, is called an **event**.

Definition 2.2.2 (Probabilities of events)**.** Let $A, B \in \mathcal{A}$ be events. Then $P(A)$ is the **probability** of the event A and $P(B)$ is the probability of the event B.
$P(A \cap B) = P(\{\omega \in A \wedge \omega \in B\})$ is the **joint probability** of events A and B.
$P(A|B) := \frac{P(A \cap B)}{P(B)}$ is the **conditional probability**.
If $P(A \cap B) = P(A)P(B)$, then A and B are said to be **independent** events. It follows for independent events A and B that $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} = P(A)$.

Definition 2.2.3 (Stochastic variable)**.** Let $(\Psi, \mathcal{E})$ be a measurable space. A **stochastic variable** is a measurable function $X : \Omega \to \Psi$. The domain Ω of X is called the **sample space** or **phase space** of X. A function value $x = X(\omega)$ with $\omega \in \Omega$ is a **realization** of X. Let $E \in \mathcal{E}$ be a subset of Ψ. The probability measure P on $(\Omega, \mathcal{A})$ introduces a probability measure P_X on $(\Psi, \mathcal{E})$: $P_X(E) := P(X^{-1}(E)) = P(\{\omega \in \Omega : X(\omega) \in E\})$.

Definition 2.2.4 (Probability density)**.** Let $X : \Omega \to \Psi, \omega \mapsto X(\omega)$ be a stochastic variable. A **probability density** is a Borel measurable function $p : \mathbb{R}^d \to [0, \infty[$ satisfying $\int_{\mathbb{R}^d} p(x)dx = 1$.
In the case of $d = 1$, i.e. $p : \mathbb{R} \to [0, \infty[$ with $\alpha, \beta \in [0, \infty[, \alpha \leq \beta$ it follows that

$$P(\alpha \leq X \leq \beta) = \int_\alpha^\beta p(\omega)d\omega. \tag{2.1}$$

Let $\{X_n\}_{n=1}^N$ be a family of stochastic variables. One can equivalently use vector notation $\boldsymbol{X} = (X_1, X_2, ...X_N)^\intercal$, $\boldsymbol{X} \in \mathbb{R}^N$, with stochastic variables $X_1, X_2, ...X_N$. Let $\{X_m\}_{m=1}^M \subseteq \{X_n\}_{n=1}^N$, $M \leq N$, be a subset. Then there exists a **joint** probability density function

$$p(x_1, x_2, ..., x_n) = p(X_1 = x_1, X_2 = x_2, ..., X_n = x_n), \tag{2.2}$$

a **marginal** probability function

$$p(x_1, x_2..., x_m) = \int p(x_1, x_2..., x_{m-1}, x_m, x_{m+1}, ..., x_n)dx_{m+1}...dx_n \tag{2.3}$$

and a **conditional** probability function

$$p(x_{m+1}...x_n|x_1, ..., x_m) = \frac{p(x_1, ..., x_{m-1}, x_m, x_{m+1}, ..., x_n)}{p(x_1, ..., x_m)}. \tag{2.4}$$

Definition 2.2.5 (Moments)**.** Let X be a stochastic variable and p its probability distribution. A function $f(X)$ of the stochastic variable X is itself a stochastic variable. The mean, expectation value or average of $f(X)$ is defined by

$$\mathbb{E}[f(X)] = \langle f(X) \rangle := \int f(x)p(x)dx. \tag{2.5}$$

We call $\mu_m = \langle X^m \rangle$, $m \in \mathbb{N}$, the m-th moment of X. In particular the first moment $\mu_1 = \langle X \rangle$ is the expectation value of X. We also define the variance of X by

$$\begin{aligned}
\sigma^2 &= \mathrm{Var}(X) \\
&= \langle (X - \langle X \rangle)^2 \rangle \\
&= \langle (X - \langle X \rangle)(X - \langle X \rangle) \rangle \\
&= \langle X^2 - \langle X \rangle X - X \langle X \rangle + \langle X \rangle \langle X \rangle \rangle \\
&= \langle X^2 \rangle - \langle X \rangle \langle X \rangle - \langle X \rangle \langle X \rangle + \langle X \rangle \langle X \rangle \\
&= \langle X^2 \rangle - \langle X \rangle \langle X \rangle \\
&= \langle X^2 \rangle - \langle X \rangle^2 \\
&= \mu_2 - \mu_1^2
\end{aligned} \tag{2.6}$$

with σ being the standart deviation. We also introduce the characteristic function

$$\phi(k) = \langle e^{ikx} \rangle. \tag{2.7}$$

For multivariate stochastic variables $\boldsymbol{X} = (X_1, X_2, ..., X_N)^\intercal$ it is suitable to define the covariance matrix $\mathbf{C}$ with coefficients

$$\begin{aligned}
C_{ij} &= \mathrm{Cov}(X_i, X_j) \\
&\equiv \langle (X_i - \langle X_i \rangle)(X_j - \langle X_j \rangle) \rangle \\
&= \langle X_i X_j - \langle X_i \rangle X_j - X_i \langle X_j \rangle + \langle X_i \rangle \langle X_j \rangle \rangle \\
&= \langle X_i X_j \rangle - \langle X_i \rangle \langle X_j \rangle - \langle X_i \rangle \langle X_j \rangle + \langle X_i \rangle \langle X_j \rangle \\
&= \langle X_i X_j \rangle - \langle X_i \rangle \langle X_j \rangle.
\end{aligned} \tag{2.8}$$

Apperently, the diagonal elements C_{ii} are identical to the variance of X_i. For a random process $X(t)$, with times s, t one can define the autocovariance

$$C_{s,t} \equiv \langle (X(t) - \langle X(t) \rangle)(X(s) - \langle X(s) \rangle) \rangle = \langle X(t) X(s) \rangle - \langle X(t) \rangle \langle X(s) \rangle. \tag{2.9}$$

Definition 2.2.6 (The Gaussian distribution)**.** For one variable the Gaussian or Normal distribution is

$$P(x) = C e^{-\frac{1}{2}Ax^2 - Bx}. \tag{2.10}$$

with $x \in \mathbb{R}$, $A > 0$, $C = \left(\frac{A}{2\pi}\right)^{1/2} e^{\frac{-B^2}{2A}}$ The expectation value of a normally distributed random variable X is $\mu_1 = \langle X \rangle = \int_{-\infty}^{\infty} x C e^{-\frac{1}{2}Ax^2 - Bx} dx = \frac{-B}{A}$ and the 2nd-moment $\mu_2 = \frac{B^2}{A^2} + \frac{1}{A}$, so the variance $\sigma^2 = \langle X^2 \rangle - \langle X \rangle^2 = \frac{1}{A}$. $P(x)$ can be rewritten as

$$P(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x - \mu_1)^2}{2\sigma^2} \right]. \tag{2.11}$$

2.3 Stochastic processes

Let $(\Omega, \mathcal{A}, P)$ be a probability space and let $(\Psi, \mathcal{E})$ be a measurable space.

Definition 2.3.1 (Stochastic process)**.** A **stochastic process** is a family $\{X_t\}_{t\in T}$ of stochastic variables defined on Ω and with values in Ψ. If the **index set** T is countable, then $\{X_t\}_{t\in T}$ is a **discrete** process. Otherwise, it is a **continuous** process. An equivalent view defines a stochastic process $Y_X(t)$ as a function $f(X,t)$ of a stochastic variable X and a second variable t. If X is specified we abbreviate $Y_X(t) = Y(t)$.

Definition 2.3.2 (Probability densities of stochastic processes)**.** Let $X : \Omega \to \Psi, t \mapsto X(t)$ be a stochastic variable with values $x_1, x_2, ..., x_{n-1}, x_n \in \Psi$ at times $t_1, t_2, ..., t_{n-1}, t_n \in T$ with $t_n \geq t_{n-1} \geq ... \geq t_2 \geq t_1$. Then there is a joint probability density function

$$p(x_n, t_n; x_{n-1}, t_{n-1}; ...; x_1, t_1) \tag{2.12}$$

and a conditional probability density function

$$p(x_n, t_n | x_{n-1}, t_{n-1}; ...; x_1, t_1) \equiv \frac{p(x_n, t_n; x_{n-1}, t_{n-1}; ...; x_1, t_1)}{p(x_{n-1}, t_{n-1}; ...; x_1, t_1)}. \tag{2.13}$$

Definition 2.3.3 (Markov process)**.** Given a conditional probability density

$$p(x_n, t_n | x_{n-1}, t_{n-1}; ...; x_1, t_1) \tag{2.14}$$

the Markov assumption is

$$p(x_n, t_n | x_{n-1}, t_{n-1}; ...; x_1, t_1) = p(x_n, t_n | x_{n-1}, t_{n-1}). \tag{2.15}$$

So the probability density of x_n at time t_n depends only on the value x_{n-1} at time t_{n-1}.

Definition 2.3.4 (Chapman-Kolmogorov equation)**.** Let $\{X_t\}_{t\in T}$ be a stochastic process. Then, it generally holds for $m \leq n$ that

$$\begin{aligned}
p(x_n, t_n | x_m, t_m) &= \frac{p(x_n, t_n; x_m, t_m)}{p(x_m, t_m)} \\
&= \int \frac{p(x_n, t_n; x_{n-1}, t_{n-1}; ...; x_{m+1}, t_{m+1}; x_m, t_m)}{p(x_m, t_m)} dx_{n-1}...dx_{m+1} \\
&= \frac{p(x_n, t_n; ...; x_m, t_m)}{p(x_{n-1}, t_{n-1}; ...; x_m, t_m)} \frac{p(x_{n-1}, t_{n-1}; ...; x_m, t_m)}{p(x_m, t_m)} dx_{n-1}...dx_{m+1} \\
&= \int p(x_n, t_n | x_{n-1}, t_{n-1}; ...; x_m, t_m) p(x_{n-1}, t_{n-1}; ...; x_{m+1}, t_{m+1} | x_m, t_m) dx_{n-1}...dx_{m+1}. \\
&= \int p(x_n, t_n | x_{n-1}, t_{n-1}; ...; x_m, t_m)...p(x_{m+1}, t_{m+1} | x_m, t_m) dx_{n-1}...dx_{m+1}
\end{aligned} \tag{2.16}$$

Let's now assume that $\{X_t\}_{t\in T}$ is a Markov Process. Using the Markov property the expression simplifies to the **Chapman-Kolmogorov equation**

$$\begin{aligned}
&\int p(x_n, t_n | x_{n-1}, t_{n-1}; ...; x_m, t_m) p(x_{n-1}, t_{n-1}; ...; x_{m+1}, t_{m+1} | x_m, t_m) dx_{n-1}...dx_{m+1} \\
&= \int p(x_n, t_n | x_{n-1}, t_{n-1}) p(x_{n-1}, t_{n-1}; ...; x_{m+1}, t_{m+1} | x_m, t_m) dx_{n-1}...dx_{m+1} \\
&= \int p(x_n, t_n | x_{n-1}, t_{n-1}) p(x_{n-1}, t_{n-1} | x_{n-2}, t_{n-2})...p(x_{m+1}, t_{m+1} | x_m, t_m) dx_{n-1}...dx_{m+1}.
\end{aligned} \tag{2.17}$$

For example

$$\begin{aligned} p(x_3, t_3|x_1, t_1) &= \frac{p(x_3, t_3; x_1, t_1)}{p(x_1, t_1)} \\ &= \int \frac{p(x_3, t_3; x_2, t_2; x_1, t_1)}{p(x_1, t_1)} dx_2 \\ &= \int \frac{p(x_3, t_3; x_2, t_2; x_1, t_1)}{p(x_2, t_2; x_1, t_1)} \frac{p(x_2, t_2; x_1, t_1)}{p(x_1, t_1)} dx_2 \\ &= \int p(x_3, t_3|x_2, t_2; x_1, t_1) p(x_2, t_2|x_1, t_1) dx_2 \\ &= \int p(x_3, t_3|x_2, t_2) p(x_2, t_2|x_1, t_1) dx_2, \end{aligned} \tag{2.18}$$

where in the last step we used the fact that $\{X_t\}_{t\in T}$ is a Markov Process. So, the full probability density factorizes into a product of conditional probabilities, where each one is only conditioned on the last time step. We call these **transition probabilities**. We can therefore compute the probability density at any step, if we know the transition probabilities and the initial probability. Using the last example:

$$\begin{aligned} p(x_3, t_3) &= p(x_1, t_1) p(x_3, t_3|x_1, t_1) \\ &= p(x_1, t_1) \int p(x_3, t_3|x_2, t_2) p(x_2, t_2|x_1, t_1) dx_2. \end{aligned} \tag{2.19}$$

Definition 2.3.5 (Continuous Markov process)**.** The sample paths of a Markov process $\{X_t\}_{t\in T}$ are a continuous function of t, if

$$\forall \varepsilon > 0 : \lim_{\Delta t \to 0} \int_{|x-z|>\varepsilon} dx \; p(x, t + \Delta t|z, t) = 0 \tag{2.20}$$

uniformly in z, t and Δt.

Chapter 3

Fundamentals of machine learning and empirical results

In this chapter we introduce relevant concepts from machine learning and perform experiments that form the empirical foundations of our hypothesis. Our nomenclature is derived from Refs. [5], [21] and [28].

3.1 Data



Figure 3.1: Grid overview of 100 sample images of handwritten digits from the MNIST dataset, see [29]. Each of the grayscale images has a size of 28×28 pixels.

Machine learning is learning from data, see for example [5]. In **supervised learning** the task of the system is the correct classification of samples from a dataset. A **dataset** $\{\boldsymbol{s}_i\}_{i=1}^{I} \subseteq \mathcal{D}$ is a finite set of sample size I and a subset of the true underlying distribution $\mathcal{D}$. The elements of a dataset are called **instances** and each instance $\boldsymbol{s}_i = (\boldsymbol{a}^{(1)}, c) \in \mathbb{R}^d \times K$ consists of a d-dimensional training **sample**, characterized by its features $\boldsymbol{a}^{(1)} \in \mathbb{R}^d$, and a classification **label** $c \in K = \{1, 2, ..., k-1, k\}$, where $k \in \mathbb{N}$ is the number of classes. Each entry $a_i^{(1)}$ of the vector $\boldsymbol{a}^{(1)}$ is a **feature** for $i \leq d$, i.e. there are a total of d features for each sample. The dataset is often split into a **training dataset** and a **test dataset**. While the training dataset is used for learning, i.e. to update the model parameters during

training, the test set must not interfere with the training process. It is there to evaluate the generalization performance of the model after training is complete.
The machine learning experiments in this work are performed on the MNIST dataset, see Ref. [29]. This dataset contains 70,000 instances of grayscale images of handwritten digits together with their corresponding labels $c \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Figure 3.1 shows an overview of 100 images from the dataset. Each image consists of 28×28 pixels with values ranging from zero (black) to 255 (white) and can be represented by a normalized vector $\boldsymbol{a}^{(1)} \in \mathbb{R}^d = \mathbb{R}^{784}$ containing $d = 28 \times 28 = 784$ features with values between 0 and 1. The MNIST dataset is then described by the set $\{\boldsymbol{s}_i\}_{i=1}^{I}$ with sample size $I = 70.000$. The number of classes is $k = 10$ and each instance consists of a sample with features $\boldsymbol{a}^{(1)}$ and its label c, for example the pixels of the image of the handwritten digit two and the label $c = 2$.

3.2 Artificial neural networks

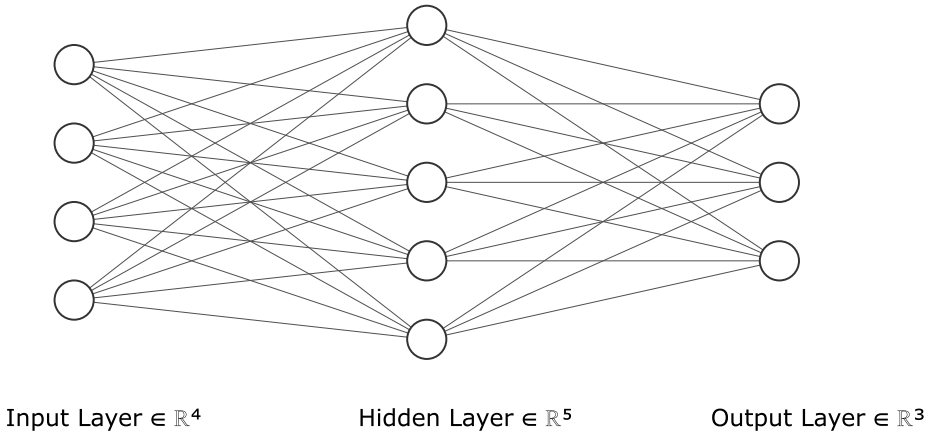


Figure 3.2: Schematic of a fully connected neural network with three layers. There are four neurons in the input layer, five in the hidden layer and three in the output layer. The figure was produced using software created by Ref. [30].

An **artificial neural network**, see Fig. 3.2, is a composite function represented by layers of neurons and connections between them. Neurons are essentially just placeholders for numerical values that result from computations performed on the functions input, while the connections are the functions parameters and are subject to optimization during the training process. Assume a neural network with $l \in \mathbb{N}$ layers of $n_1, n_2, ..., n_{l-1}, n_l \in \mathbb{N}$ neurons each. The first layer of the neural network is called the input layer. It stores the features $\boldsymbol{a}^{(1)} \in \mathbb{R}^d$ of a sample and therefore has a corresponding number of $n_1 = d$ neurons. The last layer is the output layer, which has one neuron for each of the k classes in the dataset, such that $n_l = k$. In between are the so-called hidden layers. For the MNIST dataset a sample has features $\boldsymbol{a}^{(1)} \in \mathbb{R}^{784}$ and the input layer has a corresponding number of $n_1 = 784$ neurons, while the output layer has $n_l = 10$ neurons, covering all possible classification labels. In a fully connected neural network with linear layers each neuron of one layer is connected with every neuron of the next layer. Consider two consecutive layers of neurons. The first layer has m neurons that store the values $a_i^{(1)}$ for $i \leq m$, while the

n neurons of the second layer store the values $a_j^{(2)}$ for $j \leq n$. Consider neuron number j of the second layer. It receives inputs from every neuron of the first layer in the form of a linear transformation of their stored values $a_i^{(1)} \times w_{ij}^{(1)}$ with a weight $w_{ij}^{(1)} \in \mathbb{R}$. The inputs from all neurons of the first layer are summed and a bias term $b_j^{(1)}$ is added

$$a_1^{(1)} \times w_{1j}^{(1)} + \cdots + a_m^{(1)} \times w_{mj}^{(1)} + b_j^{(1)}. \tag{3.1}$$

In a last step the aggregated input is evaluated by a non-linear activation function φ, such that the value of the j-th neuron of the second layer is

$$a_j^{(2)} = \varphi\left(a_1^{(1)} \times w_{1j}^{(1)} + \cdots + a_m^{(1)} \times w_{mj}^{(1)} + b_j^{(1)}\right). \tag{3.2}$$

Combined in vector notation, the second layer of neurons has values

$$\begin{aligned}
\boldsymbol{a}^{(2)} = \begin{bmatrix} a_1^{(2)} \\ \vdots \\ a_n^{(2)} \end{bmatrix} &= \begin{bmatrix} \varphi\left(a_1^{(1)} \times w_{11}^{(1)} + \cdots + a_m^{(1)} \times w_{m1}^{(1)} + b_1^{(1)}\right) \\ \vdots \\ \varphi\left(a_1^{(1)} \times w_{1n}^{(1)} + \cdots + a_m^{(1)} \times w_{mn}^{(0)} + b_n^{(0)}\right) \end{bmatrix} \\
&\equiv \boldsymbol{\varphi}\left(\begin{bmatrix} \left(a_1^{(1)} \times w_{11}^{(1)} + \cdots + a_m^{(1)} \times w_{m1}^{(1)} + b_1^{(1)}\right) \\ \vdots \\ \left(a_1^{(1)} \times w_{1m}^{(1)} + \cdots + a_m^{(1)} \times w_{mn}^{(0)} + b_n^{(0)}\right) \end{bmatrix}\right) \\
&= \boldsymbol{\varphi}\left(\begin{bmatrix} w_{11}^{(1)} & \cdots & w_{m1}^{(1)} \\ \vdots & \ddots & \vdots \\ w_{1n}^{(1)} & \cdots & w_{mn}^{(1)} \end{bmatrix} \cdot \begin{bmatrix} a_1^{(1)} \\ \vdots \\ a_m^{(1)} \end{bmatrix} + \begin{bmatrix} b_1^{(1)} \\ \vdots \\ a_n^{(1)} \end{bmatrix}\right) \\
&= \boldsymbol{\varphi}\left(\boldsymbol{W}^{(1)} \cdot \boldsymbol{a}^{(1)} + \boldsymbol{b}^{(1)}\right)
\end{aligned} \tag{3.3}$$

with weight matrix $\boldsymbol{W}^{(1)}$ and bias vector $\boldsymbol{b}$. In this way, the features of a sample $\boldsymbol{a}$ are propagated through the network from the input layer with values $\boldsymbol{a}^{(1)}$ to the output layer with values $\boldsymbol{a}^{(l)}$. In general, for a network of l layers with $n_1, n_2, ..., n_{l-1}, n_l \in \mathbb{N}$ neurons each, there are $p_w = \sum_{i=1}^{l-1} n_i \cdot n_{i+1}$ weights and $p_b = \sum_{i=2}^{l} n_i$ bias terms. They are collected in the parameter vector $\boldsymbol{\theta} \in \mathbb{R}^{p_w+p_b} = \mathbb{R}^p$, where $\mathbb{R}^p$ is the parameter space with $p = p_w + p_b$. The output $\boldsymbol{a}^{(l)}$ of the full network is then a vector field

$$\boldsymbol{\psi}(\boldsymbol{a}^{(1)}, \boldsymbol{\theta}) : \mathbb{R}^d \times \mathbb{R}^p \to \mathbb{R}^k, \quad \boldsymbol{\psi}(\boldsymbol{a}^{(1)}, \boldsymbol{\theta}) = \boldsymbol{a}^{(l)} \tag{3.4}$$

of the parameters $\boldsymbol{\theta}$ and the features $\boldsymbol{a}^{(1)}$ of a given input sample.
The output $\boldsymbol{a}^{(l)}$ consists of activations for each of the classes and can be normalized by the softmax function

$$\boldsymbol{\phi} : \mathbb{R}^k \to \mathbb{R}^k, \quad \boldsymbol{\phi}(\boldsymbol{a}^{(l)}) = (\phi_1(\boldsymbol{a}^{(l)}), ..., \phi_k(\boldsymbol{a}^{(l)})) \tag{3.5}$$

with

$$\phi_i(\boldsymbol{a}^{(1)}) = \frac{\exp[a_i^{(l)}]}{\sum_{i=1}^{k} \exp[a_i^{(l)}]}. \tag{3.6}$$

This can equally be written as a function of the input features $\boldsymbol{a}^{(1)}$ and parameters $\boldsymbol{\theta}$ with

$$\boldsymbol{\phi} \circ \boldsymbol{\psi} : \mathbb{R}^d \times \mathbb{R}^p \to \mathbb{R}^k, \quad \boldsymbol{\psi}(\boldsymbol{a}^{(1)}, \boldsymbol{\theta}) = (\phi_1(\boldsymbol{\psi}(\boldsymbol{a}^{(1)}, \boldsymbol{\theta})), ..., \phi_k(\boldsymbol{\psi}(\boldsymbol{a}^{(1)}, \boldsymbol{\theta}))) \tag{3.7}$$

and

$$\phi_i(\boldsymbol{\psi}(\boldsymbol{a}^{(1)}, \boldsymbol{\theta})) = \frac{\exp[\psi_i]}{\sum_{i=1}^{k} \exp[\psi_i]}, \tag{3.8}$$

where ψ_i is the i-th entry of the vector $\boldsymbol{\psi}(\boldsymbol{a}^{(1)}, \boldsymbol{\theta})$, see for example Ref. [5].
These normalized outputs are the models predictions for the classification label c of the input features $\boldsymbol{a}^{(1)}$. For training the model we now have to define a function that quantifies the quality of this prediction: the loss function.

3.3 Loss functions

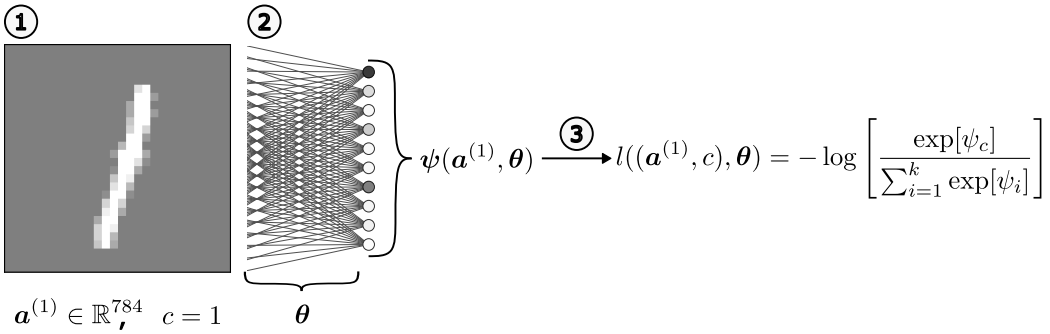


Figure 3.3: Schematic of a prediction task. (1) An image of the handwritten digit 1 from the MNIST dataset consists of 784 pixels and is represented by its feature vector $\boldsymbol{a}^{(1)}\mathbb{R}^{784}$ and its label $c = 1$. (2) The feature vector $\boldsymbol{a}^{(1)}$ is the input for an artificial neural network with parameters $\boldsymbol{\theta}$. The neural network has an output layer with 10 neurons, corresponding to the 10 digits and represents the function $\psi(\boldsymbol{a}^{(1)}, \boldsymbol{\theta})$. (3) The outputs are evaluated by a samplewise loss function called cross entropy, i.e. $l((\boldsymbol{a}^{(1)}, c), \boldsymbol{\theta}) \log\left[\frac{\exp[\psi_c]}{\sum_{i=1}^{k}\exp[\psi_i]}\right]$.

Let $\boldsymbol{\theta}$ be the networks parameters and $\boldsymbol{s}_i = (\boldsymbol{a}^{(1)}, c)$ be an instance of the dataset with network input $\boldsymbol{a}^{(1)} \in \mathbb{R}^d$ and classification label $c \in \{1, ..., k\}$. Then, the networks output is $\boldsymbol{a}^{(l)} = \boldsymbol{\psi}(\boldsymbol{a}^{(1)}, \boldsymbol{\theta}) \in \mathbb{R}^k$. The **samplewise loss function** is defined by

$$\begin{aligned} l : (\mathbb{R}^d \times K, \mathbb{R}^p) &\to [0, \infty[, \\ (\boldsymbol{s}_i, \boldsymbol{\theta}) &\mapsto l(\boldsymbol{s}_i, \boldsymbol{\theta}) \end{aligned} \tag{3.9}$$

and evaluates the discrepancy between the models predictions for a given sample and the true label. Depending on context and research field it is also called distance function or error function. A commonly used samplewise loss function is the cross entropy loss, see Ref. [5],

$$l((\boldsymbol{a}^{(1)}, c), \boldsymbol{\theta}) = -\log\left[\phi_c(\boldsymbol{\psi}(\boldsymbol{a}^{(1)}, \boldsymbol{\theta}))\right] = -\log\left[\frac{\exp[\psi_c]}{\sum_{i=1}^{k}\exp[\psi_i]}\right]. \tag{3.10}$$

In Fig. 3.3 the full process of providing input features of a sample to the neural network and then evaluating its outputs with the samplewise loss function, is illustrated. Building on this, we define the (general) dataset **loss function** or empirical risk

$$L : (\mathbb{R}^d \times K \times M, \mathbb{R}^p) \to [0, \infty[, \quad L\left(\{\boldsymbol{s}_i\}_{i=1}^M, \boldsymbol{\theta}\right) = \frac{1}{M} \sum_{i=1}^{M} l(\boldsymbol{s}_i, \boldsymbol{\theta}) \tag{3.11}$$

by the average of the samplewise loss with respect to the whole training dataset $\{\boldsymbol{s}_i\}_{i=1}^M$. We call a subset $\{\boldsymbol{s}_{\mu_i}\}_{i=1}^B \subseteq \{\boldsymbol{s}_i\}_{i=1}^M$ with $\mu_i \in \{1, 2, ..., M-1, M\}$ and $B \leq M$ of the dataset a **minibatch** with batch size B. We can then similarly define the **batchwise loss function** as the average of the samplewise loss with respect to the minibatch $\{\boldsymbol{s}_{\mu_i}\}_{i=1}^B$,

$$L^\mu : (B \times K, \mathbb{R}^p) \to [0, \infty[, \quad L^\mu(\{\boldsymbol{s}_{\mu_i}\}_{i=1}^B, \boldsymbol{\theta}) = \frac{1}{B} \sum_{i=1}^{B} l(\boldsymbol{s}_{\mu_i}; \boldsymbol{\theta}). \tag{3.12}$$

In our notation we mostly suppress the dependence of $L\left(\{\boldsymbol{s}_i\}_{i=1}^M, \boldsymbol{\theta}\right)$ on the chosen dataset and write $L(\boldsymbol{\theta})$ or even just L. For later reference, we introduce the gradient $\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta})$ of $L(\boldsymbol{\theta})$, defined by

$$\nabla_{\boldsymbol{\theta}} L = \left(\frac{\partial L}{\partial \theta_1}, ..., \frac{\partial L}{\partial \theta_p}\right)^\intercal \in \mathbb{R}^p, \tag{3.13}$$

and the **Hessian H** of $L(\boldsymbol{\theta})$, defined by

$$\mathbf{H}(L(\boldsymbol{\theta})) = \nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} L = \begin{bmatrix} \frac{\partial^2 L}{\partial \theta_1^2} & \frac{\partial^2 L}{\partial \theta_1 \partial \theta_2} & \cdots & \frac{\partial^2 L}{\partial \theta_1 \partial \theta_p} \\ \frac{\partial^2 L}{\partial \theta_2 \theta_1} & \frac{\partial^2 L}{\partial \theta_2^2} & \cdots & \frac{\partial^2 L}{\partial \theta_2 \partial \theta_p} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 L}{\partial \theta_p \partial \theta_1} & \frac{\partial^2 L}{\partial \theta_p \partial \theta_2} & \cdots & \frac{\partial^2 L}{\partial \theta_p^2} \end{bmatrix} \in \mathbb{R}^p \times \mathbb{R}^p \tag{3.14}$$

with respect to the parameters $\boldsymbol{\theta}$. Since the hessian $\mathbf{H}$ is in general a real valued symmetric matrix, there is a unitary transformation $\mathbf{U}$ such that $\mathbf{H}$ is diagonalizable by $\mathbf{H} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\dagger$, see Ref. [31]. The diagonal matrix

$$\boldsymbol{\Lambda} = \begin{bmatrix} \lambda_{11} & 0 & \dots & 0 \\ 0 & \lambda_{22} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_{pp} \end{bmatrix} \tag{3.15}$$

contains the real eigenvalues $\lambda_i \equiv \lambda_{ii} \in \mathbb{R}$ of $\mathbf{H}$. Unitary means that $\mathbf{U}\mathbf{U}^\dagger = \mathbf{U}^\dagger\mathbf{U} = \mathbf{1}$. A Hessian matrix is called degenerate if at least one of its eigenvalues is degenerate, i.e. there is at least one i, such that $\lambda_i = 0$. The hessian contains the second derivatives of the loss $L(\boldsymbol{\theta})$ and therefore its eigenvalues inform about the curvature of the loss landscape at a local point $\boldsymbol{\theta}$ in the parameter space. Reference [21] defines the flatness $F \equiv \prod_{i=1}^{10} \lambda_{ii}^{-1/20}$ as a function of the ten largest eigenvalues λ_{ii} of the Hessian in terms of their absolute value. In close analogy we define the flatness

$$F \equiv \prod_{i=1}^{10} |\lambda_{ii}|^{-1/20}, \tag{3.16}$$

where we substituted the eigenvalues with their absolute values. This ensures that the flatness F is always real valued.

3.4 Optimization

Now we are able to explain what we mean by **training** a neural network. During the training process, input samples are provided to the neural network and the samplewise loss of its output is computed. After every sample from the dataset was presented once to the neural network, the training is said to have completed one **epoch**. In the general supervised learning setup, the goal of the training process is to find a set of parameters $\boldsymbol{\theta}$, that minimizes the loss function:

$$\min_{\boldsymbol{\theta}\in\mathbb{R}^p} L(\boldsymbol{\theta}). \tag{3.17}$$

So, learning is performed by applying optimization algorithms to the loss function L and updating the model parameters $\boldsymbol{\theta}$ accordingly. There are many optimization strategies with nuanced differences. One way to categorize them is by the number of training samples provided to the model before computing the batchwise loss function and updating the parameters accordingly.
Gradient descent (GD) is an optimization technique characterized by the parameter updating rule

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t), \tag{3.18}$$

while **Stochastic gradient descent** (SGD) is based on

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\nabla_{\boldsymbol{\theta}} L^{\mu}(\boldsymbol{\theta}_t). \tag{3.19}$$

Here $\boldsymbol{\theta}_t$ are the model parameters after optimization step t and $\boldsymbol{\theta}_{t+1}$ are the parameters after one further iteration. The learning rate η modulates the optimization step sizes.
GD is based on the general loss function $L(\boldsymbol{\theta})$, evaluated on the full dataset at each iteration. In contrast, SGD updates parameters based on the batchwise loss function $L^{\mu}(\boldsymbol{\theta})$, where the minibatch is a small subset with batch size $B \ll M$. We can view GD as SGD with a batch size equal to the size of the dataset. Figures 3.4 and 3.5 provide a schematic of the training process with gradient descent and stochastic gradient descent.

3.5 Empirical results

In this section we present the empirical results from our machine learning experiments. We first explain the experimental setup, then define relevant metrics and discuss our findings. The experiments performed in this section are closely related to those performed by Ref. [21]. We added the gradient descent option with batch size $B = 60000$ samples, iterated through more combinations of batch sizes B and learning rates η and used the weights of the full model to compute the Hessian eigenvalues and the flatness. While these extensions increase the computational burden, they enable clear comparisons between SGD and GD. A non standard programming library that was relevant for the experiments is PyHessian [32]. The programmed code in this section was first implemented by myself and only afterward LLM based coding assistants were used to check and optimize the code. Implemented suggestions were carefully inspected by myself.

3.5.1 Experimental setup

We train a fully connected neural network with linear layers on the MNIST dataset, see Ref. [29], that we already introduced in section 3.1. The dataset contains 70,000 instances

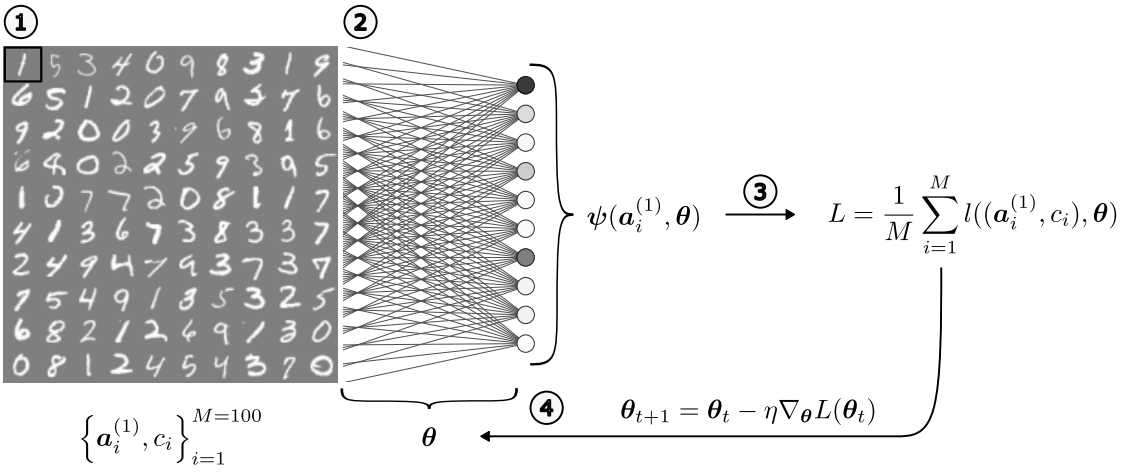


Figure 3.4: Schematic of the training process with GD. (1) A dataset $\{\boldsymbol{a}_i^{(1)}, c_i\}_{i=1}^{M} = 100$ consisting of 100 images from the MNIST dataset. (2) Input features $\boldsymbol{a}_i^{(1)}$ of the dataset are iteratively provided to a neural network and an output $\psi(\boldsymbol{a}_i^{(1)}, \boldsymbol{\theta})$ is computed. (3) The loss function L is the average over the samplewise loss of all 100 samples from the dataset. (4) The parameters $\boldsymbol{\theta}$ of the neural networks are updated depending on the loss function.

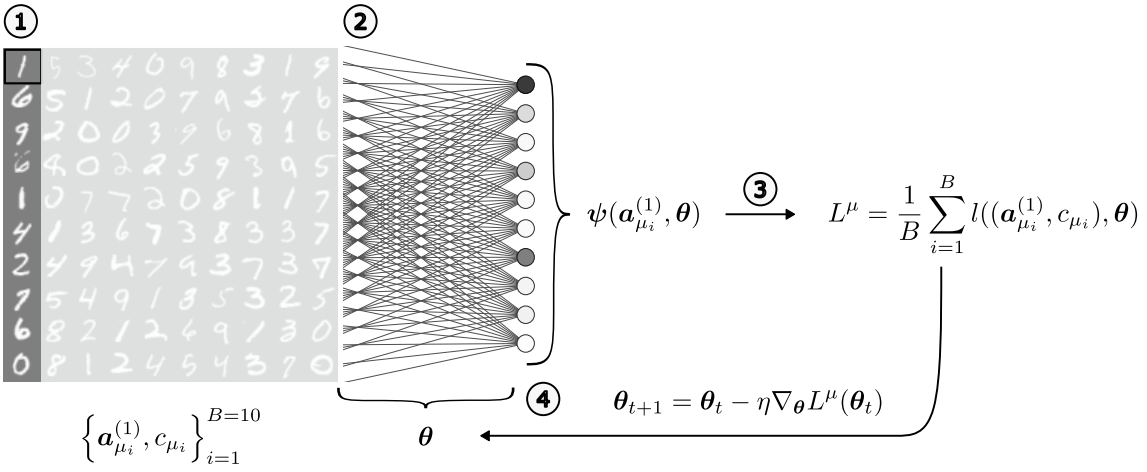


Figure 3.5: Schematic of the training process with SGD. (1) The minibatch $\{\boldsymbol{a}_i^{(1)}, c_i\}_{i=1}^{B} = 10$ is a subset with 10 samples of the full dataset consisting of 100 samples from the MNIST dataset. (2) Input features $\boldsymbol{a}_i^{(1)}$ of the minibatch are iteratively provided to a neural network and an output $\psi(\boldsymbol{a}_i^{(1)}, \boldsymbol{\theta})$ is computed. (3) The batch loss function L^{μ} is the average over the samplewise loss of all 10 samples from the minibatch. (4) The parameters $\boldsymbol{\theta}$ of the neural network are updated depending on the batch loss function L^{μ}. This process is repeated for all minibatches of the dataset.

of black and white images of handwritten digits together with their corresponding labels $c \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Each image consists of 28×28 pixels with values ranging from zero (black) to 255 (white) and is represented by the normalized vector $\boldsymbol{a}^{(1)} \in \mathbb{R}^d = \mathbb{R}^{784}$ containing $d = 28 \times 28 = 784$ features with values between 0 and 1. We split the dataset into a train dataset containing 60,000 images and a test dataset containing 10,000 images. The model is a fully connected neural network. The networks input layer stores the features of a sample and therefore has a corresponding number of $m = 784$ neurons. This is followed by a second and a third layer with $n = 50$ neurons each. They are the first and second hidden layer. The fourth and last layer is the output layer with $k = 10$ neurons, corresponding to the ten digits $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$. As explained above, in a fully connected neural network each neuron of one layer is connected with each neuron of the consecutive layer. So, there are $m \times n = 784 \times 50$ connections between the input layer and the first hidden layer, $n \times n = 50 \times 50$ connections between the hidden layers and $n \times k = 50 \times 10$ connections between the second hidden layer and the output payer. Overall there are $m \times n + n \times n + n \times k = 784 \times 50 + 50 \times 50 + 50 \times 10 = 42200$ weights and $n + n + k = 50 + 50 + 10 = 110$ biases in the parameter vector $\boldsymbol{\theta} \in \mathbb{R}^{42310}$. As non-linear activation function we apply the rectified linear unit function (ReLU) component wise, see Ref. [33],

$$\varphi(x) = max(0, x) \tag{3.20}$$

and the samplewise loss function is cross entropy loss, Eq. (3.10),

$$l(\boldsymbol{\psi}(\boldsymbol{a}, \boldsymbol{\theta}), c) = -\log\left[\frac{\exp[\psi_c]}{\sum_{i=1}^{10} \exp[\psi_i]}\right]. \tag{3.21}$$

3.5.2 Metrics

Several metrics provide information about the training dynamics. We define the number of optimization steps between consecutive recordings of the training metrics as our measure and the point of data recording as measurement. For comparability between training runs with different learning rates η, the measure is scaled by the learning rate. Two standard metrics are the training accuracy and training loss. The training accuracy is the percentage of correctly classified training samples and the training loss is the batchwise loss, both averaged over all minibatches between two data recordings. Every measurement, the test accuracy, test loss and flatness are computed. The neural network is evaluated on the full test dataset with 10,000 images. The test accuracy is the percentage of correctly classified test samples, the test loss is the average loss regarding the test dataset and the flatness is defined as in Eq. (3.16).

3.5.3 Influence of batch size and learning rate on training dynamics

In a first experiment the batch size $B \in \{50, 100, 200, 60000\}$ and the learning rate $\eta \in \{0.1, 0.05, 0.025\}$ were varied independently between training runs. Batch sizes $B = 50$, $B = 100$ and $B = 200$ correspond to SGD, while $B = 60000$ is the full training dataset, corresponding to GD. Figures 3.6 and 3.7 depict an overview of the results. The flatness, train accuracy, train loss, test accuracy and test loss are plotted against the number of iterations, i.e. the number of minibatches at the time of recording, scaled by the learning rate η. The experiments suggest that the dynamics of the training accuracy and the

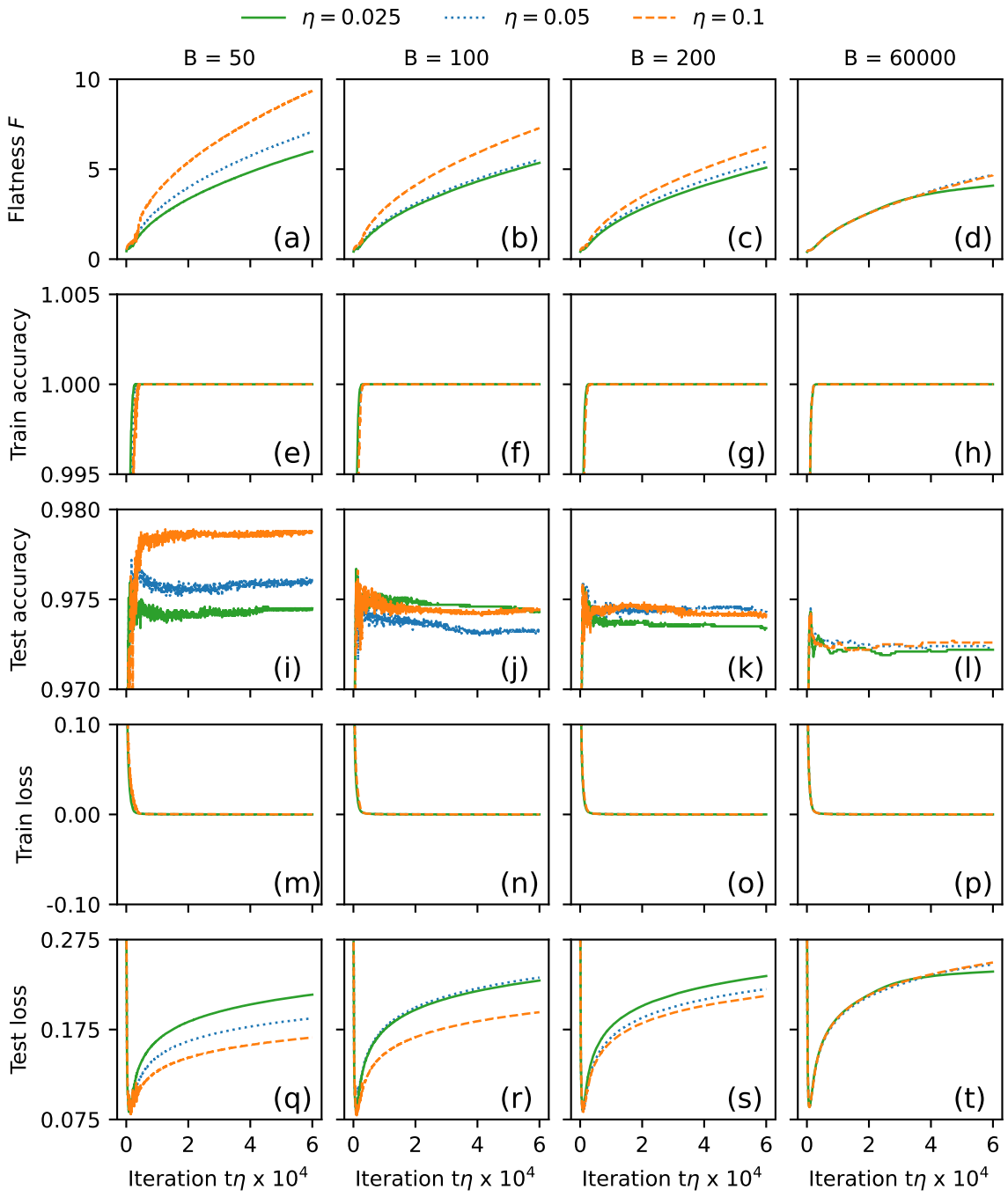


Figure 3.6: Overview of the training dynamics of a neural network with two linear layers of 50 neurons each on the MNIST dataset with 60,000 instances of training data and 10,000 instances of testing data. Flatness (a), (b), (c), (d) training accuracy (e), (f), (g), (h), training loss (i), (j), (k), (l), testing accuracy (m), (n), (o), (p) and testing loss (q), (r), (s), (t) for learning rates $\eta = 0.025$, $\eta = 0.05$ and $\eta = 0.1$ are plotted against the scaled number of iterations ηt. The batch size is $B = 50$ for (a), (e), (i), (m), (q), $B = 100$ for (b), (f), (j), (n), (r), $B = 200$ for (c), (g), (k), (o), (s) and $B = 60,000$ for (d), (h), (l), (p), (t). Training accuracy and training loss reach similar extrema for all algorithms. Larger learning rates outperform smaller ones in terms of higher flatness and test accuracy and lower test loss, irrespective of the batch size.

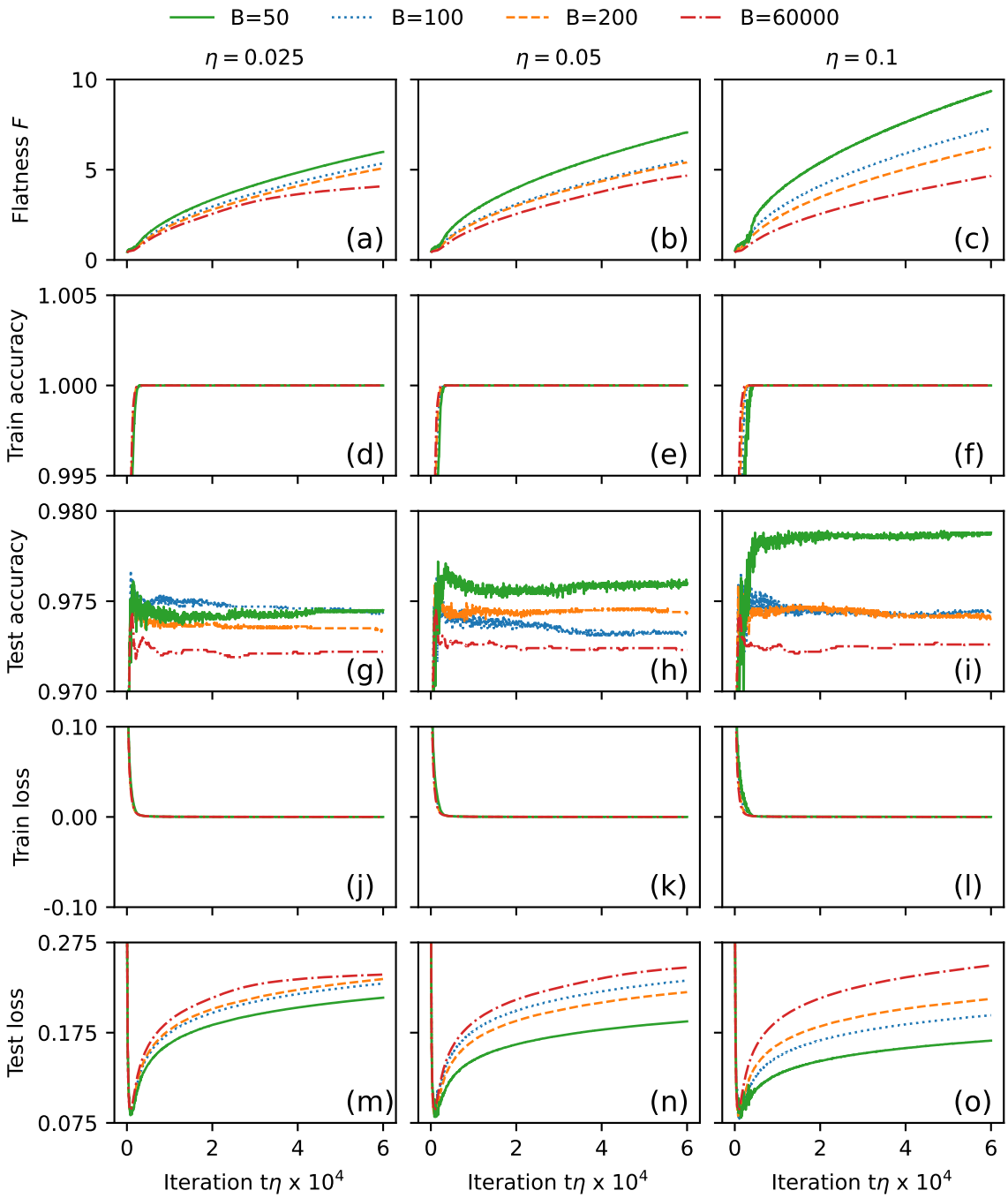


Figure 3.7: Overview of the training dynamics of a neural network with two linear layers of 50 neurons each on the MNIST dataset with 60,000 instances of training data and 10,000 instances of testing data. Flatness (a), (b), (c), training accuracy (d), (e), (f), training loss (g), (h), (i), testing accuracy (j), (k), (l) and testing loss (m), (n), (o) for batch sizes $B = 50$, $B = 100$, $B = 200$ and $B = 60000$ are plotted against the scaled number of iterations ηt. The learning rate $\eta = 0.025$ for (a), (d), (g), (j), (m), $\eta = 0.05$ for (b), (e), (h), (k), (n) and $\eta = 0.1$ for (c), (f), (i), (l), (o). Training accuracy and training loss reach similar extrema for all algorithms. Smaller batch sizes outperform larger ones in terms of higher flatness and test accuracy and lower test loss, irrespective of the learning rate.

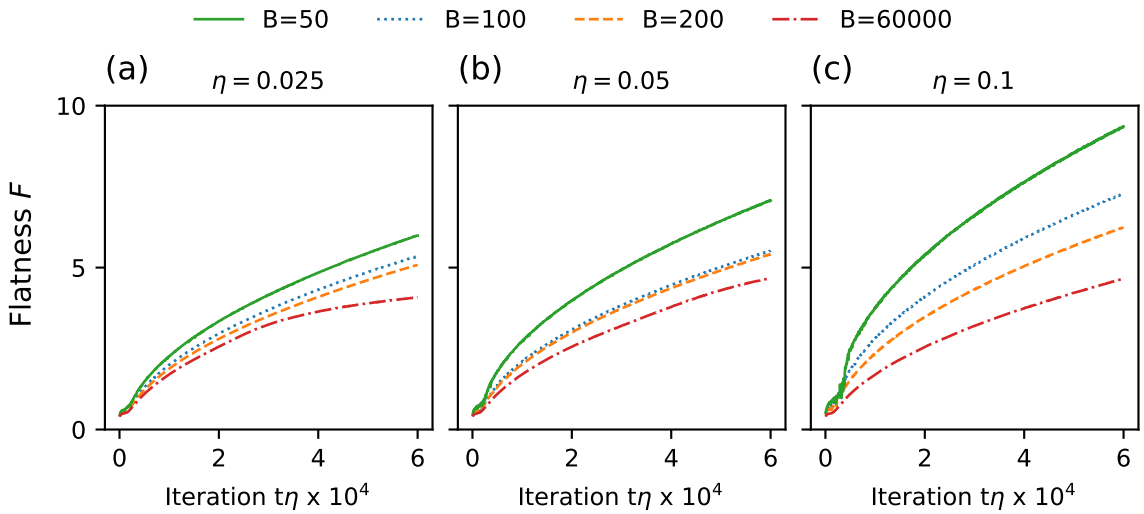


Figure 3.8: Training dynamics of a neural network with two linear layers of 50 neurons each on the MNIST dataset with 60,000 instances of training data and 10,000 instances of testing data. Flatness F for different batch sizes $B = 50$, $B = 100$, $B = 200$ and $B = 60,000$ is plotted against the scaled number of iterations. The learning rates are $\eta = 0.025$ for (a), $\eta = 0.05$ for (b) and $\eta = 0.1$ for (c).

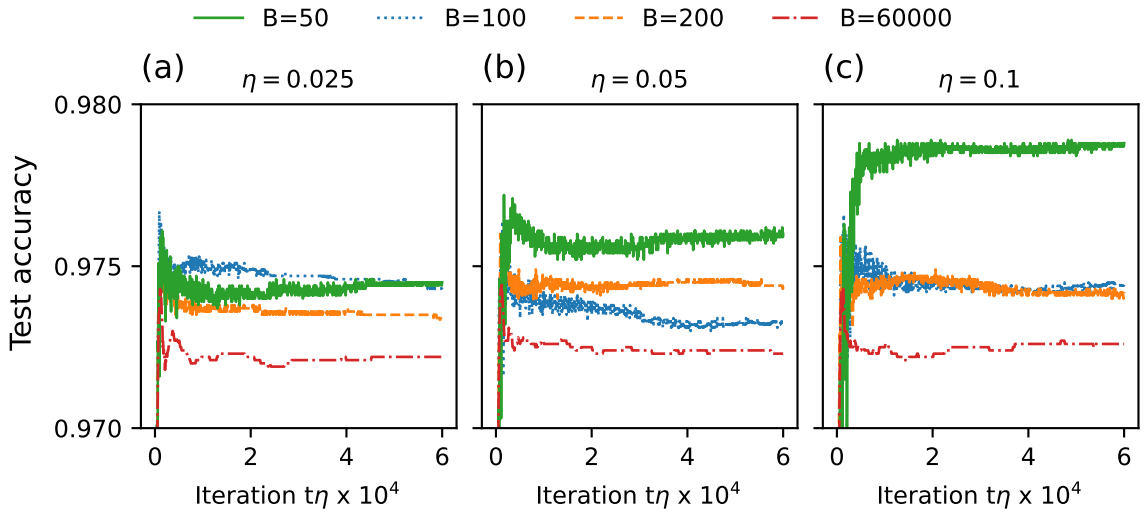


Figure 3.9: Training dynamics of a neural network with two linear layers of 50 neurons each on the MNIST dataset with 60,000 instances of training data and 10,000 instances of testing data. Test accuracy for different batch sizes $B = 50$, $B = 100$, $B = 200$ and $B = 60,000$ is plotted against the scaled number of iterations. The learning rates are $\eta = 0.025$ for (a), $\eta = 0.05$ for (b) and $\eta = 0.1$ for (c).

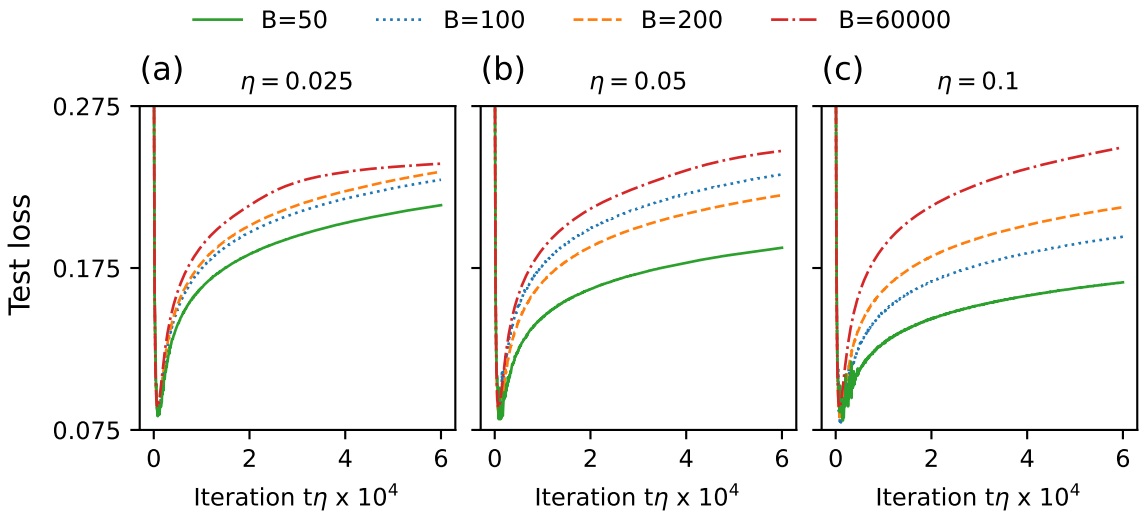


Figure 3.10: Training dynamics of a neural network with two linear layers of 50 neurons each on the MNIST dataset with 60,000 instances of training data and 10,000 instances of testing data. The test loss for different batch sizes $B = 50$, $B = 100$, $B = 200$ and $B = 60000$ is plotted against the scaled number of iterations. The learning rates are $\eta = 0.025$ for (a), $\eta = 0.05$ for (b) and $\eta = 0.1$ for (c).

training loss remain relatively unaffected by the batch size B and the learning rate η. In every investigated constellation the training accuracy quickly reaches a score of exactly or almost 1. The loss accuracy approaches a minimal value of 0 quickly, too. This suggests that for most of the training process the loss either stays at a minimum or moves along a valley of connected minima. Figures 3.8, 3.9, 3.10 are cutouts focused on the influence of the batch size at different fixed learning rates on the flatness, test accuracy and test loss. It is observed that for all learning rates smaller batch sizes lead to increased flatness and higher test accuracy as well as a lower test loss in comparison to larger batch sizes. Analogously, fig. 3.6 implies that higher learning rates lead to increased flatness and higher test accuracy as well as a lower test loss in comparison to lower learning rates. Overall, our observation is the following: Smaller batch sizes and higher learning rates outperform larger batch sizes and lower learning rates in terms of higher flatness, higher test accuracy and lower test loss, while they reach similar values for training accuracy and training loss. It should be noted, that there are slight deviations, for example in fig. 3.9 (a) and (c). For the learning rate $\eta = 0.025$ the batch sizes $B = 50$ and $B = 100$ reach similar test accuracies. The same is true for the learning rate $\eta = 0.1$ and the batch sizes $B = 100$ and $B = 200$. Yet, this does only affect batch sizes within the range of SGD and not the full GD. How should the results be interpreted? The increased testing accuracy and decreased testing loss of smaller batch sizes and larger learning rates imply a better generalization performance on unseen data. The increased flatness is directly connected with smaller eigenvalues of the Hessian. This suggests a smaller curvature of the loss function or, in other terms, a flatter loss landscape in parameter space.

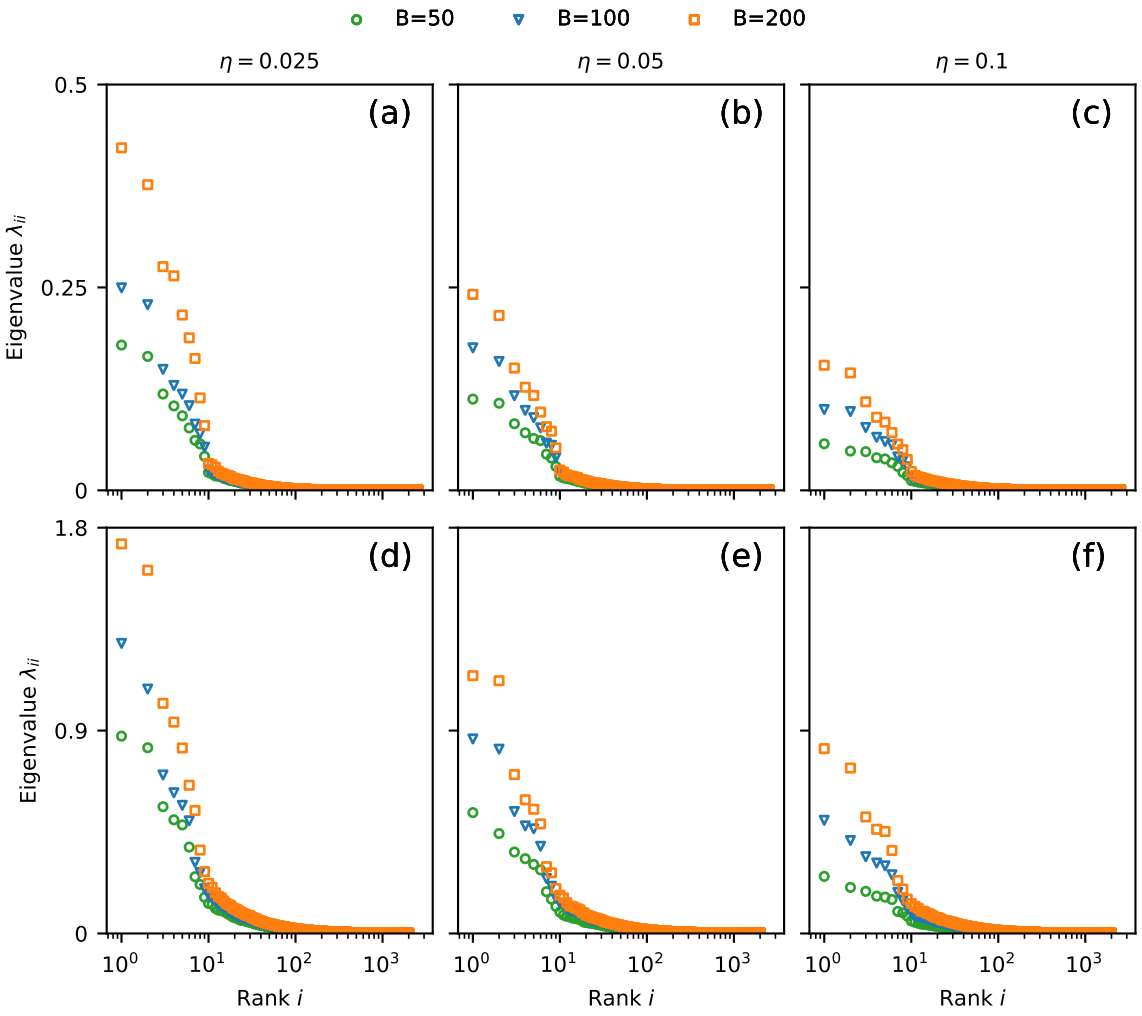


Figure 3.11: High degeneracy of the hessian spectrum. The largest eigenvalues of the Hessian of the loss function after 120,000 iterations, scaled by the learning rate, of training on 60,000 samples from the MNIST dataset. (a), (b), (c) show the exact eigenvalues for the 2550 parameters between the two hidden layers. (d), (e), (f) show numerical estimates of the 2000 largest eigenvalues for all 42310 parameters.

3.5.4 Highly degenerate hessian of the loss

We trained the neural net again for 120,000 iterations, scaled by the learning rate η, and varied the batch size $B \in \{50, 100, 200, 60000\}$ and the learning rate $\eta \in \{0.1, 0.05, 0.025\}$ independently. As Fig. 3.5.4 shows, most of the hessian eigenvalues are close to 0. The hessian is therefore (incompletely) degenerate. We further observe, that the largest eigenvalues are positively correlated with the batch size and negatively correlated with the learning rate. This is the expected behavior given the results of the other experiments regarding the flatness metric, see Fig. 3.8.

3.5.5 Conclusions

In conclusion our empirical observations suggest the following assumptions: For most of the training process the loss function is moving along a valley of connected minima of almost equally low loss. The Hessian of the loss is highly degenerate. The observations also suggest our hypothesis that training with SGD leads to flatter solutions, lower test loss and higher test accuracy than training with GD.

Chapter 4

Loss landscapes: Theoretical models and numerical simulations

In the next section we will introduce a model by Ref. [21] that is based on the assumptions of a minimal loss valley and a degenerate hessian spectrum. We will show that this model reproduces the flatness part of the hypothesis.
During the training process with stochastic gradient descent the loss function traverses a high-dimensional loss landscape. In this section, we establish a model for the time evolution of the loss during this training process. The model was introduced by Ref. [21] and is based on the assumptions of a minimal loss valley and a degenerate hessian that were established in chapter 3. We will carefully construct this model and show that it reproduces the empirically observed behavior of SGD favoring flat minima
Consider a neural network with parameters $\boldsymbol{\theta} \in \mathbb{R}^p$ in parameter space $\mathbb{R}^p$. We recall that the **loss function** $L(\boldsymbol{\theta})$ is defined as the average of the samplewise loss with respect to the whole training dataset of size M,

$$L\left(\{\boldsymbol{s}_i\}_{i=1}^M, \boldsymbol{\theta}\right) = \frac{1}{M}\sum_{i=1}^{M} l(\boldsymbol{s}_i, \boldsymbol{\theta}). \tag{4.1}$$

The Hessian $\mathbf{H}$ of the loss function $L(\boldsymbol{\theta})$ with respect to the parameters $\boldsymbol{\theta}$ is defined by

$$\mathbf{H}(L(\boldsymbol{\theta})) = \nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}} L = \begin{bmatrix} \frac{\partial^2 L}{\partial \theta_1^2} & \frac{\partial^2 L}{\partial \theta_1 \partial \theta_2} & \cdots & \frac{\partial^2 L}{\partial \theta_1 \partial \theta_p} \\ \frac{\partial^2 L}{\partial \theta_2 \theta_1} & \frac{\partial^2 L}{\partial \theta_2 \partial \theta_2} & \cdots & \frac{\partial^2 L}{\partial \theta_2 \partial \theta_p} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 L}{\partial \theta_p \partial \theta_1} & \frac{\partial^2 L}{\partial \theta_p \partial \theta_2} & \cdots & \frac{\partial^2 L}{\partial \theta_p^2} \end{bmatrix} \in \mathbb{R}^p \times \mathbb{R}^p \tag{4.2}$$

and has eigenvalues

$$\Lambda = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_p \end{bmatrix}. \tag{4.3}$$

4.1 Random shift model

We first want to derive a model of the SGD updating rule,

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \nabla_{\boldsymbol{\theta}} L^{\mu}(\boldsymbol{\theta}_t).$$

SGD introduces stochastic noise. To see this we follow [21] and first rewrite Eq. (3.19) to

$$\begin{aligned}\boldsymbol{\theta}_{t+1} &= \boldsymbol{\theta}_t - \eta\nabla_{\boldsymbol{\theta}} L^{\mu}(\boldsymbol{\theta}_t) \\ &= \boldsymbol{\theta}_t - \eta\nabla_{\boldsymbol{\theta}} L^{\mu}(\boldsymbol{\theta}_t) + [\eta\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) - \eta\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t)] \\ &= \boldsymbol{\theta}_t - \eta\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) + [\eta\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) - \eta\nabla_{\boldsymbol{\theta}} L^{\mu}(\boldsymbol{\theta}_t)] \\ &= \boldsymbol{\theta}_t - \eta\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) + \eta[\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) - \nabla_{\boldsymbol{\theta}} L^{\mu}(\boldsymbol{\theta}_t)] \\ &= \boldsymbol{\theta}_t - \eta\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) + \sqrt{\eta}\boldsymbol{\xi}(\boldsymbol{\theta}).\end{aligned} \tag{4.4}$$

Here, in a first step, we add terms summing up to zero to the equation. Then we rearrange the terms and separate η. In the last step we introduce the SGD-noise term

$$\boldsymbol{\xi}(\boldsymbol{\theta}) \equiv \sqrt{\eta}[\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) - \nabla_{\boldsymbol{\theta}} L^{\mu}(\boldsymbol{\theta}_t)]. \tag{4.5}$$

We immediately see that the updating rule of SGD, Eq. (3.19), resembles the updating rule of GD, Eq. (3.18), with an additional term $\sqrt{\eta}\boldsymbol{\xi}(\boldsymbol{\theta})$. In general the minibatch loss $L^{\mu}(\boldsymbol{\theta})$ appearing in SGD is not equal to the loss of the training set, $L(\boldsymbol{\theta})$, used for GD. This difference introduces the additional noise term $\sqrt{\eta}\boldsymbol{\xi}(\boldsymbol{\theta}) = \eta[\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) - \nabla_{\boldsymbol{\theta}} L^{\mu}(\boldsymbol{\theta}_t)]$. Yet, for the ensemble average, we can show that $\langle L^{\mu}(\boldsymbol{\theta})\rangle_{\mu} \approx L(\boldsymbol{\theta})$, because

$$\begin{aligned}\langle L^{\mu}(\boldsymbol{\theta})\rangle_{\mu} &\equiv \frac{1}{N}\sum_{\mu=1}^{N} L^{\mu}(\boldsymbol{\theta}) \\ &= \frac{1}{N}\sum_{\mu=1}^{N}\left(\frac{1}{B}\sum_{i=1}^{B} l(\boldsymbol{s}_{\mu_i};\boldsymbol{\theta})\right) \\ &= \frac{1}{NB}\sum_{n=1}^{NB} l(\boldsymbol{s}_n;\boldsymbol{\theta}) \\ &\approx \frac{1}{M}\sum_{n=1}^{M} l(\boldsymbol{s}_n;\boldsymbol{\theta}) \\ &= L(\boldsymbol{\theta}).\end{aligned} \tag{4.6}$$

Here, the constant M is the number of instances in the full dataset, constant B is the number of instances in each minibatch and constant N is the total number of minibatches. If M/B is a natural number $N \in \mathbb{N}$, then there are N minibatches of equal size B. But if $M/B = N - 1 + c$ with $c \leq 1$ then there is one minibatch of size $cB < B$. One option is to accept one batch with different sample size. Another one is to randomly redraw images from the dataset to fill the last batch to size B. For this reason Eq. (4.6) is an approximation and only an equality if $M/B = N$. In our experiments we only use batch sizes B such that M is divisible by B without rest and $\langle L^{\mu}(\boldsymbol{\theta})\rangle_{\mu} = L(\boldsymbol{\theta})$. Using Eq. (4.6) we can now compute the ensemble average of the difference between the gradient of the loss $L(\boldsymbol{\theta}_t)$ and the gradient of the minibatch loss $L^{\mu}(\boldsymbol{\theta}_t)$,

$$\langle \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) - \nabla_{\boldsymbol{\theta}} L^{\mu}(\boldsymbol{\theta}_t)\rangle_{\mu} = \begin{pmatrix} \left\langle \frac{\partial L}{\partial \theta_1} - \frac{\partial L^{\mu}}{\partial \theta_1} \right\rangle_{\mu} \\ \vdots \\ \left\langle \frac{\partial L}{\partial \theta_i} - \frac{\partial L^{\mu}}{\partial \theta_p} \right\rangle_{\mu} \end{pmatrix} \tag{4.7}$$

with elements

$$\left\langle \frac{\partial L}{\partial \theta_i} - \frac{\partial L^\mu}{\partial \theta_i} \right\rangle_\mu = \frac{\partial L}{\partial \theta_i} - \frac{\partial \langle L^\mu \rangle_\mu}{\partial \theta_i} \approx \frac{\partial L}{\partial \theta_i} - \frac{\partial L}{\partial \theta_i} = 0. \tag{4.8}$$

This implies for the ensemble average of the noise that

$$\langle \boldsymbol{\xi} \rangle_\mu = \sqrt{\eta} \left\langle \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) - \nabla_{\boldsymbol{\theta}} L^\mu(\boldsymbol{\theta}_t) \right\rangle_\mu = \mathbf{0}. \tag{4.9}$$

To further quantify the noise we compute the **covariance matrix**

$$\mathbf{C}(\boldsymbol{\theta}) = \begin{pmatrix} \left\langle \left(\frac{\partial L}{\partial \theta_1} - \frac{\partial L^\mu}{\partial \theta_1}\right)\left(\frac{\partial L}{\partial \theta_1} - \frac{\partial L^\mu}{\partial \theta_1}\right) \right\rangle_\mu & \cdots & \left\langle \left(\frac{\partial L}{\partial \theta_1} - \frac{\partial L^\mu}{\partial \theta_1}\right)\left(\frac{\partial L}{\partial \theta_p} - \frac{\partial L^\mu}{\partial \theta_p}\right) \right\rangle_\mu \\ \vdots & \ddots & \vdots \\ \left\langle \left(\frac{\partial L}{\partial \theta_p} - \frac{\partial L^\mu}{\partial \theta_p}\right)\left(\frac{\partial L}{\partial \theta_1} - \frac{\partial L^\mu}{\partial \theta_1}\right) \right\rangle_\mu & \cdots & \left\langle \left(\frac{\partial L}{\partial \theta_p} - \frac{\partial L^\mu}{\partial \theta_p}\right)\left(\frac{\partial L}{\partial \theta_p} - \frac{\partial L^\mu}{\partial \theta_p}\right) \right\rangle_\mu \end{pmatrix} \tag{4.10}$$

with elements

$$\begin{aligned} C_{ij}(\boldsymbol{\theta}) &:= \left\langle \left(\frac{\partial L}{\partial \theta_i} - \frac{\partial L^\mu}{\partial \theta_i}\right)\left(\frac{\partial L}{\partial \theta_j} - \frac{\partial L^\mu}{\partial \theta_j}\right) \right\rangle_\mu - \left\langle \left(\frac{\partial L}{\partial \theta_i} - \frac{\partial L^\mu}{\partial \theta_i}\right) \right\rangle_\mu \left\langle \left(\frac{\partial L}{\partial \theta_j} - \frac{\partial L^\mu}{\partial \theta_j}\right) \right\rangle_\mu \\ &= \left\langle \left(\frac{\partial L}{\partial \theta_i} - \frac{\partial L^\mu}{\partial \theta_i}\right)\left(\frac{\partial L}{\partial \theta_j} - \frac{\partial L^\mu}{\partial \theta_j}\right) \right\rangle_\mu - \left(\frac{\partial L}{\partial \theta_i} - \frac{\partial \langle L^\mu \rangle_\mu}{\partial \theta_i}\right)\left(\frac{\partial L}{\partial \theta_j} - \frac{\partial \langle L^\mu \rangle_\mu}{\partial \theta_j}\right) \\ &= \left\langle \left(\frac{\partial L}{\partial \theta_i} - \frac{\partial L^\mu}{\partial \theta_i}\right)\left(\frac{\partial L}{\partial \theta_j} - \frac{\partial L^\mu}{\partial \theta_j}\right) \right\rangle_\mu - \left(\frac{\partial L}{\partial \theta_i} - \frac{\partial L}{\partial \theta_i}\right)\left(\frac{\partial L}{\partial \theta_j} - \frac{\partial L}{\partial \theta_j}\right) \\ &= \left\langle \left(\frac{\partial L}{\partial \theta_i} - \frac{\partial L^\mu}{\partial \theta_i}\right)\left(\frac{\partial L}{\partial \theta_j} - \frac{\partial L^\mu}{\partial \theta_j}\right) \right\rangle_\mu, \end{aligned} \tag{4.11}$$

where we abbreviated $L := L(\boldsymbol{\theta})$ for readability and made use of Eq. (4.6) in the third equation. It follows that the covariance matrix of $\boldsymbol{\xi}(\boldsymbol{\theta})$ is $\eta \mathbf{C}(\boldsymbol{\theta})$.
To model SGD, Ref. [21] introduces a model for the stochastic noise. Given model parameters $\boldsymbol{\theta}_t \in \mathbb{R}^d$ at time step t, Ref. [21] approximate the batch loss by

$$L(\boldsymbol{\theta}_t)^\mu = L(\boldsymbol{\theta}_t - \boldsymbol{\mu}(t)) \approx L(\boldsymbol{\theta}_t) - \boldsymbol{\mu}(t) \cdot \nabla_{\boldsymbol{\theta}_t} L(\boldsymbol{\theta}_t). \tag{4.12}$$

The underlying assumption, explained in Ref. [34], is that there is an ensemble of minibatch loss landscapes $\{L^\mu\}_\mu$ with the general loss landscape as the ensemble average $L(\boldsymbol{\theta}) = \langle L^\mu(\boldsymbol{\theta}_t) \rangle_\mu$. Reference [21] makes the simplification that all minibatch loss landscapes can be expressed as shifts of the general loss landscape, i.e. the minibatch loss $L(\boldsymbol{\theta}_t)^\mu$ can be expressed by the full dataset loss $L(\tilde{\boldsymbol{\theta}})$ evaluated at a different point $\tilde{\boldsymbol{\theta}} = \boldsymbol{\theta} - \boldsymbol{\mu}(t)$ in parameter space. Then, the expression is Taylor expanded and only terms up to first order are kept. The random shift noise

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_d \end{pmatrix} \tag{4.13}$$

is a multidimensional stochastic variable, and modeled as Gaussian white noise. It has mean

$$\langle \boldsymbol{\mu} \rangle := \begin{pmatrix} \langle \mu_1 \rangle \\ \vdots \\ \langle \mu_d \rangle \end{pmatrix} = \mathbf{0} \tag{4.14}$$

and covariance $2\sigma\boldsymbol{\Sigma}$ with constant σ. We will refer to σ as the noise strength. The random variables $\mu_1, ..., \mu_d$ are independent and identically distributed, such that $\Sigma_{ij} = \delta_{ij}$. Then, the ensemble average of the modeled minibatch loss function is

$$\begin{aligned}\langle L(\boldsymbol{\theta})^{\mu}\rangle_{\mu} &\approx \langle L(\boldsymbol{\theta}) - \boldsymbol{\mu}\cdot\nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta})\rangle_{\mu} \\ &= \langle L(\boldsymbol{\theta})\rangle_{\mu} - \langle\boldsymbol{\mu}\rangle_{\mu}\cdot\langle\nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta})\rangle_{\mu} \\ &= L(\boldsymbol{\theta}) - \mathbf{0}\cdot\nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta}) \\ &= L(\boldsymbol{\theta}),\end{aligned} \tag{4.15}$$

which matches the average of the real minibatch loss, Eq. (4.6). It should be noted that we are averaging with respect to the ensemble of loss landscapes not with respect to the parameters. This explains why

$$\langle L(\boldsymbol{\theta})\rangle_{\mu} = L(\boldsymbol{\theta}). \tag{4.16}$$

Because of the linearity of the gradient operator, the same applies to the ensemble average of the gradient,

$$\langle\nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta})\rangle_{\mu} = \nabla_{\boldsymbol{\theta}}\langle L(\boldsymbol{\theta})\rangle_{\mu} = \nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta}) \tag{4.17}$$

and the ensemble average of the Hessian,

$$\langle\mathbf{H}(\boldsymbol{\theta})\rangle_{\mu} = \langle\nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta})\rangle_{\mu} = \nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}\langle L(\boldsymbol{\theta})\rangle_{\mu} = \nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta}). \tag{4.18}$$

Further, we multiply the random shift noise $\boldsymbol{\mu}$ with the gradient of the general loss function,

$$\boldsymbol{\mu}\cdot\nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta}) = \begin{pmatrix}\mu_1\\ \vdots \\ \mu_p\end{pmatrix}\cdot\begin{pmatrix}\frac{\partial L(\boldsymbol{\theta})}{\partial\theta_1}\\ \vdots \\ \frac{\partial L(\boldsymbol{\theta})}{\partial\theta_p}\end{pmatrix} = \sum_{i=1}^{p}\mu_i\frac{\partial L(\boldsymbol{\theta})}{\partial\theta_i} \tag{4.19}$$

and can now derive the model for the SGD updating rule. The gradient of the minibatch loss is given by

$$\nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta})^{\mu} \approx \nabla_{\boldsymbol{\theta}}(L(\boldsymbol{\theta}) - \boldsymbol{\mu}\cdot\nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta})). \tag{4.20}$$

We rearrange terms to

$$\nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta})^{\mu} \approx \nabla_{\boldsymbol{\theta}}(L(\boldsymbol{\theta}) - \nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta})\cdot\boldsymbol{\mu}). \tag{4.21}$$

From the linearity of the gradient, follows

$$\nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta})^{\mu} \approx \nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta}) - \nabla_{\boldsymbol{\theta}}\nabla_{\boldsymbol{\theta}}L(\boldsymbol{\theta})\boldsymbol{\mu} \tag{4.22}$$

Inserting the definition of gradient $\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta})$, Eq. (3.13), and Hessian $\mathbf{H}(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta})$ of the loss, Eq. (3.14), gives

$$\begin{aligned}
\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta})^{\mu} &\approx \begin{pmatrix} \frac{\partial L(\boldsymbol{\theta})}{\partial \theta_1} \\ \vdots \\ \frac{\partial L(\boldsymbol{\theta})}{\partial \theta_p} \end{pmatrix} - \begin{pmatrix} H_{11} & \dots & H_{1p} \\ \vdots & \ddots & \vdots \\ H_{p1} & \dots & H_{pp} \end{pmatrix} \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_p \end{pmatrix} \\
&= \begin{pmatrix} \frac{\partial L(\boldsymbol{\theta})}{\partial \theta_1} \\ \vdots \\ \frac{\partial L(\boldsymbol{\theta})}{\partial \theta_p} \end{pmatrix} - \begin{pmatrix} \frac{\partial^2 L}{\partial \theta_1^2} & \cdots & \frac{\partial^2 L}{\partial \theta_1 \partial \theta_p} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 L}{\partial \theta_p \partial \theta_1} & \cdots & \frac{\partial^2 L}{\partial \theta_p^2} \end{pmatrix} \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_p \end{pmatrix} \\
&= \begin{pmatrix} \frac{\partial L(\boldsymbol{\theta})}{\partial \theta_1} \\ \vdots \\ \frac{\partial L(\boldsymbol{\theta})}{\partial \theta_p} \end{pmatrix} - \begin{pmatrix} \frac{\partial}{\partial \theta_1} \left(\sum_{i=1}^{p} \frac{\partial L(\boldsymbol{\theta})}{\partial \theta_i} \mu_i \right) \\ \vdots \\ \frac{\partial}{\partial \theta_p} \left(\sum_{i=1}^{p} \frac{\partial L(\boldsymbol{\theta})}{\partial \theta_i} \mu_i \right) \end{pmatrix}.
\end{aligned} \tag{4.23}$$

The elements of this vector are

$$\frac{\partial L(\boldsymbol{\theta})^{\mu}}{\partial \theta_i} = \frac{\partial L(\boldsymbol{\theta})}{\partial \theta_i} - \sum_{j=1}^{p} H_{ij}(\boldsymbol{\theta}) \mu_j = \frac{\partial L(\boldsymbol{\theta})}{\partial \theta_i} - \frac{\partial}{\partial \theta_i} \left(\sum_{j=1}^{p} \frac{\partial L(\boldsymbol{\theta})}{\partial \theta_j} \mu_j \right), \tag{4.24}$$

where p is the number of parameters. We can now restate the SGD-noise term within the random-shift model. We start from Eq. (4.5),

$$\boldsymbol{\xi}(\boldsymbol{\theta}) = \sqrt{\eta} \left[\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) - \nabla_{\boldsymbol{\theta}} L^{\mu}(\boldsymbol{\theta}_t) \right]. \tag{4.25}$$

Then, inserting the gradient of the minibatch loss function, Eq. (4.20), the SGD-noise is given by

$$\begin{aligned}
\boldsymbol{\xi}(\boldsymbol{\theta}) &= \sqrt{\eta} \left[\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) - \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) + \mathbf{H}(\boldsymbol{\theta}_t) \boldsymbol{\mu} \right] \\
&= \sqrt{\eta}\ \mathbf{H}(\boldsymbol{\theta}_t) \boldsymbol{\mu}.
\end{aligned} \tag{4.26}$$

This relation between the SGD-noise and the Hessian matrix introduces anisotropy in our model of SGD, i.e. a strong dependence on the direction. The ensemble average of the SGD noise term is given by

$$\langle \boldsymbol{\xi}(\boldsymbol{\theta}) \rangle_{\mu} = \langle \sqrt{\eta}\ \mathbf{H}(\boldsymbol{\theta}_t) \boldsymbol{\mu} \rangle_{\mu}. \tag{4.27}$$

Since we are averaging over minibatch loss landscapes, the general loss function and in connection the Hessian is not affected, see 4.18. The ensemble average is therefor given by

$$\langle \boldsymbol{\xi}(\boldsymbol{\theta}) \rangle_{\mu} = \sqrt{\eta}\ \mathbf{H}(\boldsymbol{\theta}_t) \langle \boldsymbol{\mu} \rangle_{\mu} = \mathbf{0}, \tag{4.28}$$

since $\langle \boldsymbol{\mu} \rangle_{\mu} = \mathbf{0}$. The covariance matrix $\mathbf{C}$ of the difference $[\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) - \nabla_{\boldsymbol{\theta}} L^{\mu}(\boldsymbol{\theta}_t)]$ between the gradient of the loss $L(\boldsymbol{\theta}_t)$ and the gradient of the minibatch loss $L^{\mu}(\boldsymbol{\theta}_t)$ is given by Eq.

(4.10). In the random shift model the elements of the covariance matrix evaluate to

$$\begin{aligned}
C_{ij} &= \left\langle \left(\frac{\partial L(\boldsymbol{\theta})}{\partial \theta_i} - \frac{\partial L(\boldsymbol{\theta})^\mu}{\partial \theta_i} \right) \left(\frac{\partial L(\boldsymbol{\theta})}{\partial \theta_j} - \frac{\partial L(\boldsymbol{\theta})^\mu}{\partial \theta_j} \right) \right\rangle_\mu \\
&= \left\langle \left(\frac{\partial L(\boldsymbol{\theta})}{\partial \theta_i} - \left[\frac{\partial L(\boldsymbol{\theta})}{\partial \theta_i} - \sum_{k=1}^{p} H_{ik}(\boldsymbol{\theta}) \mu_k \right] \right) \left(\frac{\partial L(\boldsymbol{\theta})}{\partial \theta_j} - \left[\frac{\partial L(\boldsymbol{\theta})}{\partial \theta_j} - \sum_{l=1}^{p} H_{jl}(\boldsymbol{\theta}) \mu_l \right] \right) \right\rangle_\mu \\
&= \left\langle \left[\sum_{k=1}^{p} H_{ik}(\boldsymbol{\theta}) \mu_k \right] \left[\sum_{l=1}^{p} H_{jl}(\boldsymbol{\theta}) \mu_l \right] \right\rangle_\mu \\
&= \left\langle \sum_{kl}^{p} H_{ik}(\boldsymbol{\theta}) H_{jl}(\boldsymbol{\theta}) \mu_k \mu_l \right\rangle_\mu .
\end{aligned} \tag{4.29}$$

Again, we are averaging over minibatch loss landscapes, such that the Hessian is not affected. The expression therefor simplifies to

$$\begin{aligned}
C_{ij} &= \sum_{kl}^{p} H_{ik}(\boldsymbol{\theta}) H_{jl}(\boldsymbol{\theta}) \left\langle \mu_k \mu_l \right\rangle_\mu \\
&= 2\sigma \sum_{kl}^{p} H_{ik}(\boldsymbol{\theta}) H_{jl}(\boldsymbol{\theta}) \Sigma_{kl}(\boldsymbol{\theta}) \\
&= 2\sigma \sum_{kl}^{p} H_{ik}(\boldsymbol{\theta}) H_{jl}(\boldsymbol{\theta}) \delta_{kl} \\
&= 2\sigma \sum_{k}^{p} H_{ik}(\boldsymbol{\theta}) H_{jk}(\boldsymbol{\theta}) \\
&= 2\sigma \sum_{k}^{p} H_{ik}(\boldsymbol{\theta}) H_{kj}(\boldsymbol{\theta}),
\end{aligned} \tag{4.30}$$

where $H_{jk} = H_{kj}$ follows from the symmetry of the Hessian. This implies that the covariance of the SGD noise term $\boldsymbol{\xi}(\boldsymbol{\theta})$ is

$$\eta \mathbf{C} = 2\eta\sigma \mathbf{H}(\boldsymbol{\theta})^2. \tag{4.31}$$

Since the first moment of the SGD noise is $\langle \boldsymbol{\xi}(\boldsymbol{\theta}) \rangle_\mu = \mathbf{0}$, the second moment is equal to the covariance, i.e.

$$\langle \boldsymbol{\xi}(\boldsymbol{\theta})^2 \rangle_\mu = \eta \mathbf{C} = \eta 2\sigma \mathbf{H}(\boldsymbol{\theta})^2. \tag{4.32}$$

The model for the full SGD algorithm can now be given by,

$$\begin{aligned}
\boldsymbol{\theta}_{t+1} &= \boldsymbol{\theta}_t - \eta \nabla_{\boldsymbol{\theta}} L^\mu(\boldsymbol{\theta}_t) \\
&= \boldsymbol{\theta}_t - \eta \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) + \sqrt{\eta} \boldsymbol{\xi}(\boldsymbol{\theta}). \\
&= \boldsymbol{\theta}_t - \eta \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) + \eta \mathbf{H}(\boldsymbol{\theta}_t) \boldsymbol{\mu}.
\end{aligned} \tag{4.33}$$

We want to derive some of its properties and define the parameter increment

$$\Delta \boldsymbol{\theta}_t = \boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t. \tag{4.34}$$

Then, the ensemble average is given by

$$\langle \Delta \boldsymbol{\theta}_t \rangle_\mu = \langle \boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t \rangle_\mu \,. \tag{4.35}$$

Inserting the SGD algorithm, Eq. (4.33), this gives

$$\begin{aligned} \langle \Delta \boldsymbol{\theta}_t \rangle_\mu &= \langle \boldsymbol{\theta}_t - \eta \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) + \sqrt{\eta} \boldsymbol{\xi}(\boldsymbol{\theta}_t) - \boldsymbol{\theta}_t \rangle_\mu \\ &= \langle -\eta \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) + \sqrt{\eta} \boldsymbol{\xi}(\boldsymbol{\theta}_t) \rangle_\mu \\ &= -\eta \langle \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) \rangle_\mu + \sqrt{\eta} \langle \boldsymbol{\xi}(\boldsymbol{\theta}_t) \rangle_\mu \\ &= -\eta \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t), \end{aligned} \tag{4.36}$$

that is, as expected, deterministic gradient descent. The second moment is given by

$$\left\langle \Delta \boldsymbol{\theta}_t^2 \right\rangle_\mu = \langle (\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t)(\boldsymbol{\theta}_{t+1} - \boldsymbol{\theta}_t) \rangle_\mu \,. \tag{4.37}$$

We again insert the SGD algorithm, Eq. (4.33),

$$\left\langle \Delta \boldsymbol{\theta}_t^2 \right\rangle_\mu = \left\langle \Big(-\eta \nabla_{\boldsymbol{\theta}} L(\boldsymbol{x}) + \sqrt{\eta} \boldsymbol{\xi}(\boldsymbol{x}) \Big) \Big(-\eta \nabla_{\boldsymbol{\theta}} L(\boldsymbol{x}) + \sqrt{\eta} \boldsymbol{\xi}(\boldsymbol{x}) \Big)_\mu \right\rangle \tag{4.38}$$

Multiplying out the brackets, gives

$$\begin{aligned} \left\langle \Delta \boldsymbol{\theta}_t^2 \right\rangle_\mu = \eta^2 &\left\langle \left(\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}) \right)^2 \right\rangle_\mu \\ &+ \eta^{3/2} \left\langle (\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}))(\boldsymbol{\xi}(\boldsymbol{\theta})) \right\rangle_\mu \\ &+ \eta^{3/2} \left\langle (\boldsymbol{\xi}(\boldsymbol{\theta}))(\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta})) \right\rangle_\mu \\ &+ \eta \left\langle \boldsymbol{\xi}(\boldsymbol{\theta})^2 \right\rangle_\mu \end{aligned} \tag{4.39}$$

We go through each of the terms. First, the general loss functions are independent of the minibatch energy landscape and therefore,

$$\eta^2 \left\langle \left(\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}) \right)^2 \right\rangle_\mu = \eta^2 L(\boldsymbol{\theta})^2. \tag{4.40}$$

With the average of the SGD noise, $\langle \boldsymbol{\xi}(\boldsymbol{\theta}) \rangle_\mu = \mathbf{0}$, the second term is given by

$$\eta^{3/2} \left\langle (\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}))(\boldsymbol{\xi}(\boldsymbol{\theta})) \right\rangle_\mu = \eta^{3/2} \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}) \left\langle \boldsymbol{\xi}(\boldsymbol{\theta}) \right\rangle_\mu = \mathbf{0}. \tag{4.41}$$

The same applies to the third term, such that

$$\eta^{3/2} \left\langle (\boldsymbol{\xi}(\boldsymbol{\theta}))(\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta})) \right\rangle_\mu = \eta^{3/2} \left\langle \boldsymbol{\xi}(\boldsymbol{\theta}) \right\rangle_\mu \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}) = \mathbf{0}. \tag{4.42}$$

And for the fourth term we need the second moment of the SGD noise, Eq. (4.32). Then,

$$\eta \left\langle \boldsymbol{\xi}(\boldsymbol{\theta})^2 \right\rangle_\mu = \eta^2 \mathbf{C} = 2\eta^2 \sigma \mathbf{H}(\boldsymbol{\theta})^2. \tag{4.43}$$

Overall, the second moment of the parameter increment is given by

$$\left\langle \Delta \boldsymbol{\theta}_t^2 \right\rangle_\mu = \eta^2 L(\boldsymbol{\theta})^2 + \eta^2 \mathbf{C} = \eta^2 L(\boldsymbol{\theta})^2 + 2\eta^2 \sigma \mathbf{H}(\boldsymbol{\theta})^2. \tag{4.44}$$

4.2 Loss in a minimum valley

To further model the time evolution of the loss function during training with SGD, we need to establish a model of the loss function that is independent of data. Our empirical results suggest that for most of the training process the loss function is at or near a local minimum. To investigate the behavior of the loss function in the vicinity of a minimum, we follow Ref. [5] and perform a Taylor expansion to the second degree around a minimum $L^* = L(\boldsymbol{\theta}_0)$,

$$L(\boldsymbol{\theta}) \approx L(\boldsymbol{\theta}_0) + (\boldsymbol{\theta} - \boldsymbol{\theta}_0)\ \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta})\big|_{\boldsymbol{\theta}_0} + \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)^2\ \mathbf{H}(L(\boldsymbol{\theta}))\big|_{\boldsymbol{\theta}_0} + \mathcal{O}((\boldsymbol{\theta} - \boldsymbol{\theta}_0)^3). \tag{4.45}$$

Since the loss function is at a minimum, the gradient $\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta})\big|_{\boldsymbol{\theta}_0} = \mathbf{0}$. The expression $\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta})\big|_{\boldsymbol{\theta}_0}$ signals that the gradient is first formed and afterward evaluated at the point $\boldsymbol{\theta}_0$. Neglecting terms of order $\mathcal{O}((\boldsymbol{\theta} - \boldsymbol{\theta}_0)^3)$, we can approximate

$$L(\boldsymbol{\theta}) \approx L(\boldsymbol{\theta}_0) + \frac{1}{2}(\boldsymbol{\theta} - \boldsymbol{\theta}_0)^2\ \mathbf{H}(\boldsymbol{\theta})\big|_{\boldsymbol{\theta}_0}. \tag{4.46}$$

Reference [35] shows that under these approximations there exists a transformation Q and a translation vector $\boldsymbol{b}$ with $\boldsymbol{\varphi} = Q\boldsymbol{\theta} - \boldsymbol{b}$, such that

$$L(\boldsymbol{\varphi}) = L^* + \frac{1}{2}\sum_{i=1}^{S} \varphi_i^2 \lambda_i(\varphi_{S+1}, ..., \varphi_N). \tag{4.47}$$

Since L^* is constant, and we are only interested in the dynamics of L, we can set $L^* = 0$, such that

$$L(\boldsymbol{\varphi}) = \frac{1}{2}\sum_{i=1}^{S} \varphi_i^2 \lambda_i(\varphi_{S+1}, ..., \varphi_N). \tag{4.48}$$

For consistency with the rest of this work we relabel $\boldsymbol{\varphi} \to \boldsymbol{\theta}$, such that

$$L(\boldsymbol{\theta}) = \frac{1}{2}\sum_{i=1}^{S} \theta_i^2 \lambda_i(\theta_{S+1}, ..., \theta_N) = \frac{1}{2}\sum_{i=1}^{N_s} \lambda^{(i)}(\boldsymbol{x}) y_i^2. \tag{4.49}$$

Here, the entries of the parameter vector $\boldsymbol{\theta}$ are separated into two classes. The parameters corresponding to the degenerate eigenvalues of the hessian are called **flat variables**,

$$\boldsymbol{x} = (\theta_{N_S+1}, ..., \theta_{N_S+N_F})^\intercal \in \mathbb{R}^{N_F}. \tag{4.50}$$

The parameters corresponding to the non-degenerate eigenvalues of the hessian are called **sharp variables**,

$$\boldsymbol{y} = (\theta_1, ..., \theta_{N_S})^\intercal \in \mathbb{R}^{N_S}. \tag{4.51}$$

Accordingly, the parameter space $\mathbb{R}^p$ has the degenerate subspace $\mathbb{R}^{N_F}$ and the non-degenerate subspace $\mathbb{R}^{N_S}$ and overall

$$\boldsymbol{\theta} = (\boldsymbol{x}, \boldsymbol{y}) \in \mathbb{R}^{N_F+N_S} = \mathbb{R}^p. \tag{4.52}$$

The non-degenerate eigenvalues of the Hessian can be expanded in terms of the flat variables by $\lambda_i(\boldsymbol{x})$.

4.2.1 General model

Reference [21] proposes a two-dimensional model with one non-degenerate sharp variable y and one degenerate flat variable x, implying $N_S = 1$ and $N_F = 1$, $\boldsymbol{\theta} = (x, y)$ and

$$L(\boldsymbol{\theta}) = L(x, y) = \frac{1}{2}\lambda(x)y^2. \tag{4.53}$$

Here, the eigenvalue corresponding to x is degenerate, i.e. the eigenvalue is zero, and the eigenvalue corresponding to y is expanded in terms of the flat variable x by

$$\lambda(x) = \sum_{n=0}^{\infty} \lambda_n x^{2n}. \tag{4.54}$$

For the case that x is not completely degenerate, i.e. the eigenvalue is only close to but not equal to zero, Ref. [21] extend the model to

$$L(x, y) = \frac{1}{2}\lambda_0 \varepsilon x^2 + \frac{1}{2}\lambda(x)y^2 \tag{4.55}$$

with $0 < \varepsilon \ll 1$.
The first and second derivative of the eigenvalue expansion $\lambda(x)$ are

$$\lambda(x)' = \frac{\partial}{\partial x}\lambda(x) = \frac{\partial}{\partial x}\sum_{n=0}^{\infty} \lambda_n x^{2n} = \sum_{n=0}^{\infty} \lambda_n \frac{\partial}{\partial x} x^{2n} = \sum_{n=0}^{\infty} 2n\lambda_n x^{2n-1}, \tag{4.56}$$

$$\lambda(x)'' = \frac{\partial^2}{\partial x^2}\lambda(x) = \frac{\partial}{\partial x}\lambda(x)' = \sum_{n=0}^{\infty} 2n\lambda_n \frac{\partial}{\partial x} x^{2n-1} = \sum_{n=0}^{\infty} 2n(2n-1)\lambda_n x^{2n-2}. \tag{4.57}$$

The first partial derivatives of the loss are then

$$\frac{\partial}{\partial x}L(x, y) = \lambda_0 \varepsilon x + \frac{1}{2}\lambda'(x)y^2, \;\; \frac{\partial}{\partial y}L(x, y) = \lambda(x)y \tag{4.58}$$

and the gradient of the loss is

$$\nabla_{\boldsymbol{\theta}} L = \begin{pmatrix} \frac{\partial L(x,y)}{\partial x} \\ \frac{\partial L(x,y)}{\partial y} \end{pmatrix} = \begin{pmatrix} \lambda_0 \varepsilon x + \frac{1}{2}\lambda'(x)y^2 \\ \lambda(x)y \end{pmatrix}. \tag{4.59}$$

The second partial derivatives are the entries of the Hessian matrix of the loss,

$$\mathbf{H}(x, y) = \begin{bmatrix} H_{xx} & H_{xy} \\ H_{yx} & H_{yy} \end{bmatrix} = \begin{bmatrix} \frac{\partial^2 L}{\partial x^2} & \frac{\partial^2 L}{\partial x \partial y} \\ \frac{\partial^2 L}{\partial y \partial x} & \frac{\partial^2 L}{\partial y^2} \end{bmatrix} = \begin{bmatrix} \lambda_0 \varepsilon + \frac{1}{2}\lambda''(x)y^2 & \lambda'(x)y \\ \lambda'(x)y & \lambda(x) \end{bmatrix}. \tag{4.60}$$

For the completely degenerate loss the Hessian follows from setting $\varepsilon = 0$:

$$\mathbf{H}(x, y) = \begin{bmatrix} \frac{1}{2}\lambda''(x)y^2 & \lambda'(x)y \\ \lambda'(x)y & \lambda(x) \end{bmatrix}. \tag{4.61}$$

4.2.2 Simplifications

For a further analysis we follow Ref. [21] and set $\lambda_n = 0$ for $n > 2$, such that

$$\lambda(x) = \sum_{n=0}^{\infty} \lambda_n x^{2n} = \lambda_0 + \lambda_1 x^2 + \lambda_2 x^4. \tag{4.62}$$

Then the first derivative is

$$\lambda'(x) = 2\lambda_1 x + 4\lambda_2 x^3 \tag{4.63}$$

and the second derivative is

$$\lambda''(x) = 2\lambda_1 + 12\lambda_2 x^2. \tag{4.64}$$

This implies for the loss

$$L(x,y) = \frac{1}{2}\lambda_0 \varepsilon x^2 + \frac{1}{2}\lambda(x) y^2 = \frac{1}{2}\lambda_0 \varepsilon x^2 + \frac{1}{2}(\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4) y^2 \tag{4.65}$$

and for the gradient

$$\nabla_{\boldsymbol{\theta}} L = \begin{pmatrix} \frac{\partial L(x,y)}{\partial x} \\ \frac{\partial L(x,y)}{\partial y} \end{pmatrix} = \begin{pmatrix} \lambda_0 \varepsilon x + \frac{1}{2}\lambda'(x) y^2 \\ \lambda(x) y \end{pmatrix} = \begin{pmatrix} \lambda_0 \varepsilon x + \left(\lambda_1 x + 2\lambda_2 x^3\right) y^2 \\ \left(\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4\right) y \end{pmatrix}. \tag{4.66}$$

4.2.3 Loss analysis

Next, we want to analyze the structure of this simplified loss function. We first need to find the roots of the partial derivative of $L(x,y)$ with respect to x.

$$\begin{aligned} \frac{\partial L(x,y)}{\partial x} = 0 &\iff \lambda_0 \varepsilon x + \left(\lambda_1 x + 2\lambda_2 x^3\right) y^2 = 0 \\ &\iff x(\lambda_0 \varepsilon + y^2 \lambda_1 + 2\lambda_2 x^2 y^2) = 0 \end{aligned} \tag{4.67}$$

and the first solution is $x_0 = 0$. Let $x \neq 0$, then

$$\begin{aligned} \lambda_0 \varepsilon + y^2 \lambda_1 + 2\lambda_2 x^2 y^2 = 0, \\ x^2 = -\frac{\lambda_0 \varepsilon + y^2 \lambda_1}{2\lambda_2 y^2} \end{aligned} \tag{4.68}$$

with solutions $x_{1,2} = \pm\sqrt{-\frac{\lambda_0 \varepsilon + y^2 \lambda_1}{2\lambda_2 y^2}}$. We assumed $y \neq 0$, $\lambda_2 \neq 0$ and $\frac{\lambda_0 \varepsilon + y^2 \lambda_1}{2\lambda_2} < 0$, since the loss function is real valued and $x \neq 0$. Thus, $x_0 = 0$ is the only solution that holds for all $y \in \mathbb{R}$.
For the derivative of $L(x,y)$ with respect to y, we get

$$\frac{\partial L(x,y)}{\partial y} = 0 \iff \left(\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4\right) y = 0. \tag{4.69}$$

The only solution that solves the equation for all x is $y_0 = 0$.
In summary, $(x_0, y_0) = (0,0) \in \mathbb{R}^2$ is the only critical point of the loss function L.
The Hessian of the loss function L is

$$\mathbf{H}(x,y) = \begin{bmatrix} \frac{\partial^2 L}{\partial x^2} & \frac{\partial^2 L}{\partial x \partial y} \\ \frac{\partial^2 L}{\partial y \partial x} & \frac{\partial^2 L}{\partial y^2} \end{bmatrix} = \begin{bmatrix} \lambda_0 \varepsilon + \frac{1}{2}\lambda''(x) y^2 & \lambda'(x) y \\ \lambda'(x) y & \lambda(x) \end{bmatrix}. \tag{4.70}$$

For our choice of coefficients this gives

$$\mathbf{H}(x,y) = \begin{bmatrix} \lambda_0\varepsilon + y^2(\lambda_1 + 6\lambda_2 x^2) & 2y(\lambda_1 x + 2\lambda_2 x^3) \\ 2y(\lambda_1 x + 2\lambda_2 x^3) & \lambda_0 + \lambda_1 x^2 + \lambda_2 x^4 \end{bmatrix}. \tag{4.71}$$

At the critical point $(x_0, y_0) = (0, 0)$ the Hessian evaluates to

$$\mathbf{H}(x_0,y_0) = \begin{bmatrix} \lambda_0\varepsilon & 0 \\ 0 & \lambda_0 \end{bmatrix} \tag{4.72}$$

with determinant

$$\det(\mathbf{H}) = \lambda_0^2\varepsilon \geq 0, \tag{4.73}$$

since $\varepsilon > 0$ and $\lambda_0 \in \mathbb{R}$. Let's assume $\lambda_0 > 0$. Then, the eigenvalues $\lambda_0\varepsilon$ and λ_0 of the hessian are positive, and the hessian is positive definite. This implies, that $L(x, y)$ has a local minimum at $(x_0, y_0) = (0, 0)$, if $\lambda_0 > 0$. For the relationship between hessian eigenvalues and local extrema see for example [36]. To decide if $(x_0, y_0) = (0, 0)$ is a global minimum, we need to explore the limiting behavior of L.

$$\begin{aligned} \lim_{|x|\to\infty} L(x,y) &= \lim_{|x|\to\infty} \left[\frac{1}{2}\lambda_0\varepsilon x^2 + \frac{1}{2}\left(\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4\right) y^2\right] \\ &= \lim_{|x|\to\infty} \left[\frac{1}{2}x^4\left(\lambda_2 y^2 + \frac{\lambda_1 y^2 + \lambda_0\varepsilon}{x^2} + \frac{\lambda_0 y^2}{x^4}\right)\right] \\ &= \infty \cdot \operatorname{sign}(\lambda_2 y^2). \end{aligned} \tag{4.74}$$

$$\begin{aligned} \lim_{|y|\to\infty} L(x,y) &= \lim_{|y|\to\infty} \left[\frac{1}{2}\lambda_0\varepsilon x^2 + \frac{1}{2}\left(\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4\right) y^2\right] \\ &= \lim_{|y|\to\infty} \left[\frac{1}{2}y^2\left(\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4 + \frac{\lambda_0\varepsilon x^2}{y^2}\right)\right] \\ &= \infty \cdot \operatorname{sign}(\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4). \end{aligned} \tag{4.75}$$

So, in combination, sufficient conditions for a global minimum of $L(x, y)$ at the critical point $(x_0, y_0) = (0, 0)$ are $\lambda_0 > 0$, $\lambda_2 \geq 0$ and $\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4 \geq 0$ for all $x \in \mathbb{R}$.

4.2.4 Flatness analysis

Next, we want to inspect the curvature of the loss landscape. Reference [21] defines the flatness $F(x)$ by

$$\begin{aligned} F(x) &= 1/\sqrt{\lambda(x)} \\ &= 1/\sqrt{\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4}. \end{aligned} \tag{4.76}$$

For $F(x)$ to be a well-defined real valued function for all $x \in \mathbb{R}$, we must have $\lambda(x) > 0$ for all $x \in \mathbb{R}$. Particularly, this sets the constraint $\lambda_0 > 0$. $F(x)$ has its maximum values at the values of x that minimize $\lambda(x) = \lambda_0 + \lambda_1 x^2 + \lambda_2 x^4$ with $\lambda(x) > 0$. We now compute the roots of the first derivative.

$$\frac{d\lambda(x)}{dx} = 2\lambda_1 x + 4\lambda_2 x^3 = x(2\lambda_1 + 4\lambda_2 x^2) \stackrel{!}{=} 0 \tag{4.77}$$

has solution $x_0 = 0$. Let $x \neq 0$, then $x_{1,2} = \pm\sqrt{-\frac{\lambda_1}{2\lambda_2}}$ with $\frac{\lambda_1}{\lambda_2} < 0$ and $\lambda_2 \neq 0$. The second derivative is

$$\frac{d^2\lambda(x)}{dx^2} = 2\lambda_1 + 12\lambda_2 x^2. \tag{4.78}$$

At $x_0 = 0$ the second derivative is $\frac{d^2\lambda}{dx^2}(x_0) = 2\lambda_1$ and at $x_{1,2} = \pm\sqrt{-\frac{\lambda_1}{2\lambda_2}}$ the second derivative is $\frac{d^2\lambda}{dx^2}(x_{1,2}) = -4\lambda_1$.
First, let $\lambda_1 > 0$. Then $\frac{d^2\lambda}{dx^2}(x_0) > 0$, $\frac{d^2\lambda}{dx^2}(x_{1,2}) < 0$ and $\lambda(x)$ has its only local minimum at $x_0 = 0$. This implies, the flatness $F(x)$ has its only local maximum

$$F(0) = 1/\sqrt{\lambda(0)} = 1/\sqrt{\lambda_0} \tag{4.79}$$

at $x_0 = 0$ with $\lambda_0 > 0$. From $\lambda_1 \neq 0$ follows the limiting behavior

$$\lim_{|x|\to\infty} F(x) = \lim_{|x|\to\infty} \frac{1}{\sqrt{\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4}} = 0. \tag{4.80}$$

In conclusion, $F(x)$ has a global maximum $F(x_0) = 1/\sqrt{\lambda_0}$ at $x_0 = 0$, if $\lambda_0 > 0$ and $\lambda_1 > 0$. Now, let $\lambda_1 < 0$. Then, $\frac{d^2\lambda}{dx^2}(x_0) < 0$, $\frac{d^2\lambda}{dx^2}(x_{1,2}) > 0$ and $\lambda(x)$ has its only local minima at $x_{1,2} = \pm\sqrt{-\frac{\lambda_1}{2\lambda_2}}$. At $x_{1,2}$ the flatness has local maxima

$$F(x_{1,2}) = \left(\lambda_0 - \lambda_1\frac{\lambda_1}{2\lambda_2} + \lambda_2\frac{\lambda_1^2}{4\lambda_2^2}\right)^{-\frac{1}{2}} = \left(\lambda_0 - \frac{\lambda_1^2}{2\lambda_2} + \frac{\lambda_1^2}{4\lambda_2}\right)^{-\frac{1}{2}} = \left(\lambda_0 - \frac{\lambda_1^2}{4\lambda_2}\right)^{-\frac{1}{2}} \tag{4.81}$$

with constraints $\lambda_0 > 0$, $\lambda_1 < 0$ and $\lambda_0 - \frac{\lambda_1^2}{4\lambda_2} > 0$.

4.2.5 Coinciding extrema

The analysis above enables us to find conditions where $(x_0, y_0) = (0, 0)$ is the global minimum of the loss function $L(x, y)$ and simultaneously $x_0 = 0$ is the global maximum of the Flatness $F(x)$, i.e. the loss function is flattest at its global minimum. We follow Ref. [21] and set $\lambda_0 = 8$, $\lambda_1 = 2$, $\lambda_2 = 0$. We first discuss the loss function. With the given parameters it is expressed by

$$L_1(x, y) = \frac{1}{2}\lambda_0\varepsilon x^2 + \frac{1}{2}(\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4)y^2 = 4\varepsilon x^2 + 4y^2 + x^2y^2. \tag{4.82}$$

We thus have $\lambda_0 = 8 > 0$, $\text{sign}(\lambda_2 y^2) = \text{sign}(0) = 0$ for all y and $\text{sign}(\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4) = \text{sign}(8 + 2x^2) = 1$ for all x. Accordingly, the loss has its global minimum $L(x, y) = 0$ at $(x_L, y_L) = (0, 0)$. For completeness, we show this explicitly. The partial derivates are

$$\frac{\partial L_1}{\partial x}(x, y) = 8\varepsilon x + 2y^2 x, \;\; \frac{\partial L_1}{\partial y}(x, y) = 8y + 2x^2 y \tag{4.83}$$

with roots $x_L = 0$ and $y_L = 0$. The Hessian

$$\mathbf{H}(x_L, y_L) = \begin{bmatrix} 8\varepsilon & 0 \\ 0 & 8 \end{bmatrix} \tag{4.84}$$

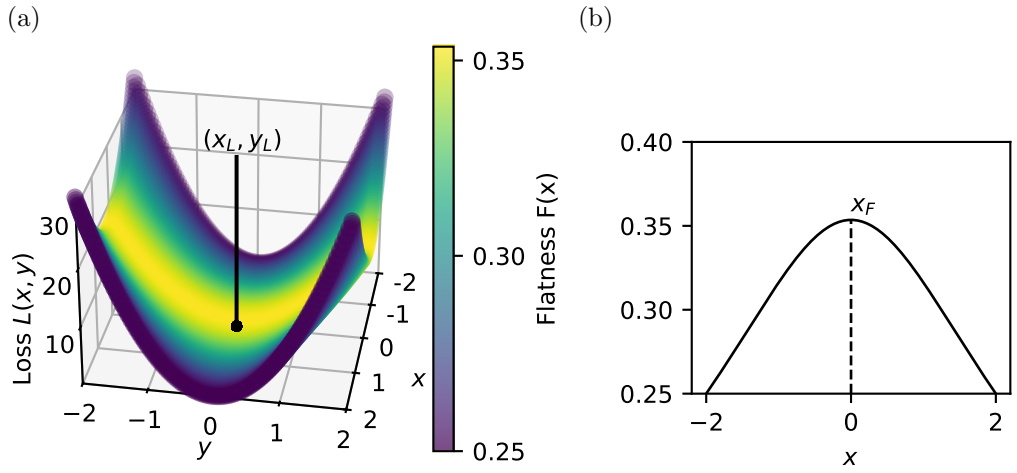


Figure 4.1: Coinciding extrema of loss and flatness at model parameters $\lambda_0 = 8$, $\lambda_1 = 2$, $\lambda_2 = 0$. (a) The loss $L(x, y)$ is plotted against the flat variable x and the sharp variable y. The color bar visualizes the flatness $F(x)$. The global minimum $(x_L, y_L) = (0, 0)$ of the loss coincides with the global maximum of the flatness. (b) The flatness $F(x)$ is plotted against the flat variable x. The curve has global maximum $F(x_F) = 1/\sqrt{\lambda_0} \approx 0.354$ at $x_F = 0$.

is positive definite at $(x_L, y_L) = (0, 0)$, since $\varepsilon > 0$. Analogously, for the given parameters the flatness

$$F(x) = 1/\sqrt{\lambda(x)} = 1/\sqrt{8 + x^2} \tag{4.85}$$

has the global maximum $F(x_F) = 1/\sqrt{\lambda_0} = 1/\sqrt{8} \approx 0.354$ at $x_F = x_L = 0$, since $\lambda_0 = 8 > 0$ and $\lambda_1 = 2 > 0$. It follows that the maximum $(x_L, y_L) = (0, 0)$ of the loss coincides with the maximum $x_F = 0$. A visualization of this loss landscape is shown in Fig. 4.1.

4.2.6 Distinct extrema

Another interesting case is a situation where the minimum of the loss function does not coincide with a minimum of the flatness. Reference [21] suggest the parameters $\lambda_0 = 8$, $\lambda_1 = -2$ and $\lambda_2 = 1$. Then, $\lambda_0 = 8 > 0$, $\lambda_2 = 1 > 0$ and $\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4 > 0$ for all $x \in \mathbb{R}$, and it follows that the loss $L(x, y)$ has a global minimum at $(x_L, y_L) = (0, 0)$. For the flatness $F(x)$ we have $\lambda_0 = 8 > 0$, $\lambda_1 = -2 < 0$ and

$$\lambda_0 - \frac{\lambda_1^2}{4\lambda_2} = 8 - \frac{(-2)^2}{4 \cdot 1} = 8 - \frac{4}{4} = 7 > 0. \tag{4.86}$$

Under these constraints the flatness has two global maxima

$$F(x_{1,2}) = \left(\lambda_0 - \frac{\lambda_1^2}{4\lambda_2}\right)^{-\frac{1}{2}} = 1/\sqrt{7} \approx 0.38 \tag{4.87}$$

at the positions

$$x_{F1} = -\sqrt{-\frac{\lambda_1}{\lambda_2}} = -\sqrt{-\frac{-2}{2 \cdot 1}} = -1 \neq x_L \tag{4.88}$$

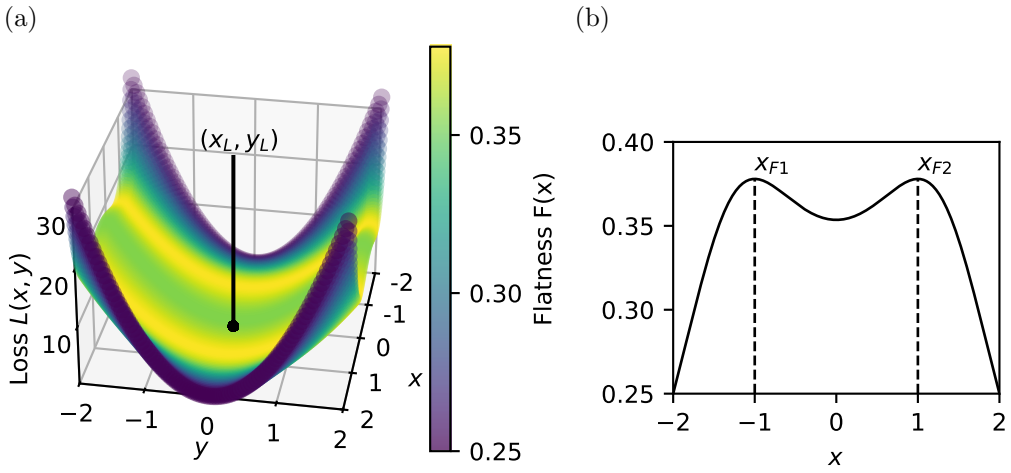


Figure 4.2: Coinciding extrema of loss and flatness at model parameters $\lambda_0 = 8$, $\lambda_1 = -2$, $\lambda_2 = 1$. (a) The loss $L(x, y)$ is plotted against the flat variable x and the sharp variable y. The color bar visualizes the flatness $F(x)$. The global minimum $(x_L, y_L) = (0, 0)$ of the loss does not coincide with the global maxima of the flatness. (b) The flatness $F(x)$ is plotted against the flat variable x. The curve has global maxima $F(x_{1,2}) = \left(\lambda_0 - \frac{\lambda_1^2}{4\lambda_2}\right)^{-\frac{1}{2}} \approx 0.38$ at $x_{F1} = -1$ and $x_{F1} = 1$.

and

$$x_{F2} = \sqrt{-\frac{\lambda_1}{\lambda_2}} = \sqrt{-\frac{-2}{2 \cdot 1}} = 1 \neq x_L. \tag{4.89}$$

In summary, the global minimum of the loss does not coincide with a global maximum of the flatness. The flatness $F(x)$ does however have a local minimum

$$F(x_{F0}) = 1/\sqrt{\lambda_0} = 1/\sqrt{8} \approx 0.354 \tag{4.90}$$

at $x_{F0} = x_L = 0$. A visualization of this loss landscape is shown in Fig. 4.2.

4.3 Combined model for SGD

We now have data independent models for the SGD updating rule, namely the random-shift model, as well as for the loss function. Combining both models, in particular the results Eq. (4.33), Eq. (4.59) and Eq. (4.60), provides a data independent model for the time evolution of the parameters and thus of the loss function during the training process with SGD, i.e. the dynamics of stochastic gradient descent. We first recognize that

$$\boldsymbol{\theta}_t = \begin{pmatrix} x_t \\ y_t \end{pmatrix}, \tag{4.91}$$

such that the parameter updating rule can be expressed as

$$\begin{pmatrix} x_{t+1} \\ y_{t+1} \end{pmatrix} = \begin{pmatrix} x_t \\ y_t \end{pmatrix} - \eta \nabla_{\boldsymbol{\theta}} L^{\mu}(x_t, y_t). \tag{4.92}$$

With Eq. (4.33) from the random shift model, the gradient of the minibatch loss is

$$\nabla_{\boldsymbol{\theta}} L^{\mu}(x,y) = \nabla_{\boldsymbol{\theta}} L(x,y) - \mathbf{H}(x,y) \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}. \tag{4.93}$$

Inserting the gradient, Eq. (4.59), and the Hessian, Eq. (4.60), from the loss model gives

$$\begin{aligned}
\nabla_{\boldsymbol{\theta}} L^{\mu}(x,y) &= \begin{pmatrix} \lambda_0\varepsilon x + \frac{1}{2}\lambda'(x)y^2 \\ \lambda(x)y \end{pmatrix} - \begin{bmatrix} \lambda_0\varepsilon + \frac{1}{2}\lambda''(x)y^2 & \lambda'(x)y \\ \lambda'(x)y & \lambda(x) \end{bmatrix} \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix} \\
&= \begin{pmatrix} \lambda_0\varepsilon x + \frac{1}{2}\lambda'(x)y^2 \\ \lambda(x)y \end{pmatrix} - \begin{pmatrix} \mu_x(\lambda_0\varepsilon + \frac{1}{2}\lambda''(x)y^2) + \mu_y\lambda'(x)y \\ \mu_x\lambda'(x)y + \mu_y\lambda(x) \end{pmatrix} \\
&= \begin{pmatrix} \lambda_0\varepsilon x + \frac{1}{2}\lambda'(x)y^2 - \mu_x(\lambda_0\varepsilon + \frac{1}{2}\lambda''(x)y^2) - \mu_y\lambda'(x)y \\ \lambda(x)y - \mu_x\lambda'(x)y - \mu_y\lambda(x) \end{pmatrix}.
\end{aligned} \tag{4.94}$$

In conclusion, the combined model for the updating rule is

$$\begin{aligned}
\begin{pmatrix} x_{t+1} \\ y_{t+1} \end{pmatrix} &= \begin{pmatrix} x_t \\ y_t \end{pmatrix} - \eta \begin{pmatrix} \lambda_0\varepsilon x_t + \frac{1}{2}\lambda'(x_t)y_t^2 - \mu_{x_t}(\lambda_0\varepsilon + \frac{1}{2}\lambda''(x_t)y_t^2) - \mu_{y_t}\lambda'(x_t)y_t \\ \lambda(x_t)y_t - \mu_{x_t}\lambda'(x_t)y_t - \mu_{y_t}\lambda(x_t) \end{pmatrix} \\
&= \begin{pmatrix} x_t - \eta\left[\lambda_0\varepsilon x_t + \frac{1}{2}\lambda'(x_t)y_t^2 - \mu_{x_t}(\lambda_0\varepsilon + \frac{1}{2}\lambda''(x_t)y_t^2) - \mu_{y_t}\lambda'(x_t)y_t\right] \\ y_t - \eta\left[\lambda(x_t)y_t - \mu_{x_t}\lambda'(x_t)y_t - \mu_{y_t}\lambda(x_t)\right] \end{pmatrix}.
\end{aligned} \tag{4.95}$$

For the simplified model with $\lambda_n = 0$ for $n > 2$ the terms $\lambda(x)$, $\lambda'(x)$ and $\lambda''(x)$ are given by Eq. (4.62), Eq. (4.63) and Eq. (4.64). We thus have for the minibatch loss function

$$\begin{aligned}
L^{\mu}(x,y) &= L((x,y) - \boldsymbol{\mu}) \\
&\approx L(x,y) - \boldsymbol{\mu} \cdot \nabla_{\boldsymbol{\theta}} L(x,y) \\
&= \frac{1}{2}\lambda_0\varepsilon x^2 + \frac{1}{2}\lambda(x)y^2 - \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix} \cdot \begin{pmatrix} \lambda_0\varepsilon x + \frac{1}{2}\lambda'(x)y^2 \\ \lambda(x)y \end{pmatrix} \\
&= \frac{1}{2}\lambda_0\varepsilon x^2 + \frac{1}{2}\lambda(x)y^2 - \mu_x(\lambda_0\varepsilon x + \frac{1}{2}\lambda'(x)y^2) - \mu_y(\lambda(x)y) \\
&= \frac{1}{2}\lambda_0\varepsilon x^2 + \frac{1}{2}y^2\left(\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4\right) - \mu_x\left(\lambda_0\varepsilon x + y^2\left(\lambda_1 x + 2\lambda_2 x^3\right)\right) \\
&\quad - \mu_y y\left(\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4\right).
\end{aligned} \tag{4.96}$$

The parameter update of the flat variable can be expressed by

$$\begin{aligned}
x_{t+1} &= x_t - \eta\left[\lambda_0\varepsilon x_t + \frac{1}{2}\lambda'(x_t)y_t^2 - \mu_{x_t}(\lambda_0\varepsilon + \frac{1}{2}\lambda''(x_t)y_t^2) - \mu_{y_t}\lambda'(x_t)y_t\right] \\
&= x_t - \eta[\lambda_0\varepsilon x_t + (\lambda_1 x_t + 2\lambda_2 x_t^3)y_t^2 - \mu_{x_t}(\lambda_0\varepsilon + (\lambda_1 + 6\lambda_2 x_t^2)y_t^2) \\
&\quad - 2\mu_{y_t}(\lambda_1 x_t + 2\lambda_2 x_t^3)y_t].
\end{aligned} \tag{4.97}$$

The parameter update of the sharp variable can be expressed by

$$\begin{aligned}
y_{t+1} &= y_t - \eta\left[\lambda(x_t)y_t - \mu_{x_t}\lambda'(x_t)y_t - \mu_{y_t}\lambda(x_t)\right] \\
&= y_t - \eta\left[(\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4)y_t - 2\mu_{x_t}(\lambda_1 x_t + 2\lambda_2 x_t^3)y_t - \mu_{y_t}(\lambda_0 + \lambda_1 x^2 + \lambda_2 x^4)\right].
\end{aligned} \tag{4.98}$$

In this model the effect of learning rate η aligns with its effect in SGD. On the contrary, the noise strength σ works in the opposite direction as the batch size B in SGD. Increasing the noise strength σ models decreasing the batch size B.

4.4 Numerical simulations

We want to understand how GD and SGD traverse the loss landscape in the model, Eq. (4.95). For this we implement the numerical experiment performed by [21] for the simplified case with $\lambda_n = 0$ for $n > 2$. The programmed code in this section was first implemented by myself and only afterward LLM based coding assistants were used to check and optimize the code. Implemented suggestions were carefully inspected by myself.

4.4.1 Experimental setup

In the experiment an initial position is given by x_0 and y_0 and has loss

$$L(x_0, y_0) = \frac{1}{2}\lambda_0 \varepsilon x_0^2 + \frac{1}{2}\lambda(x_0) y_0^2. \tag{4.99}$$

The time evolution of the loss function is governed by the time evolution of the parameters (x, y) that is given by Eq. (4.95). The experiment runs until it fulfills the stationarity condition at iteration T with

$$\left| \frac{1}{T+1} \sum_{t=1}^{T+1} x(t) - \frac{1}{T} \sum_{t=1}^{T} x(t) \right| < 10^{-6}\eta. \tag{4.100}$$

The steady state solution is defined as the average

$$x_{ss} = \frac{1}{T} \sum_{t=1}^{T} x(t) \tag{4.101}$$

and the flatness of the steady state solution is defined by

$$F_{ss} = \frac{1}{\sqrt{\lambda(x_{ss})}}. \tag{4.102}$$

Reference [21] further defines

$$\langle x^2 \rangle_{ss} = \frac{1}{T} \sum_{t=1}^{T} x(t)^2, \tag{4.103}$$

$$\langle y^2 \rangle_{ss} = \frac{1}{T} \sum_{t=1}^{T} y(t)^2 \tag{4.104}$$

as well as the variance

$$V_x = \langle x^2 \rangle_{ss} - x_{ss}^2 \tag{4.105}$$

and

$$V_y = \langle y^2 \rangle_{ss} - y_{ss}^2. \tag{4.106}$$

Once the steady state x_{ss} is established, we estimate the first passage time $\tilde{t}$ of iterations to first reach it. For this the simulation is run a number of $N = 100$ times and $\tilde{t}$ is the time, where

$$x_{\tilde{t}} = x_{ss}((x_{\tilde{t}} - x_{ss})(x_{\tilde{t}+1} - x_{ss})) < 0. \tag{4.107}$$

Then, the search time or convergence time τ is defined as the average of the first passage times scaled by the learning rate,

$$\tau = \eta\langle\tilde{t}\rangle = \eta \sum_{n=1}^{N=100} \tilde{t}_n. \tag{4.108}$$

We further specify maximum numbers of iterations to stop divergent experiments. For the estimation of the steady state x_{ss} we set the maximum number of iterations to $N_{ss} = 10,000,000$ and for the estimation of passage times we set the limit of iterations to $N_{\tilde{t}} = 1,000,000$. The experiment depends on several constants that have to be specified. That is the initial position (x_0, y_0), the expansion coefficients λ_0, λ_1 and λ_2, the constant regulating incomplete degeneracy, ε, as well as the learning rate η and the noise strength σ that influences the variance of the noise term $\boldsymbol{\mu}$. For our experiments we set $(x_0, y_0) = (1.5, 1.5)$ and $\varepsilon = 0.001$ as specified by Ref. [21]. We are interested in the effects of learning rate η and noise strength σ on the experimental results. We therefor repeat the experiment, while varying noise strength

$$\sigma \in \{10^{-2.5}, 10^{-2.45}, \ldots, 10^{0.95}, 10^{1}\} \tag{4.109}$$

as well as learning rate

$$\eta \in \{10^{-2.5}, 10^{-2.45}, \ldots, 10^{0.95}, 10^{1}\} \tag{4.110}$$

independently. Before we discuss the simulation results for different choices of coefficients, we want to remind of the connection between curves in logarithmic scale to those in with a linear scaling. Suppose we plot our data points in logarithmic scale and fit them with a curve that that is linear with slope $\alpha = 0.5$ in logarithmic scale. This means we have a relation of the form

$$\log_{10}[\sigma] = a + b \cdot \log_{10}[\eta] \tag{4.111}$$

which can be transformed to

$$\sigma = 10^{a+b\cdot\log_{10}[\eta]} = 10^{a} \cdot \eta^{b}. \tag{4.112}$$

This means that a linear function in logarithmic space translates to an exponential function in linear space.

4.4.2 Simulation of coinciding extrema

In a first setup, we choose the parameters $\varepsilon = 0.001$, $\lambda_0 = 8$, $\lambda_1 = 2$, $\lambda_2 = 0$ and $\lambda_n = 0$ for all $n > 2$. This produces the loss landscape explored in section 4.2.5. In Fig. 4.3 the largest values of the noise strength σ for which the experiment converged are plotted for all learning rates η. In the figure the relation between the largest still converging noise strength and the respective learning rate is estimated by $\sigma_{div} \approx 10^{-1.79}\eta^{-1.28}$ or equivalently by

$$\sigma_{div} \cdot \eta^{1.28} \approx 10^{-1.79} \approx 0.016. \tag{4.113}$$

The heatmaps of our results, e.g. Fig. 4.4, visualize that there is a region of converging and a separated region of diverging experiments. The region of convergence is roughly bound by the vertical line at $\eta = 10^{-0.65}$ from the right and by the relation $\sigma_{div}\eta^{1.28} \approx 0.016$

from the top. This sets an approximated upper boundary $\Delta s_{div} \approx 0.016$ for the maximal effective noise strength before experiments begin to diverge. The heatmap in Fig. 4.4 shows almost constant steady state flatness $F(x)$ within the convergent region. There is a small subregion with slightly lower flatness values parallel to the divergence boundary. We also notice that the flatness is relatively close to the global maximum, $F(x_F) \approx 0.354$ at $x_F = 0$, that we established in 4.2.5. Additionally, we learn from Fig. 4.5 and Fig. 4.6 that most steady state solutions are at or close to $x_{ss} = 0$ and $y_{ss} = 0$ compared to the initial position $(x_0, y_0) = (1.5, 1.5)$. Our interpretation is that as long as the experiment converges, the steady state solution will be close to the minimum of the loss landscape and therefor also at the point of maximal flatness, irrespective of the noise strength and learning rate. This is expected, since global minimum and flattest minimum are coinciding. The effects of varying the noise strength and the learning rate are however observable in the variance V_x, the variance Vy and the convergence time τ, see Fig. 4.7, Fig. 4.8 and Fig. 4.9. In the heatmaps, the variances V_x and V_y are increasing from the origin of the coordinate system at $(\eta, \sigma) = (10^{-2.5}, 10^{-2.5})$ towards the boundary of the convergence region. This increase is much smoother for V_y compared to V_x. Conversely, the convergence time τ is decreasing smoothly from the coordinate origin towards the boundary. Overall, we interpret the results in the following way. Approaching $\sigma_{div}\eta^{1.28} \approx 0.016$ from the coordinate origin leads to larger variances V_x and Vy of the steady state solutions x_{ss} and y_{ss} and faster convergence times τ, while no systematical effect on the flatness $F(x)$ and the steady state solution x_{ss} is noticeable.

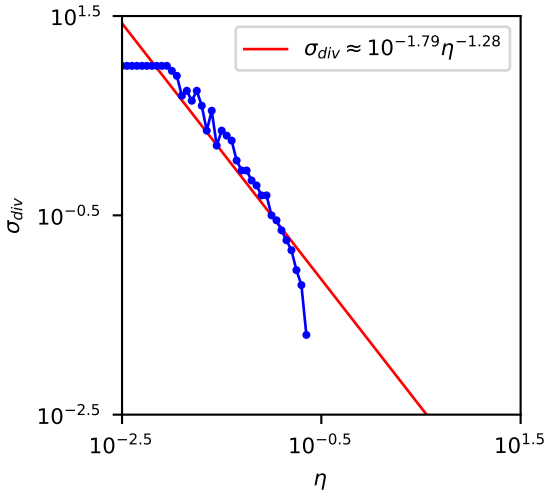


Figure 4.3: Numerical simulation of the two-dimensional loss model with $\lambda_0 = 8$, $\lambda_1 = 2$, $\lambda_2 = 0$. The largest still converging noise strength σ is plotted against the respective learning rate η. The curve is approximated by $\sigma_{div} \approx 10^{-1.79}\eta^{-1.28}$.

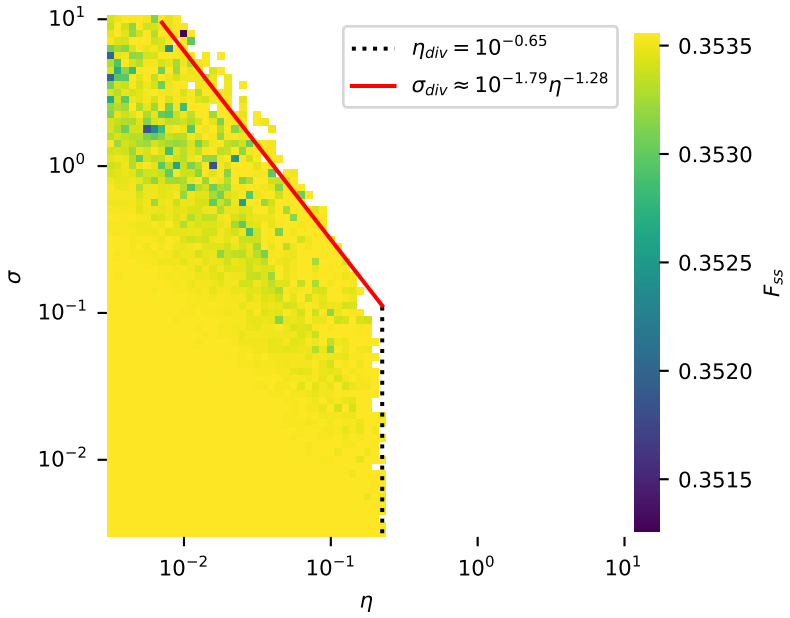


Figure 4.4: Flatness F_{ss} at the steady state solution in the numerical simulation of the two-dimensional loss model with $\lambda_0 = 8$, $\lambda_1 = 2$, $\lambda_2 = 0$. The heatmap of F_{ss} depends on the noise strength σ and the learning rate η. The convergence region has boundaries $\sigma_{div} = 10^{-1.79}\eta^{-1.28}$ and $\eta = 10^{-0.65}$.

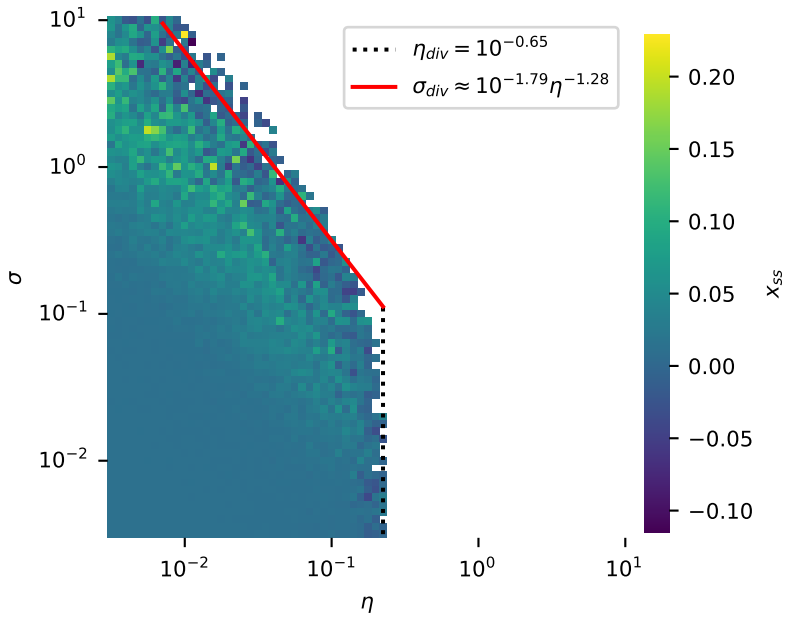


Figure 4.5: Steady state solution x_{ss} in the numerical simulation of the two-dimensional loss model with $\lambda_0 = 8$, $\lambda_1 = 2$, $\lambda_2 = 0$. The heatmap of x_{ss} depends on the noise strength σ and the learning rate η. The convergence region has boundaries $\sigma_{div} = 10^{-1.79}\eta^{-1.28}$ and $\eta = 10^{-0.65}$.

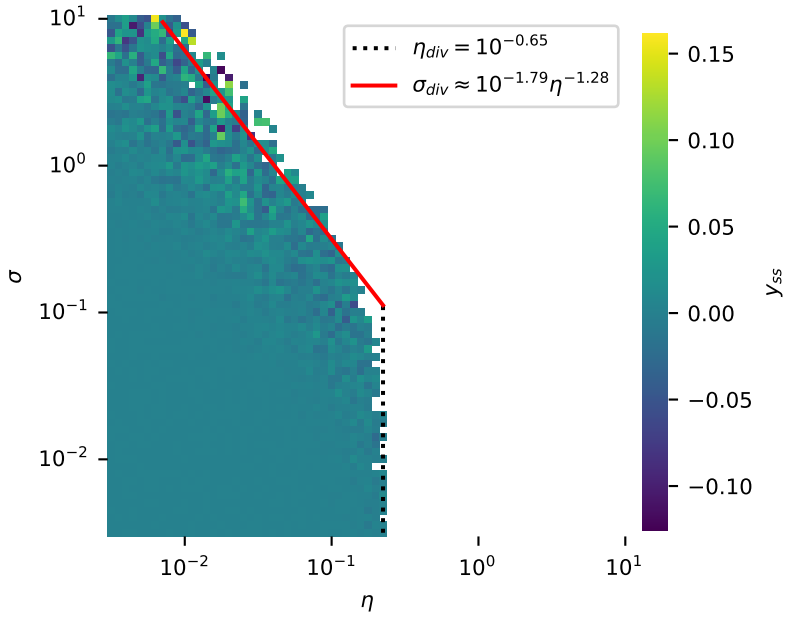


Figure 4.6: Steady state solution y_{ss} in the numerical simulation of the two-dimensional loss model with $\lambda_0 = 8$, $\lambda_1 = 2$, $\lambda_2 = 0$. The heatmap of y_{ss} depends on the noise strength σ and the learning rate η. The convergence region has boundaries $\sigma_{div} = 10^{-1.79}\eta^{-1.28}$ and $\eta = 10^{-0.65}$.

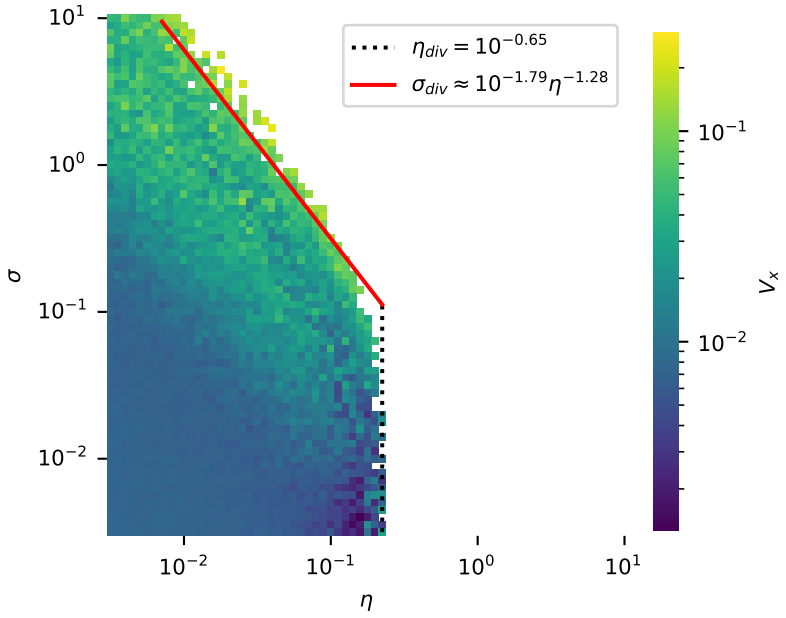


Figure 4.7: Variance V_x of the steady state solution x_{ss} in the numerical simulation of the two-dimensional loss model with $\lambda_0 = 8$, $\lambda_1 = 2$, $\lambda_2 = 0$. The heatmap of V_x depends on the noise strength σ and the learning rate η. The convergence region has boundaries $\sigma_{div} = 10^{-1.79}\eta^{-1.28}$ and $\eta = 10^{-0.65}$.

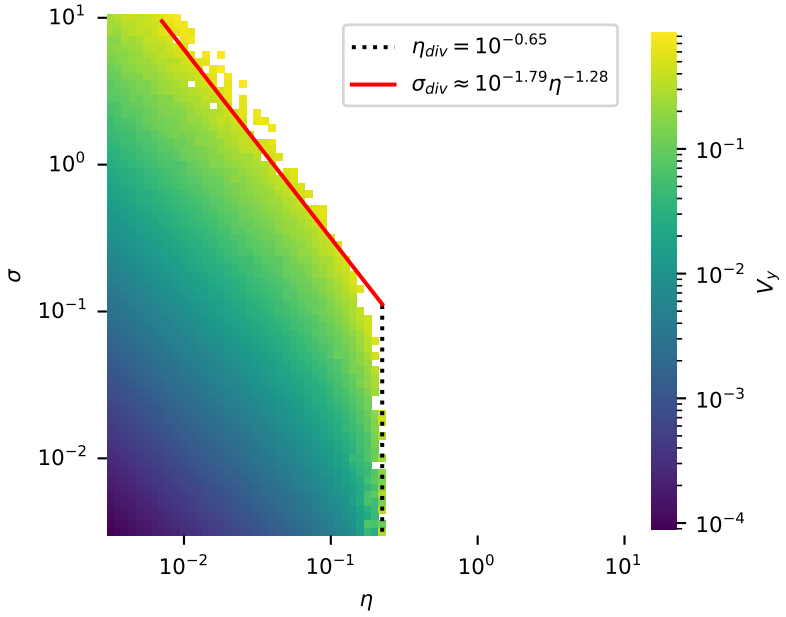


Figure 4.8: Variance V_y of the steady state solution x_{ss} in the numerical simulation of the two-dimensional loss model with $\lambda_0 = 8$, $\lambda_1 = 2$, $\lambda_2 = 0$. The heatmap of V_y depends on the noise strength σ and the learning rate η. The convergence region has boundaries $\sigma_{div} = 10^{-1.79}\eta^{-1.28}$ and $\eta = 10^{-0.65}$.

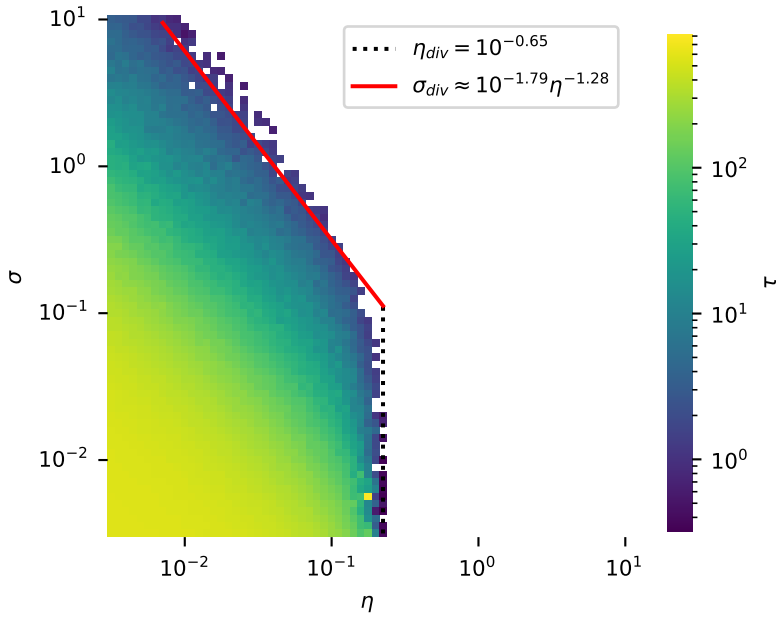


Figure 4.9: Convergence time τ of the steady state solution x_{ss} in the numerical simulation of the two-dimensional loss model with $\lambda_0 = 8$, $\lambda_1 = 2$, $\lambda_2 = 0$. The heatmap of τ depends on the noise strength σ and the learning rate η. The convergence region has boundaries $\sigma_{div} = 10^{-1.79}\eta^{-1.28}$ and $\eta = 10^{-0.65}$.

4.4.3 Simulation of distinct extrema

We now discuss the more interesting case of not coinciding extrema. As parameters, we have $\varepsilon = 0.001$, $\lambda_0 = 8$, $\lambda_1 = -2$, $\lambda_2 = 1$ and $\lambda_n = 0$ for all $n > 2$ as in section 4.2.6. We again plot the largest noise strength σ for each η where the experiment does still converge, see 4.10, and estimate the relation $\sigma_{div} \approx 10^{-2.27}\eta^{-1.36}$ or

$$\sigma_{div} \cdot \eta^{1.36} \approx 10^{-2.27} \approx 0.0054. \tag{4.114}$$

Now, the region of convergence is approximately bound by the vertical line at $\eta = 10^{-1.60}$ from the right and by the relation $\sigma_{div} \cdot \eta^{1.36} \approx 0.0054$ from the top. For the flatness $F(x)$, we recognize a phase transition. Figure 4.11 visualizes our procedure for identifying the critical points $\sigma^*_{F_{ss}}$ for order parameter F_{ss}. In panel (a) the Flatness F_{ss} is plotted against the noise strength σ at constant learning rate $\eta = 10^{-1.4}$. The critical point is the noise strength $\sigma^*_{F_{ss}}$ where the curve has maximal curvature. To automatically identify this point for all learning rates η we applied an implementation of the kneedle algorithm, see refs. [37] and [38]. In panel (b) the critical points are plotted against the respective learning rates and a linear fit was performed within logarithmic scale. The corresponding relation between critical point $\sigma^*_{F_{ss}}$ and learning rate η is given by

$$\sigma^*_{F_{ss}} \approx 10^{-3.47}\eta^{-1.05}. \tag{4.115}$$

In panel (c) we can observe that this relation marks a visible phase transition. As noticed by [21] the convergence region is divided by this transition line into a region of lower and higher flatness $F(x)$. We particularly find that before the phase transition, the flatness is mostly constant and close to the local minimum $F(x_F) \approx 0.354$ at $x_F = 0$. Starting at the coordinate origin and increasing the noise strength, corresponding to decreasing the batch size, and increasing the learning rate, leads to an increase of flatness at the transition line and within the second phase until the boundary of the convergence region is reached. Near this boundary we find flatness values close to $F(x_F) \approx 0.38$ that is the global maximum of the flatness. We apply a similar procedure to find the phase transition line for the steady state solution x_{ss}, see Fig. 4.12, and derive the equation

$$\sigma^*_{x_{ss}} \approx 10^{-3.36}\eta^{-0.98} \tag{4.116}$$

that relates critical points $\sigma^*_{x_{ss}}$ with the respective learning rates $\eta^{-0.98}$. For x_{ss}, we again notice this structure of almost constant $x_{ss} = 0$ within the first phase before the phase transition and then steadily increasing x_{ss} until the boundary of the convergence region is approached. Interestingly we find both at this boundary, values of x_{ss} close to 1 and those close to -1. If we compare the steady state solutions with the flatness, we find the following. The steady state solutions are $x_{ss} \approx 0$ in the same region where the flatness is close to the local minimum $F(x_F) \approx 0.354$. Additionally, we have $x_{ss} \approx 1$, $x_{ss} \approx -1$ and $F(x_F) \approx 0.38$ at the boundary. This is in agreement with the global maxima at $x_{F1} = 1$ and $x_{F2} = -1$ of the flatness, derived in section 4.2.6. The region before the phase transition is characterized by low learning rates and low noise strengths and therefor models variants of SGD with low learning rates and high batch sizes, i.e. SGD that is close to GD. At the boundary of the convergence region we have high learning rates and high noise strengths, i.e. we are modelling SGD with small batch sizes. The model is therefor in agreement with the hypothesis that SGD favors flat minima in contrast to GD. Within a valley of almost

similar low loss, our model of SGD arrives at flatter solutions than the model of GD, in favor of reaching the global minimum of the loss. In addition, the phase transition implies that there might be a clear threshold of learning rate η and noise strength σ after which this flatness inducing behavior emerges.
For both, the coinciding and the not coinciding case the boundary of the convergence region hints at another threshold of learning rate η and noise strength σ after which finding a sufficient minimum is not possible anymore. We now want to discuss the variances V_x and V_y as well as the convergence time τ. If we plot the V_x against the noise strength σ for varied learning rates η, see Fig. 4.14, we consistently find a local maximum at $\sigma^*_{V_x}$ that follows the relation

$$\sigma^*_{V_x} \approx 10^{-3.5}\eta^{-1.14}. \tag{4.117}$$

For V_y, see Fig. 4.15, we find smoothly increasing values from the origin towards the boundary of the convergence region. The exponential growth is of the form

$$V_y \approx 10^{-0.5}\eta^{0.96}. \tag{4.118}$$

For the convergence time τ we can plot τ against σ and for varies learning rates η, see Fig. 4.16. Again, we consistently find a local maximum at $\sigma^*_{V_x}$ with the relation

$$\sigma^*_{\tau} \approx 10^{-3.55}\eta^{-1.11}. \tag{4.119}$$

Starting from the origin the convergence time τ is increasing towards this phase transition. But, in contrast to the behavior observed in F_{ss}, x_{ss} and V_x, τ is decreasing again after the phase transition and towards the boundary of the convergence region. We notice that the equation governing the phase transitions, Eq. 4.115, 4.116, 4.117 and 4.119, are quite similar for F_{ss}, x_{ss}, V_x and τ. Therefor, we follow [21] and introduce the effective noise

$$\Delta s = \sigma \cdot \eta. \tag{4.120}$$

For the phase transition we make the approximation $\sigma^* \approx 10^{-3.3}\eta^{-1}$ or

$$\Delta s^* = \sigma^* \cdot \eta \approx 10^{-3.3} \approx 0.0005 = -\lambda_1 \varepsilon \tag{4.121}$$

with our model parameters $\lambda_1 = -2$ and $\varepsilon = 0.001$.

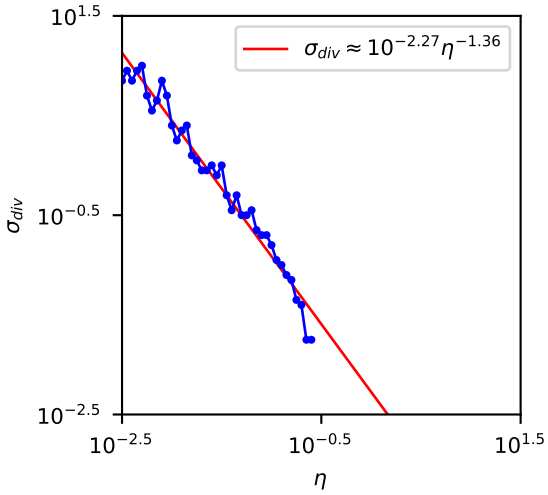


Figure 4.10: Numerical simulation of the two-dimensional loss model with $\lambda_0 = 8$, $\lambda_1 = 2$, $\lambda_2 = 0$. The largest not diverging noise strength σ is plotted against the respective learning rate η. The curve is approximated by $\sigma_{div} \approx 10^{-2.27}\eta^{-1.36}$.

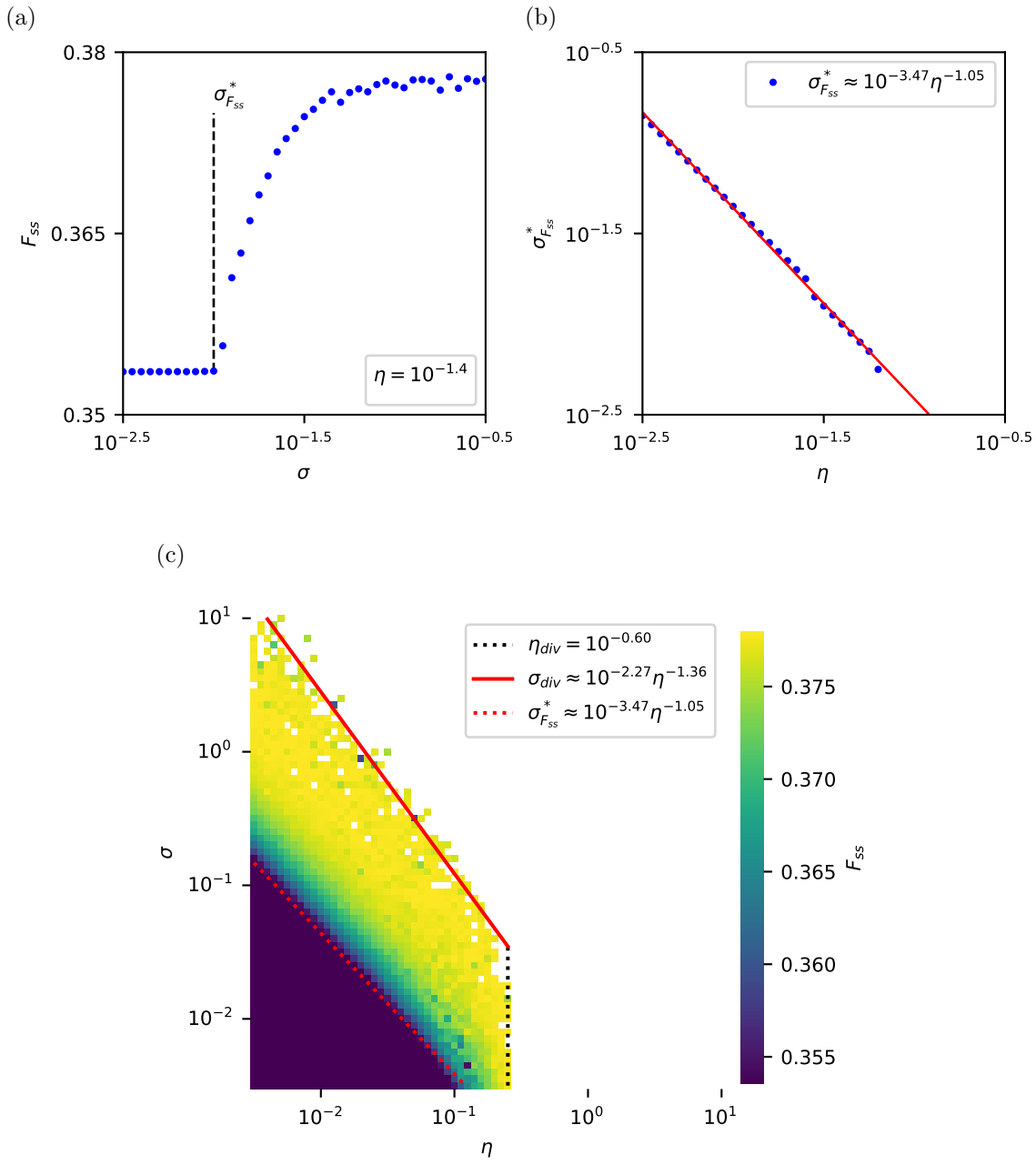


Figure 4.11: Flatness F_{ss} at the steady state solution in the numerical simulation of the two-dimensional loss model with $\lambda_0 = 8$, $\lambda_1 = -2$, $\lambda_2 = 1$. (a) The flatness F_{ss} is plotted against the noise strength σ at fixed learning rate $\eta = 10^{-1.4}$. The critical point $\sigma^*_{F_{ss}}$ marks the point of largest curvature. (b) The noise strength at critical points $\sigma^*_{F_{ss}}$ is plotted against learning rate η. The relation is $\sigma^*_{F_{ss}} \approx 10^{-3.47}\eta^{-1.05}$. (c) The Heatmap of F_{ss} depends on the noise strength σ and the learning rate η. The convergence region has boundaries $\sigma_{div} = 10^{-2.27}\eta^{-1.36}$ and $\eta = 10^{-0.60}$. There is a phase transition, characterized by the relation from (b).

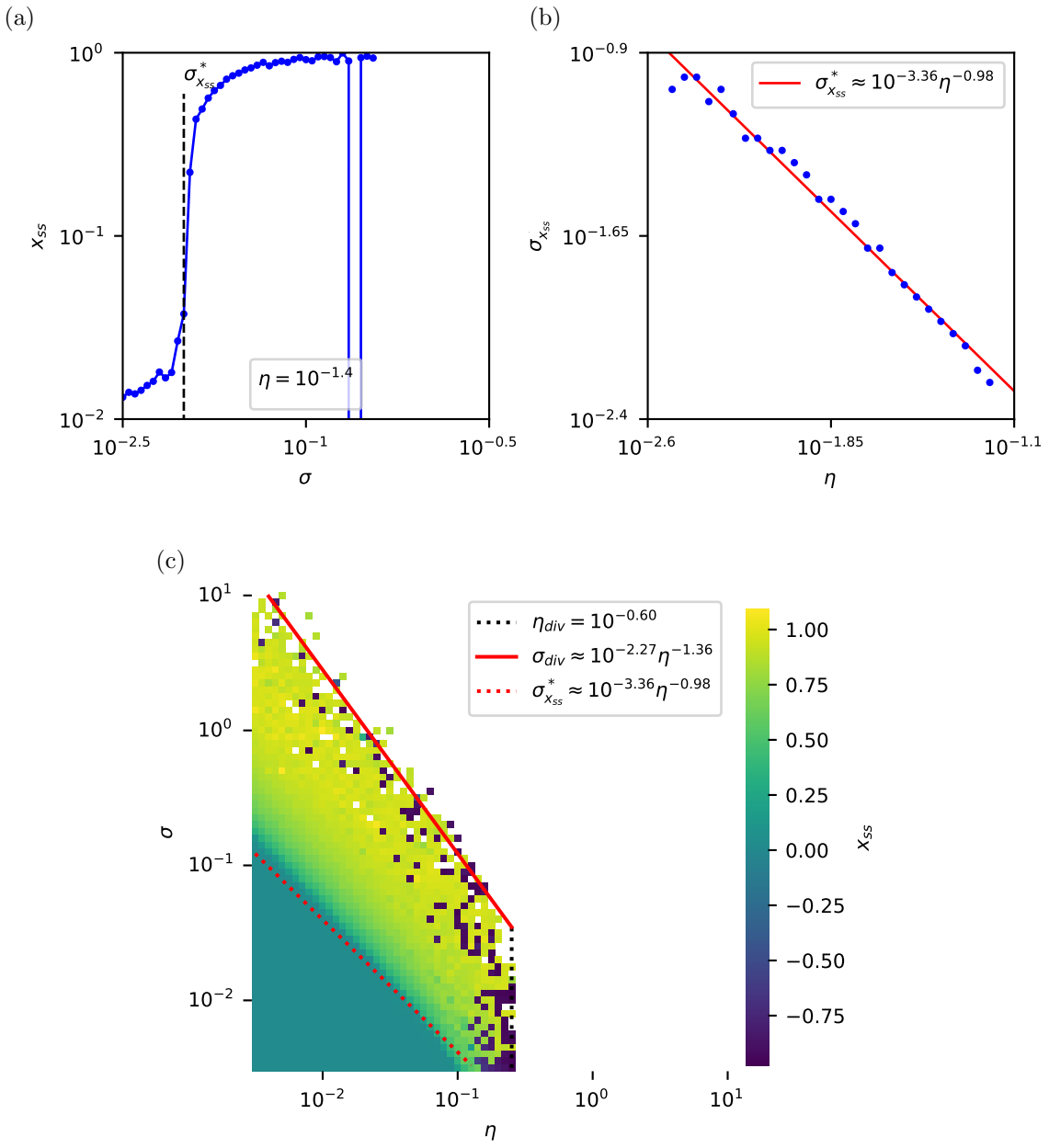


Figure 4.12: Steady state solution x_{ss} in the numerical simulation of the two-dimensional loss model with $\lambda_0 = 8$, $\lambda_1 = -2$, $\lambda_2 = 1$. (a) The steady state x_{ss} is plotted against the noise strength σ at fixed learning rate $\eta = 10^{-1.4}$. The critical point $\sigma^*_{x_{ss}}$ marks the point of largest curvature. (b) The noise strength at critical points $\sigma^*_{x_{ss}}$ is plotted against learning rate η. The relation is $\sigma^*_{x_{ss}} \approx 10^{-3.36}\eta^{-0.98}$. (c) The heatmap of x_{ss} depends on the noise strength σ and the learning rate η. The convergence region has boundaries $\sigma_{div} = 10^{-2.27}\eta^{-1.36}$ and $\eta = 10^{-0.60}$. There is a phase transition, characterized by the relation from (b).

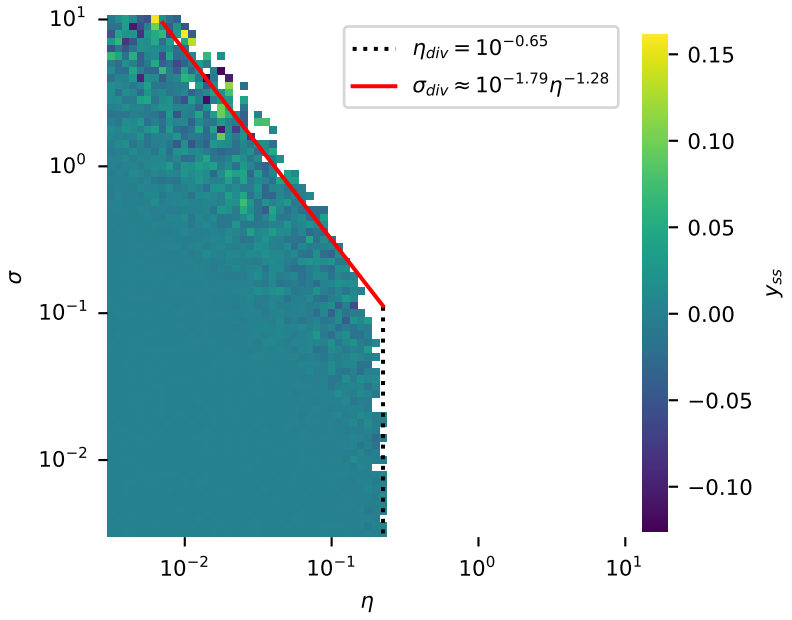


Figure 4.13: Steady state solution y_{ss} in the numerical simulation of the two-dimensional loss model with $\lambda_0 = 8$, $\lambda_1 = -2$, $\lambda_2 = 1$. The heatmap of y_{ss} depends on the noise strength σ and the learning rate η. The convergence region has boundaries $\sigma_{div} = 10^{-2.27}\eta^{-1.36}$ and $\eta = 10^{-0.60}$.

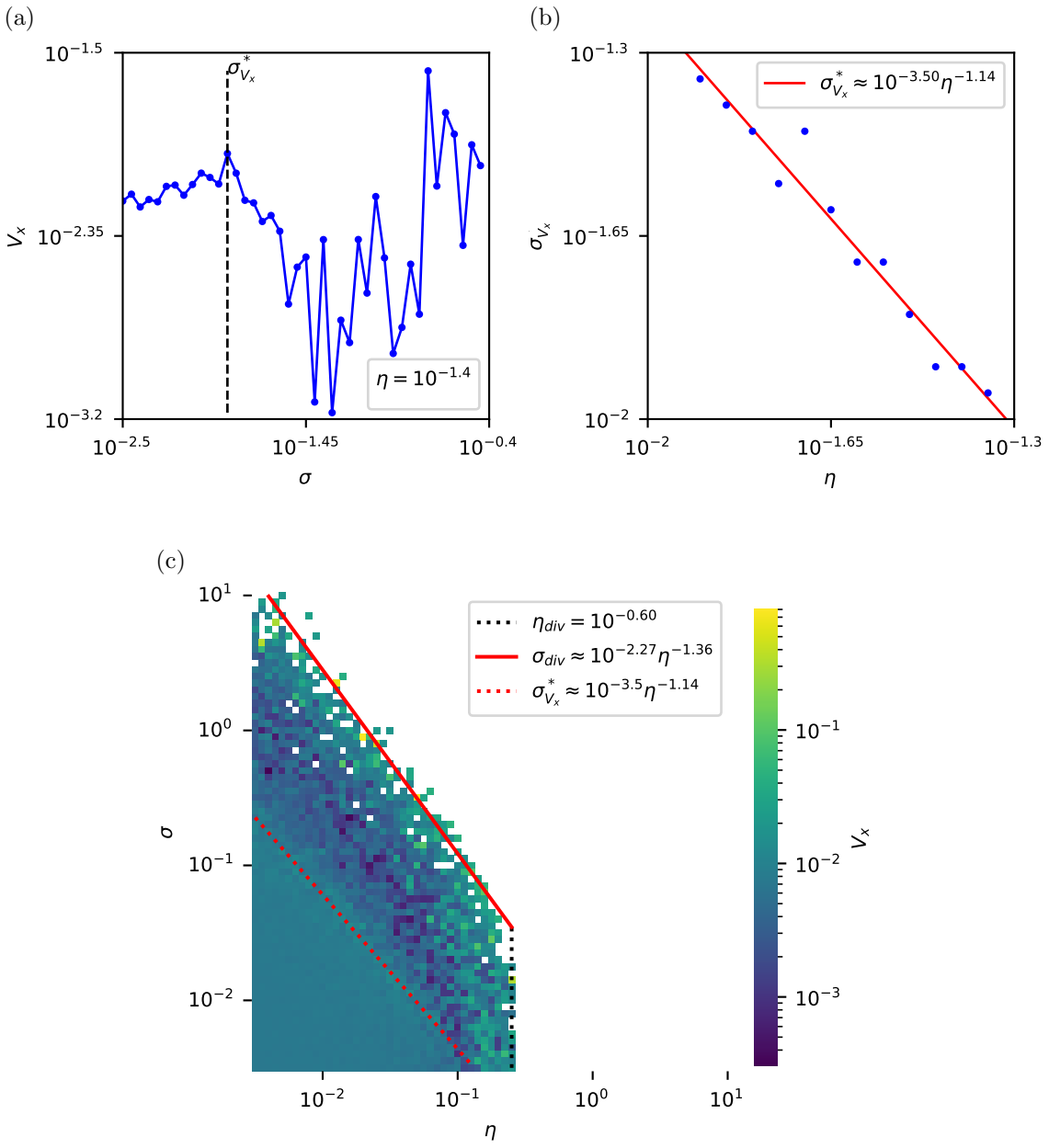


Figure 4.14: Variance V_x of the steady state solution x_{ss} in the numerical simulation of the two-dimensional loss model with $\lambda_0 = 8$, $\lambda_1 = -2$, $\lambda_2 = 1$. (a) The steady state x_{ss} is plotted against the noise strength σ at fixed learning rate $\eta = 10^{-1.4}$. The critical point $\sigma^*_{x_{ss}}$ marks a local maximum. (b) The noise strength at critical points $\sigma^*_{V_x}$ is plotted against learning rate η. The relation is $\sigma^*_{V_x} \approx 10^{-3.5}\eta^{-1.14}$. (c) The heatmap of V_x depends on the noise strength σ and the learning rate η. The convergence region has boundaries $\sigma_{div} = 10^{-2.27}\eta^{-1.36}$ and $\eta = 10^{-0.60}$. There is a phase transition, characterized by the relation from (b).

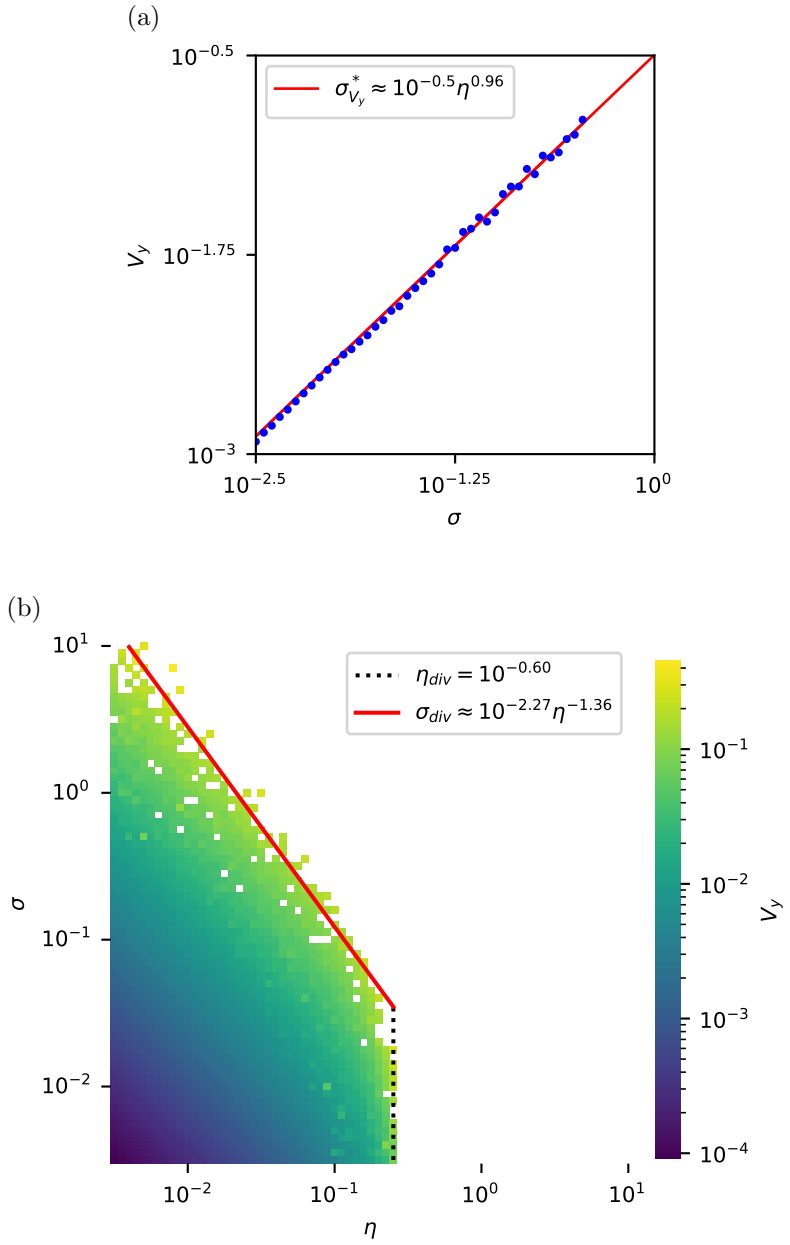


Figure 4.15: Variance V_y of the steady state solution y_{ss} in the numerical simulation of the two-dimensional loss model with $\lambda_0 = 8$, $\lambda_1 = -2$, $\lambda_2 = 1$. (a) The Variance V_y of steady state solution y_{ss} is plotted against the noise strength σ at fixed learning rate $\eta = 10^{-1.4}$. The curve follows the relation $V_y = 10^{-0.5}\eta^{0.96}$. (c) The heatmap of V_y depends on the noise strength σ and the learning rate η. The convergence region has boundaries $\sigma_{div} = 10^{-2.27}\eta^{-1.36}$ and $\eta = 10^{-0.60}$.

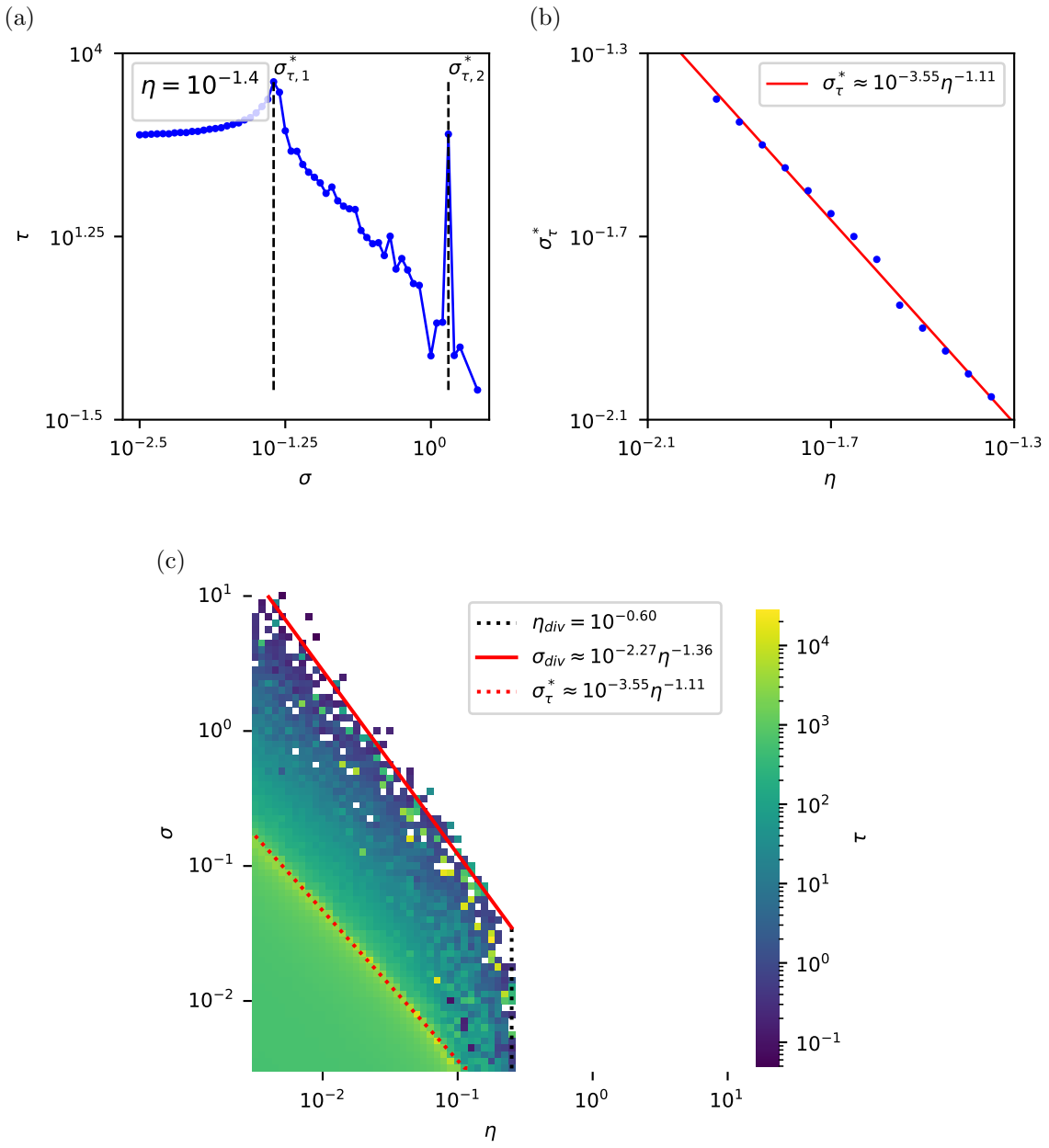


Figure 4.16: Convergence time τ at the steady state solution in the numerical simulation of the two-dimensional loss model with $\lambda_0 = 8$, $\lambda_1 = -2$, $\lambda_2 = 1$. (a) The τ is d against the noise strength σ at fixed learning rate $\eta = 10^{-1.4}$. The critical points $\sigma^*_{\tau,1}$ and $\sigma^*_{\tau,2}$ mark the point of largest curvature. (b) The noise strength at critical points $\sigma^*_\tau = \sigma^*_{\tau,1}$ is plotted against learning rate η. The relation is $\sigma^*_\tau \approx 10^{-3.55}\eta^{-1.11}$. (c) The Heatmap of τ depends on the noise strength σ and the learning rate η. The convergence region has boundaries $\sigma_{div} = 10^{-2.27}\eta^{-1.36}$ and $\eta = 10^{-0.60}$. There is a phase transition, characterized by the relation from (b).

Chapter 5

Introduction to the stochastic dynamics of diffusion processes

Having introduced a model for SGD, we want to derive theoretical results based on this model. For this we need to establish relevant concepts from statistical physics and in particular stochastic dynamics.

5.1 Brownian motion

In 1827 Robert Brown described the vivid random motion of pollen in liquids [39]. Among others, Einstein [40], Smoluchowski [41] and Langevin [42] developed theoretical explanations of this phenomenon known as Brownian motion. Following the steps in Ref. [43], we will explain the framework developed by Ref. [42] that is particularly relevant to our investigations. For an intuitive approach, see [44]. Consider a Brownian particle, i.e. pollen suspended in a liquid, with momentum $\boldsymbol{p}$, velocity $\boldsymbol{v}$ and constant mass m at position $\boldsymbol{x}$. The particle's movement is governed by Newton's second law

$$\boldsymbol{F} = \frac{d\boldsymbol{p}}{dt} = \frac{d(m\boldsymbol{v})}{dt} = m\frac{d(\boldsymbol{v})}{dt} = m\dot{\boldsymbol{v}} = m\ddot{\boldsymbol{x}}, \tag{5.1}$$

where $\boldsymbol{F}$ is the Force acting on the particle. In the case of a Brownian particle, the force

$$\boldsymbol{F} = \boldsymbol{F}_{ext} + \boldsymbol{F}_f \tag{5.2}$$

is composed of an external force $\boldsymbol{F}_{ext} = -\nabla V + \boldsymbol{\xi}(t)$ and a velocity dependent friction force $\boldsymbol{F}_f = -\gamma\dot{\boldsymbol{x}}$. V is any external potential affecting the particle. In contrast, the random force or noise $\boldsymbol{\xi}(t)$ is caused by the random collisions of small molecules from the liquid with the Brownian particle, due to the kinetic theory of temperature [40]. So, the Langevin equation of a Brownian particle is

$$m\ddot{\boldsymbol{x}} = -\gamma\dot{\boldsymbol{x}} - \nabla V + \boldsymbol{\xi}(t). \tag{5.3}$$

We simplify this equation by setting the potential $V = 0$, such that

$$m\ddot{\boldsymbol{x}} = -\gamma\dot{\boldsymbol{x}} + \boldsymbol{\xi}(t). \tag{5.4}$$

This is solved by the Ansatz

$$\frac{d}{dt}\left(\dot{\boldsymbol{x}}e^{\gamma t/m}\right) = \frac{1}{m}\boldsymbol{\xi}(t)e^{\gamma t/m}, \tag{5.5}$$

since

$$\frac{d}{dt}\left(\dot{\boldsymbol{x}}e^{\gamma t/m}\right) = \frac{1}{m}\boldsymbol{\xi}(t)e^{\gamma t/m}, \tag{5.6}$$

$$\ddot{\boldsymbol{x}}e^{\gamma t/m} + \dot{\boldsymbol{x}}\frac{\gamma}{m}e^{\gamma t/m} = \frac{1}{m}\boldsymbol{\xi}(t)e^{\gamma t/m}, \tag{5.7}$$

$$m\ddot{\boldsymbol{x}} = -\gamma\dot{\boldsymbol{x}} + \boldsymbol{\xi}(t). \tag{5.8}$$

This implies that

$$\dot{\boldsymbol{x}}(t)e^{\gamma t/m} = \dot{\boldsymbol{x}}(0) + \frac{1}{m}\int_0^t dt'\boldsymbol{\xi}(t')e^{\gamma t'/m},$$

$$\dot{\boldsymbol{x}}(t) = \dot{\boldsymbol{x}}(0)e^{-\gamma t/m} + \frac{1}{m}\int_0^t dt'\boldsymbol{\xi}(t')e^{\gamma(t'-t)/m}. \tag{5.9}$$

In addition, we assume the random force $\boldsymbol{\xi}(t)$ to be white noise. Since the random molecular collisions are equally likely to occur from any direction, we demand that the ensemble average of many realizations at the same time t is

$$\langle\boldsymbol{\xi}(t)\rangle = \mathbf{0}. \tag{5.10}$$

We also assume that observed collisions are fast and uncorrelated leading to

$$\langle\xi_i(t_1)\xi_j(t_2)\rangle = 2D\gamma^2\delta_{ij}\delta(t_1 - t_2) \tag{5.11}$$

for times t_1 and t_2 and force components ξ_i and ξ_j along directions i and j. Here, the factor $2D\gamma^2$ quantifies the correlation strength.

5.2 Diffusion processes

In this section we follow lectures at LMU by Prof. Erwin Frey, Ref. [45], as well as Ref. [26]. Brownian motion is the quintessential **diffusion process**. Diffusion processes are stochastic processes that have a continuous state space and a continuous trajectory. The first moment of a diffusion process $\boldsymbol{X}(t)$ with realizations $\boldsymbol{x}$ and $\boldsymbol{x'}$ at times t and $t + \Delta t$ is the drift vector $\boldsymbol{A}(\boldsymbol{x}, t)$ with coefficients

$$A_i(\boldsymbol{x}, t) := \lim_{\Delta t\to 0}\frac{1}{\Delta t}\int d\boldsymbol{x}'\ (x_i' - x_i)p(\boldsymbol{x}', t + \Delta t|\boldsymbol{x}, t) = \lim_{\Delta t\to 0}\frac{\langle\Delta x_i\rangle}{\Delta t} \tag{5.12}$$

and the second moment is the diffusion matrix $\mathbf{B}$ with coefficients

$$B_{ij}(\boldsymbol{x}, t) := \lim_{\Delta t\to 0}\frac{1}{\Delta t}\int d\boldsymbol{x}'\ (x_i' - x_i)(x_j' - x_j)p(\boldsymbol{x}', t + \Delta t|\boldsymbol{x}, t) = \lim_{\Delta t\to 0}\frac{\langle\Delta x_i\Delta x_j\rangle}{\Delta t} \tag{5.13}$$

with $\Delta x_i := x' - x$. Higher moments are equal to zero, see e.g. [26].

5.2.1 Fokker-Planck equation

In this chapter we want to derive the **Fokker-Planck equation**. Consider a suitable test function $f(\boldsymbol{x})$, where $\boldsymbol{x}$ is a realization of the diffusion process $\boldsymbol{X}(t)$ and the initial

conditions at time t_0 are $\boldsymbol{X}(t_0) = \boldsymbol{x}_0$. Then, the time evolution of the average of $f(\boldsymbol{x})$, given the initial conditions, is

$$\frac{\partial \langle f(\boldsymbol{x})\rangle_{x_0,t_0}}{\partial t} = \frac{\partial}{\partial t}\int d\boldsymbol{x} f(\boldsymbol{x}) p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0). \tag{5.14}$$

We rewrite the derivative with respect to time as the limit of a difference quotient,

$$\frac{\partial \langle f(\boldsymbol{x})\rangle_{x_0,t_0}}{\partial t} = \lim_{\Delta t\to 0}\frac{1}{\Delta t}\int d\boldsymbol{x} f(\boldsymbol{x})\left[(p(\boldsymbol{x}, t+\Delta t|\boldsymbol{x}_0, t_0) - p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0))\right]. \tag{5.15}$$

Expanding the brackets leads to

$$\frac{\partial \langle f(\boldsymbol{x})\rangle_{x_0,t_0}}{\partial t} = \lim_{\Delta t\to 0}\frac{1}{\Delta t}\Bigg[\int d\boldsymbol{x} f(\boldsymbol{x}) p(\boldsymbol{x}, t+\Delta t|\boldsymbol{x}_0, t_0) - \int d\boldsymbol{x} f(\boldsymbol{x}) p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0)\Bigg]. \tag{5.16}$$

From the Chapman-Kolmogorov equation, Eq. 2.17, it follows that

$$p(\boldsymbol{x}, t+\Delta t|\boldsymbol{x}_0, t_0) = \int d\boldsymbol{x}'\ p(\boldsymbol{x}, t+\Delta t|\boldsymbol{x}', t) p(\boldsymbol{x}', t|\boldsymbol{x}_0, t_0). \tag{5.17}$$

Inserting this expression into Eq. (5.16) gives

$$\frac{\partial \langle f(\boldsymbol{x})\rangle_{x_0,t_0}}{\partial t} = \lim_{\Delta t\to 0}\frac{1}{\Delta t}\Bigg[\int d\boldsymbol{x} f(\boldsymbol{x})\int d\boldsymbol{x}'\ p(\boldsymbol{x}, t+\Delta t|\boldsymbol{x}', t) p(\boldsymbol{x}', t|\boldsymbol{x}_0, t_0) - \int d\boldsymbol{x} f(\boldsymbol{x}) p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0)\Bigg]. \tag{5.18}$$

Since probability densities are normalized, we can always write

$$\int d\boldsymbol{x}'\ p(\boldsymbol{x}', t+\Delta t|\boldsymbol{x}, t) = 1. \tag{5.19}$$

We apply this to the second term in the brackets and derive

$$\frac{\partial \langle f(\boldsymbol{x}')\rangle_{x_0,t_0}}{\partial t} = \lim_{\Delta t\to 0}\frac{1}{\Delta t}\Bigg[\int d\boldsymbol{x} f(\boldsymbol{x})\int d\boldsymbol{x}'\ p(\boldsymbol{x}, t+\Delta t|\boldsymbol{x}', t) p(\boldsymbol{x}', t|\boldsymbol{x}_0, t_0) - \int d\boldsymbol{x} f(\boldsymbol{x}) p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0)\int d\boldsymbol{x}'\ p(\boldsymbol{x}', t+\Delta t|\boldsymbol{x}, t)\Bigg]. \tag{5.20}$$

Because the test function $f(\boldsymbol{x})$ as well as the density $p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0)$ are independent of the variable $\boldsymbol{x'}$, the equation can be rewritten as

$$\frac{\partial \langle f(\boldsymbol{x}')\rangle_{x_0,t_0}}{\partial t} = \lim_{\Delta t\to 0}\frac{1}{\Delta t}\Bigg[\int d\boldsymbol{x}\int d\boldsymbol{x}'\ f(\boldsymbol{x}) p(\boldsymbol{x}, t+\Delta t|\boldsymbol{x}', t) p(\boldsymbol{x}', t|\boldsymbol{x}_0, t_0) - \int d\boldsymbol{x}\int d\boldsymbol{x}'\ f(\boldsymbol{x}) p(\boldsymbol{x}', t+\Delta t|\boldsymbol{x}, t) p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0)\Bigg]. \tag{5.21}$$

Relabeling the dummy-indexes $\boldsymbol{x} \leftrightarrow \boldsymbol{x}'$ of the first term in the brackets gives

$$\frac{\partial \langle f(\boldsymbol{x}') \rangle_{x_0,t_0}}{\partial t} = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \Bigg[\int d\boldsymbol{x}' \int d\boldsymbol{x} \; f(\boldsymbol{x}') p(\boldsymbol{x}', t+\Delta t | \boldsymbol{x}, t) p(\boldsymbol{x}, t | \boldsymbol{x}_0, t_0) - \int d\boldsymbol{x} \int d\boldsymbol{x}' \; f(\boldsymbol{x}) p(\boldsymbol{x}', t+\Delta t | \boldsymbol{x}, t) p(\boldsymbol{x}, t | \boldsymbol{x}_0, t_0) \Bigg]. \tag{5.22}$$

We now change the order of integration in the first term, $\int dx' \int dx \leftrightarrow \int dx \int dx'$, such that

$$\frac{\partial \langle f(\boldsymbol{x}') \rangle_{x_0,t_0}}{\partial t} = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \Bigg[\int d\boldsymbol{x} \int d\boldsymbol{x}' \; f(\boldsymbol{x}') p(\boldsymbol{x}', t+\Delta t | \boldsymbol{x}, t) p(\boldsymbol{x}, t | \boldsymbol{x}_0, t_0) - \int d\boldsymbol{x} \int d\boldsymbol{x}' \; f(\boldsymbol{x}) p(\boldsymbol{x}', t+\Delta t | \boldsymbol{x}, t) p(\boldsymbol{x}, t | \boldsymbol{x}_0, t_0) \Bigg]. \tag{5.23}$$

Invoking the linearity of the integral, this gives

$$\frac{\partial \langle f(\boldsymbol{x}') \rangle_{x_0,t_0}}{\partial t} = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int d\boldsymbol{x} \int d\boldsymbol{x}' \; \Big[f(\boldsymbol{x}') p(\boldsymbol{x}', t+\Delta t | \boldsymbol{x}, t) p(\boldsymbol{x}, t | \boldsymbol{x}_0, t_0) - f(\boldsymbol{x}') p(\boldsymbol{x}', t+\Delta t | \boldsymbol{x}, t) p(\boldsymbol{x}, t | \boldsymbol{x}_0, t_0) \Big]. \tag{5.24}$$

Then, factorizing the term $\tilde{p}(\boldsymbol{x}', \boldsymbol{x}, t) := p(\boldsymbol{x}', t+\Delta t | \boldsymbol{x}, t) p(\boldsymbol{x}, t | \boldsymbol{x}_0, t_0)$, leads to

$$\frac{\partial \langle f(\boldsymbol{x}') \rangle_{x_0,t_0}}{\partial t} = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int d\boldsymbol{x} \int d\boldsymbol{x}' \; \tilde{p}(\boldsymbol{x}', \boldsymbol{x}, t) \Big[f(\boldsymbol{x}') - f(\boldsymbol{x}) \Big]. \tag{5.25}$$

We rewrite $f(\boldsymbol{x}') = f(\boldsymbol{x} + \boldsymbol{u})$, with difference $\boldsymbol{u} = \boldsymbol{x}' - \boldsymbol{x}$. We then Taylor expand $f(\boldsymbol{x}') = f(\boldsymbol{x} + \boldsymbol{u})$ to second order around $\boldsymbol{x}$,

$$f(\boldsymbol{x}') = f(\boldsymbol{x}+\boldsymbol{u}) \approx f(\boldsymbol{x}) + \sum_i (x_i' - x_i) \frac{\partial f(\boldsymbol{x})}{\partial x_i}\bigg|_{\boldsymbol{x}} + \frac{1}{2} \sum_{ij} (x_i' - x_i)(x_j' - x_j) \frac{\partial^2 f(\boldsymbol{x})}{\partial x_i \partial x_j}\bigg|_{\boldsymbol{x}}. \tag{5.26}$$

Then the $f(\boldsymbol{x})$ in the difference cancel,

$$f(\boldsymbol{x}') - f(\boldsymbol{x}) = \sum_i (x_i' - x_i) \frac{\partial f(\boldsymbol{x})}{\partial x_i}\bigg|_{\boldsymbol{x}} + \frac{1}{2} \sum_{ij} (x_i' - x_i)(x_j' - x_j) \frac{\partial^2 f(\boldsymbol{x})}{\partial x_i \partial x_j}\bigg|_{\boldsymbol{x}} \tag{5.27}$$

and, with the linearity of the integral, we have

$$\frac{\partial \langle f(\boldsymbol{x}') \rangle_{x_0,t_0}}{\partial t} = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int d\boldsymbol{x} \int d\boldsymbol{x}' \; \tilde{p}(\boldsymbol{x}', \boldsymbol{x}, t) \sum_i (x_i' - x_i) \frac{\partial f(\boldsymbol{x})}{\partial x_i} + \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int d\boldsymbol{x} \int d\boldsymbol{x}' \; \tilde{p}(\boldsymbol{x}', \boldsymbol{x}, t) \sum_{ij} (x_i' - x_i)(x_j' - x_j) \frac{\partial^2 f(\boldsymbol{x})}{\partial x_i \partial x_j}. \tag{5.28}$$

We assume sufficiently well-behaved functions, such that limes and integral are interchangeable. Then we have,

$$\frac{\partial \langle f(\boldsymbol{x}') \rangle_{x_0,t_0}}{\partial t} = \int d\boldsymbol{x} \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int d\boldsymbol{x}' \; \tilde{p}(\boldsymbol{x}', \boldsymbol{x}, t) \sum_i (x_i' - x_i) \frac{\partial f(\boldsymbol{x})}{\partial x_i} + \int d\boldsymbol{x} \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int d\boldsymbol{x}' \; \tilde{p}(\boldsymbol{x}', \boldsymbol{x}, t) \sum_{ij} (x_i' - x_i)(x_j' - x_j) \frac{\partial^2 f(\boldsymbol{x})}{\partial x_i \partial x_j}. \tag{5.29}$$

We re-substitute $\tilde{p}(\boldsymbol{x}', \boldsymbol{x}, t) := p(\boldsymbol{x}', t + \Delta t|\boldsymbol{x}, t)p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0)$ and define the terms

$$T_1 = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int d\boldsymbol{x}' \; p(\boldsymbol{x}', t + \Delta t|\boldsymbol{x}, t)p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) \sum_i (x_i' - x_i) \frac{\partial f(\boldsymbol{x})}{\partial x_i} \tag{5.30}$$

and

$$T_2 := \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int d\boldsymbol{x}' \; p(\boldsymbol{x}', t + \Delta t|\boldsymbol{x}, t)p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) \sum_{ij} (x_i' - x_i)(x_j' - x_j) \frac{\partial^2 f(\boldsymbol{x})}{\partial x_i \partial x_j}, \tag{5.31}$$

such that

$$\frac{\partial \langle f(\boldsymbol{x}') \rangle_{x_0, t_0}}{\partial t} = \int d\boldsymbol{x} \; (T_1 + T_2) = \int d\boldsymbol{x} \; T_1 + \int d\boldsymbol{x} \; T_2. \tag{5.32}$$

We focus on the first term, T_1. The integral is linear, and we can therefor exchange a finite summation with the integral, i.e. expand the equation to

$$T_1 = \sum_i \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int d\boldsymbol{x}' \; (x_i' - x_i) \; p(\boldsymbol{x}', t + \Delta t|\boldsymbol{x}, t) \; p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) \frac{\partial f(\boldsymbol{x})}{\partial x_i}. \tag{5.33}$$

Since the test function $f(\boldsymbol{x})$ as well as the density $p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0)$ are independent of $\boldsymbol{x}'$ and Δt, they may be placed in front of the limit. For the first term we then have

$$T_1 = \sum_i p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) \left(\frac{\partial f(\boldsymbol{x})}{\partial x_i} \right) \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int d\boldsymbol{x}' \; (x_i' - x_i) \; p(\boldsymbol{x}', t + \Delta t|\boldsymbol{x}, t). \tag{5.34}$$

Applying the definition of the drift vector $\boldsymbol{A}$, Eq. (5.12), we rewrite this to

$$T_1 = \sum_i \; p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) \left(\frac{\partial f(\boldsymbol{x})}{\partial x_i} \right) A_i. \tag{5.35}$$

Taking the integral of this expression and rearranging terms, we get

$$\int d\boldsymbol{x} \; T_1 = \int d\boldsymbol{x} \; \sum_i \left(\frac{\partial f(\boldsymbol{x})}{\partial x_i} \right) A_i \; p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0). \tag{5.36}$$

This equation can be partially integrated to

$$\begin{aligned} \int d\boldsymbol{x} \; T_1 = & \left[f(\boldsymbol{x}) \sum_i A_i \; p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) \right]_{-\infty}^{\infty} \\ & - \int d\boldsymbol{x} \; f(\boldsymbol{x}) \sum_i \frac{\partial}{\partial x_i} A_i \; p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0). \end{aligned} \tag{5.37}$$

We assume sufficient boundary conditions of the probability densities and the test function $f(\boldsymbol{x})$, such that

$$\left[f(\boldsymbol{x}) \sum_i A_i \; p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) \right]_{-\infty}^{\infty} = 0. \tag{5.38}$$

Then, the term simplifies to

$$\int d\boldsymbol{x} \; T_1 = - \int d\boldsymbol{x} \; f(\boldsymbol{x}) \sum_i \frac{\partial}{\partial x_i} A_i \; p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0). \tag{5.39}$$

For the second term we proceed analogously and derive

$$T_2 = \sum_{ij} p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) \left(\frac{\partial^2 f(\boldsymbol{x})}{\partial x_i \partial x_j} \right) \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int d\boldsymbol{x}' \; (x_i' - x_i) \; (x_j' - x_j) \; p(\boldsymbol{x}', t+\Delta t|\boldsymbol{x}, t). \quad (5.40)$$

We now substitute the diffusion matrix, Eq. (5.13), and restate the equation as

$$T_2 = \sum_{ij} p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) \left(\frac{\partial^2 f(\boldsymbol{x})}{\partial x_i \partial x_j} \right) B_{ij}. \quad (5.41)$$

Then, we integrate this expression and rearrange it to

$$\int d\boldsymbol{x} \; T_2 = \int d\boldsymbol{x} \sum_{ij} \left(\frac{\partial^2 f(\boldsymbol{x})}{\partial x_i \partial x_j} \right) p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) B_{ij}. \quad (5.42)$$

Partially integrating once yields

$$\begin{aligned} \int d\boldsymbol{x} \; T_2 = & \left[\sum_{ij} \left(\frac{\partial f(\boldsymbol{x})}{\partial x_i} \right) p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) B_{ij} \right]_{-\infty}^{\infty} \\ & - \int d\boldsymbol{x} \sum_{ij} \left(\frac{\partial f(\boldsymbol{x})}{\partial x_i} \right) \frac{\partial}{\partial x_j} p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) B_{ij}. \end{aligned} \quad (5.43)$$

We assume sufficient boundary conditions of the derivatives of $f(\boldsymbol{x})$, such that

$$\left[\sum_{ij} \left(\frac{\partial f(\boldsymbol{x})}{\partial x_i} \right) p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) B_{ij} \right]_{-\infty}^{\infty} = 0. \quad (5.44)$$

The term now simplifies to

$$\int d\boldsymbol{x} \; T_2 = - \int d\boldsymbol{x} \sum_{ij} \left(\frac{\partial f(\boldsymbol{x})}{\partial x_i} \right) \frac{\partial}{\partial x_j} p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) B_{ij}. \quad (5.45)$$

Next, we perform partial integration again. This gives

$$\begin{aligned} \int d\boldsymbol{x} \; T_2 = & - \left[f(\boldsymbol{x}) \sum_{ij} \frac{\partial}{\partial x_j} p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) B_{ij} \right]_{-\infty}^{\infty} \\ & + \int d\boldsymbol{x} \; f(\boldsymbol{x}) \sum_{ij} \frac{\partial^2}{\partial x_i \partial x_j} p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) B_{ij}. \end{aligned} \quad (5.46)$$

With the boundary condition of the test function, we have

$$\left[f(\boldsymbol{x}) \sum_{ij} \frac{\partial}{\partial x_j} p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) B_{ij} \right]_{-\infty}^{\infty} = 0, \quad (5.47)$$

and therefore

$$\int d\boldsymbol{x}\ T_2 = \int d\boldsymbol{x}\ f(\boldsymbol{x}) \sum_{ij} \frac{\partial^2}{\partial x_i \partial x_j} p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) B_{ij}. \tag{5.48}$$

Overall, with Eq. (5.32), Eq. (5.39) and Eq. (5.48), the full expression of the time evolution of the average of the test function is given by

$$\frac{\partial \langle f(\boldsymbol{x})\rangle_{x_0,t_0}}{\partial t} = \int d\boldsymbol{x}\ f(\boldsymbol{x}) \left[-\sum_i \frac{\partial}{\partial x_i} A_i\ p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) + \sum_{ij} \frac{\partial^2}{\partial x_i \partial x_j} p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0) B_{ij} \right]. \tag{5.49}$$

This is true for any suitable test function $f(\boldsymbol{x})$. Together with Eq. (5.14) we therefore arrive at the **Fokker-Planck equation**

$$\frac{\partial p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0)}{\partial t} = -\sum_i \frac{\partial}{\partial x_i} \left[A_i(\boldsymbol{x}, t) p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0)\right] + \frac{1}{2} \sum_{ij} \frac{\partial^2}{\partial x_i \partial x_j} [B_{ij}(\boldsymbol{x}, t) p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0)]. \tag{5.50}$$

It describes the time evolution of the transition probability $p(\boldsymbol{x}, t|\boldsymbol{x}_0, t_0)$ in terms of spatial derivatives of the drift and diffusion coefficients. If $\boldsymbol{x}_0$ and t_0 can be inferred from context, the expression simplifies to

$$\frac{\partial p(\boldsymbol{x}, t)}{\partial t} = -\sum_i \frac{\partial}{\partial x_i} \left[A_i(\boldsymbol{x}, t) p(\boldsymbol{x}, t)\right] + \frac{1}{2} \sum_{ij} \frac{\partial^2}{\partial x_i \partial x_j} [B_{ij}(\boldsymbol{x}, t) p(\boldsymbol{x}, t)]. \tag{5.51}$$

This is often rewritten as

$$\frac{\partial p(\boldsymbol{x}, t)}{\partial t} = -\sum_i \frac{\partial}{\partial x_i} \left[A_i(\boldsymbol{x}, t) p(\boldsymbol{x}, t) - \frac{1}{2} \sum_{ij} \frac{\partial}{\partial x_j} B_{ij}(\boldsymbol{x}, t) p(\boldsymbol{x}, t) \right]. \tag{5.52}$$

Equivalent representations of the Fokker-Planck equation are given by the operator equation

$$\frac{\partial p(\boldsymbol{x}, t)}{\partial t} = L_{FP}(\boldsymbol{x}, t) p(\boldsymbol{x}, t) \tag{5.53}$$

and the continuity equation

$$\frac{\partial p(\boldsymbol{x}, t)}{\partial t} = -\sum_i \frac{\partial}{\partial x_i} J_i(\boldsymbol{x}, t). \tag{5.54}$$

The Fokker-Planck operator

$$L_{FP}(x, t) = -\sum_i \frac{\partial}{\partial x_i} A_i(\boldsymbol{x}, t) + \frac{1}{2} \sum_{ij} \frac{\partial^2}{\partial x_i \partial x_j} B_{ij}(\boldsymbol{x}, t) \tag{5.55}$$

acts on the probability density $p(\boldsymbol{x}, t)$ and the probability flux $\boldsymbol{J}$ has elements

$$J_i(\boldsymbol{x}, t) = A_i(\boldsymbol{x}, t) p(\boldsymbol{x}, t) - \frac{1}{2} \sum_j \frac{\partial}{\partial x_j} \left[B_{ij}(\boldsymbol{x}, t) p(\boldsymbol{x}, t)\right]. \tag{5.56}$$

For a single variable x, the Fokker-Planck equation reduces to

$$\begin{aligned}\frac{\partial p(x,t)}{\partial t} &= -\frac{\partial}{\partial x}\left[A(x,t)p(x,t)\right] + \frac{1}{2}\frac{\partial^2}{\partial x^2}[B(x,t)p(x,t)] \\ &= -\frac{\partial}{\partial x}\left(A(x,t)p(x,t) - \frac{1}{2}\frac{\partial}{\partial x}\left[B(x,t)p(x,t)\right]\right).\end{aligned} \tag{5.57}$$

For the one dimensional Fokker-Planck operator this is

$$\frac{\partial p(x,t)}{\partial t} = L_{FP}\; p(x,t) \tag{5.58}$$

with

$$L_{FP}(x,t) = -\frac{\partial}{\partial x}A(x,t) + \frac{1}{2}\frac{\partial^2}{\partial x^2}B(x,t). \tag{5.59}$$

The representation as a continuity equation now reads as

$$\frac{\partial p(x,t)}{\partial t} = -\frac{\partial}{\partial x}J(x,t) \tag{5.60}$$

with

$$J(x,t) = A(x,t)p(x,t) - \frac{1}{2}\frac{\partial}{\partial x}[B(x,t)p(x,t)]. \tag{5.61}$$

5.2.2 Stationary solutions to the Fokker-Planck equation with one variable

Of particular interest is the stationary solution to the Fokker-Planck equation. We follow the steps in [46]. A stationary state is characterized by a constant probability density $p_{ss}(x,t)$ with

$$\frac{\partial p_{ss}(x,t)}{\partial t} = 0. \tag{5.62}$$

Therefor $p_{ss}(x,t)$ does not explicitly depend on time and we write $p_{ss}(x)$. In combination with Eq. (5.60) and Eq. (5.61) this implies that $\frac{\partial}{\partial x}J(x) = 0$ and $J(x) = J$ is a constant, given by

$$\begin{aligned}J &= A(x)p_{ss}(x) - \frac{d}{dx}[B(x)p_{ss}(x)] \\ &= A(x)p_{ss}(x) - p_{ss}(x)\frac{d}{dx}B(x,t) - B(x)\frac{d}{dx}p_{ss}(x) \\ &= p_{ss}\left[A(x) - \frac{d}{dx}B(x,t)\right] - B(x)\frac{d}{dx}p_{ss}(x).\end{aligned} \tag{5.63}$$

This is an inhomogeneous ordinary differential equation of first order, better expressed by

$$\frac{d}{dx}p_{ss}(x) - \frac{p_{ss}\left[A(x) - \frac{d}{dx}B(x,t)\right]}{B(x)} = -\frac{J}{B(x)} \tag{5.64}$$

or, equivalently,

$$\frac{d}{dx}p_{ss}(x) + f(x)p_{ss} = g(x) \tag{5.65}$$

with $f(x) = -\frac{A(x)-\frac{d}{dx}B(x)}{B(x)}$ and $g(x) = -\frac{J}{B(x)}$. The general solution can be separated into a homogenous solution and a particular solution,

$$p_{ss}(x) = p_h(x) + p_p(x) \tag{5.66}$$

with

$$\frac{d}{dx}p_{ss}(x) = \frac{d}{dx}p_h(x) + \frac{d}{dx}p_p(x). \tag{5.67}$$

We first set $J = 0$ to find the homogenous solution $p_h(x,t)$ that solves the equation

$$\frac{d}{dx}p_h(x) + f(x)p_h(x) = 0 \tag{5.68}$$

or

$$\frac{d}{dx}p_h(x) - \frac{p_h(x)\left[A(x) - \frac{d}{dx}B(x,t)\right]}{B(x)} = 0 \tag{5.69}$$

by separation of variables. We first rearrange terms to

$$\frac{d}{dx}p_h(x) = \frac{p_h(x)\left[A(x) - \frac{d}{dx}B(x,t)\right]}{B(x)}. \tag{5.70}$$

Then, we separate the variables and integrate both sides

$$\int_{p_h(x_0)}^{p_h(x)} \frac{1}{p_h(x')}dp_h(x') = \int_{x_0}^{x} \frac{A(x') - \frac{d}{dx'}B(x')}{B(x')}dx'. \tag{5.71}$$

This leads to

$$\begin{aligned}
\ln[p_h(x)] &= \int_{x_0}^{x} \frac{A(x') - \frac{d}{dx'}B(x')}{B(x')}dx' - \ln[p_h(x_0)] \\
&= \int_{x_0}^{x} \frac{A(x')}{B(x')}dx' - \int_{x_0}^{x} \frac{\frac{d}{dx'}B(x')}{B(x')}dx' - \ln[p_h(x_0)] \\
&= \int_{x_0}^{x} \frac{A(x')}{B(x')}dx' - \int_{B(x_0)}^{B(x)} \frac{dB(x')}{B(x')} - \ln[p_h(x_0)] \\
&= \int_{x_0}^{x} \frac{A(x')}{B(x')}dx' - \ln[B(x)] + \ln[B(x_0)] - \ln[p_h(x_0)] \\
&= \int_{x_0}^{x} \frac{A(x')}{B(x')}dx' - \ln[B(x)] + C,
\end{aligned} \tag{5.72}$$

where we included the initial conditions $\ln[B(x_0)]$ and $\ln[p_h(x_0)]$ in the normalization constant C. It follows that

$$p_h(x) = \exp\left[\int_{x_0}^{x} \frac{A(x')}{B(x')}dx' - \ln[B(x)] + C\right] = N\exp\left[\int_{x_0}^{x} \frac{A(x')}{B(x')}dx' - \ln[B(x)]\right] \tag{5.73}$$

with constant term $N = e^C$. Next, we introduce the potential

$$\phi(x) = \ln[B(x)] - \int_{x_0}^{x} \frac{A(x')}{B(x')}dx'. \tag{5.74}$$

The potential has the derivative

$$\begin{aligned}\frac{d}{dx}\phi(x) &= \frac{d}{dx}\left(\ln[B(x)] - \int_{x_0}^{x}\frac{A(x')}{B(x')}dx'\right)\\ &= \frac{1}{B(x)}\frac{d}{dx}B(x) - \frac{A(x)}{B(x)}\\ &= -\frac{A(x) - \frac{d}{dx}B(x)}{B(x)}.\end{aligned} \tag{5.75}$$

By substituting this potential we can state the homogenous equation, Eq. (5.69), as

$$\frac{d}{dx}p_h(x) + p_h\frac{d}{dx}\phi(x) = 0 \tag{5.76}$$

and its homogenous solution can be expressed by

$$p_h(x) = Ne^{-\phi(x)}. \tag{5.77}$$

In addition, the initial inhomogeneous equation can also be expressed in terms of the potential, by

$$-\frac{J}{B(x)} = \frac{d}{dx}p_{ss}(x) - \frac{p_{ss}\left[A(x) - \frac{d}{dx}B(x,t)\right]}{B(x)} = \frac{d}{dx}p_{ss}(x) + p_{ss}\frac{d}{dx}\phi(x). \tag{5.78}$$

The Ansatz for the solution of the inhomogeneous equation is to introduce an x dependence in the normalization factor N, i.e.

$$p_{ss}(x) = N(x)e^{-\phi(x)}. \tag{5.79}$$

Then, the derivative is given by

$$\begin{aligned}\frac{d}{dx}p_{ss}(x) &= \frac{d}{dx}N(x)e^{-\phi(x)}\\ &= e^{-\phi(x)}\frac{d}{dx}N(x) + N(x)\frac{d}{dx}e^{-\phi(x)}\\ &= e^{-\phi(x)}\frac{d}{dx}N(x) - N(x)e^{-\phi(x)}\frac{d}{dx}\phi(x).\end{aligned} \tag{5.80}$$

Then, the inhomogeneous equation transforms to

$$\begin{aligned}-\frac{J}{B(x)} &= \frac{d}{dx}p_{ss}(x) + p_{ss}\frac{d}{dx}\phi(x)\\ &= e^{-\phi(x)}\frac{d}{dx}N(x) - N(x)e^{-\phi(x)}\frac{d}{dx}\phi(x) + N(x)e^{-\phi(x)}\frac{d}{dx}\phi(x)\\ &= e^{-\phi(x)}\frac{d}{dx}N(x).\end{aligned} \tag{5.81}$$

Rearranging terms gives

$$\frac{d}{dx}N(x) = -e^{\phi(x)}\frac{J}{B(x)} \tag{5.82}$$

which can be integrated to the expression

$$N(x) = -\int_{x_0}^{x} e^{\phi(x')} \frac{J}{B(x')} dx' - N(x_0) = -N_0 - J \int_{x_0}^{x} \frac{e^{\phi(x')}}{B(x')} dx' \tag{5.83}$$

with $N_0 = N(x_0)$.
In conclusion, the stationary state solution to the single variable Fokker-Planck equation is given by

$$p_{ss}(x) = N(x)e^{-\phi(x)} = -N_0 e^{-\phi(x)} - Je^{-\phi(x)} \int_{x_0}^{x} \frac{e^{\phi(x')}}{B(x')} dx'. \tag{5.84}$$

5.2.3 Eigenvalue expansion

We now want to find non-stationary solutions and follow the steps in Refs. [46] and [47]. Consider again the one variable Fokker-Planck equation in operator form, Eq. 5.59,

$$\frac{\partial p(x,t)}{\partial t} = L_{FP}\, p(x,t). \tag{5.85}$$

A possible solution is given by

$$p(x,t) = e^{L_{FP}(t-t_0)} p(x,t_0) \tag{5.86}$$

with initial distribution $p(x,t_0)$. The exponential function of an operator is given by its power series, i.e.

$$e^{L_{FP}} = \sum_{k=0}^{\infty} \frac{(L_{FP})^k}{k!}. \tag{5.87}$$

For the non-stationary solutions we make the separation Ansatz

$$p(x,t) = e^{-\mu t}\varphi(x). \tag{5.88}$$

Then it follows for the Fokker-Planck equation that

$$\frac{\partial p(x,t)}{\partial t} = -\mu e^{-\mu t}\varphi(x). \tag{5.89}$$

With Eq. 5.59, this gives

$$e^{-\mu t}\,(-\mu)\,\varphi(x) = L_{FP}\,[p(x,t)] = L_{FP}\left[e^{-\mu t}\varphi(x)\right] = e^{-\mu t} L_{FP}\,[\varphi(x)]\,. \tag{5.90}$$

It follows that $\varphi(x)$ is an eigenfunction of the operator L_{FP} with eigenvalue $-\mu$, i.e.

$$L_{FP}\varphi(x) = -\mu\varphi(x). \tag{5.91}$$

With the potential $\phi(x)$, Eq. (5.74), the Fokker-Planck operator L_{FP} can be expressed by

$$L_{FP} = \frac{d}{dx}\left[B(x)e^{-\phi(x)}\frac{d}{dx}e^{\phi(x)}\right]. \tag{5.92}$$

This can be understood by a transformation to the original expression. We first have

$$\begin{aligned} L_{FP} &= \frac{d}{dx}\left[B(x)e^{-\phi(x)}\frac{d}{dx}e^{\phi(x)}\right] \\ &= \frac{d}{dx}\left[B(x)e^{-\phi(x)}e^{\phi(x)}\frac{d}{dx}\phi(x)\right] \\ &= \frac{d}{dx}\left[B(x)\frac{d}{dx}\phi(x)\right]. \end{aligned} \tag{5.93}$$

Inserting the derivative of the potential, Eq. (5.75), gives

$$\begin{aligned} L_{FP} &= -\frac{d}{dx}\left[B(x)\frac{A(x)-\frac{d}{dx}B(x)}{B(x)}\right] \\ &= -\frac{d}{dx}A(x)+\frac{d^2}{dx^2}B(x). \end{aligned} \tag{5.94}$$

So, we retrieved the old representation. We now define the modified Fokker-Planck operator by

$$L = e^{\phi(x)/2}L_{FP}e^{-\phi(x)/2}. \tag{5.95}$$

This operator is Hermitian. To see this, assume two probability distributions $p_1(x,t)$ and $p_2(x,t)$. Then, we can show that

$$\begin{aligned} \int_{-\infty}^{\infty} p_1(x,t)Lp_2(x,t) &= \int_{-\infty}^{\infty} p_1(x,t)e^{\phi(x)/2}L_{FP}e^{-\phi(x)/2}p_2(x,t) \\ &= \int_{-\infty}^{\infty} p_1(x,t)e^{\phi(x)/2}\frac{d}{dx}\left(B(x)e^{-\phi(x)}\frac{d}{dx}e^{\phi(x)}e^{-\phi(x)/2}p_2(x,t)\right) \\ &= \int_{-\infty}^{\infty} p_1(x,t)e^{\phi(x)/2}\frac{d}{dx}\left(B(x)e^{-\phi(x)}\frac{d}{dx}e^{\phi(x)/2}p_2(x,t)\right) \\ &\overset{p.i}{=} -\int_{-\infty}^{\infty}\left(\frac{d}{dx}e^{\phi(x)/2}p_2(x,t)\right)B(x)e^{-\phi(x)}\frac{d}{dx}e^{\phi(x)/2}p_1(x,t) \\ &\overset{p.i}{=} \int_{-\infty}^{\infty} p_2(x,t)e^{\phi(x)/2}\frac{d}{dx}\left(B(x)e^{-\phi(x)}\frac{d}{dx}e^{\phi(x)/2}p_1(x,t)\right) \\ &\overset{p.i}{=} \int_{-\infty}^{\infty} p_2(x,t)e^{\phi(x)/2}\frac{d}{dx}\left(B(x)e^{-\phi(x)}\frac{d}{dx}e^{\phi(x)}e^{-\phi(x)/2}p_1(x,t)\right) \\ &= \int_{-\infty}^{\infty} p_2(x,t)e^{\phi(x)/2}L_{FP}e^{-\phi(x)/2}p_1(x,t), \end{aligned} \tag{5.96}$$

where $p.i$ indicated partial integration with vanishing boundary terms. It follows that the operator L is Hermitian. Background regarding the properties of Hermitian operators can be found in introductory books on quantum mechanics, e.g. [48] and [49]. Since L is a Hermitian operator, there is a complete set of orthogonal normalized eigenfunctions $\psi(x)_n$ and real eigenvalues $-\mu_n$ fulfilling the eigenvalue equation

$$L\psi_n(x) = -\mu_n\psi_n(x). \tag{5.97}$$

The eigenfunctions $\varphi_n(x)$ of the operator L_{FP} and the eigenfunctions $\psi_n(x)$ of operator L correspond to the same eigenvalue $-\mu_n$ and are related by the equations

$$\psi_n = e^{\phi(x)/2}\varphi_n(x). \tag{5.98}$$

This can be derived by

$$\begin{aligned} L\psi_n &= Le^{\phi(x)/2}\varphi_n(x) \\ &= e^{\phi(x)/2}L_{FP}e^{-\phi(x)/2}e^{\phi(x)/2}\varphi_n(x) \\ &= e^{\phi(x)/2}L_{FP}\varphi_n(x) = -\mu_n e^{\phi(x)/2}\varphi_n(x) = -\mu_n\psi_n. \end{aligned} \tag{5.99}$$

The orthonormality relation is

$$\int_{-\infty}^{\infty} dx\ \overline{\psi_n(x)}\psi_m(x) = \int_{-\infty}^{\infty} dx\ e^{\phi(x)}\overline{\varphi_n(x)}\varphi_m(x) = \delta_{nm} \tag{5.100}$$

and the completeness relation is

$$\delta(x-x') = \sum_n \overline{\psi_n(x)}\psi_n(x') = e^{\phi(x)/2}e^{\phi(x')/2}\sum_n \overline{\varphi_n(x)}\varphi_n(x'). \tag{5.101}$$

We also introduce the eigenvalue equation of the adjoint Fokker Planck operator,

$$L_{FP}^{\dagger}\overline{\varphi_n(x)} = -\mu_n(x)\overline{\varphi_n(x)}. \tag{5.102}$$

For a stationary solution we require that $L\psi_0 = 0$, i.e. that the eigenvalue $\mu_0 = 0$. The corresponding eigenfunction is

$$\psi_0 = Ne^{-\phi(x)/2}. \tag{5.103}$$

The operator L acts on it by

$$\begin{aligned} L\psi_0 &= Le^{-\phi(x)/2}N \\ &= e^{\phi(x)/2}L_{FP}e^{-\phi(x)/2}e^{-\phi(x)/2}N \\ &= e^{\phi(x)/2}L_{FP}e^{-\phi(x)}N \\ &= e^{\phi(x)/2}L_{FP}N \\ &= e^{\phi(x)/2}\frac{d}{dx}\left[B(x)e^{-\phi(x)}\frac{d}{dx}e^{\phi(x)}e^{-\phi(x)}N\right] \\ &= e^{\phi(x)/2}\frac{d}{dx}\left[B(x)e^{-\phi(x)}\frac{d}{dx}N\right] \\ &= 0, \end{aligned} \tag{5.104}$$

since N is a constant. With Eq. (5.97) it follows that

$$\varphi_0 = Ne^{-\phi(x)} \tag{5.105}$$

and we can thus rewrite Eq. (5.97) as

$$\varphi_n = \psi_n\sqrt{\frac{\varphi_0}{N}}. \tag{5.106}$$

This is where we stop for now. We will use this eigenfunction expansion in the next chapter to explore the theoretical behavior of the model of SGD.

5.2.4 Wiener process

Next, we want to connect the Fokker-Planck equation with Brownian motion. We again follow the steps in [45]. We first introduce the one dimensional **Wiener process** $W(t)$. This is a diffusion process with no drift term,

$$A = \lim_{\Delta t \to 0} \frac{\langle \Delta w \rangle}{\Delta t} = 0, \tag{5.107}$$

and diffusion coefficient

$$B = \lim_{\Delta t \to 0} \frac{\langle (\Delta w)^2 \rangle}{\Delta t} = 2D, \tag{5.108}$$

where $\Delta w \equiv w - w_0$ and D is some constant quantifying the correlation strength. The probability density of the Wiener process is therefore the solution to the Fokker-Planck equation

$$\frac{\partial p(w,t|w_0,t_0)}{\partial t} = D \frac{\partial^2}{\partial w^2} p(w,t|w_0,t_0). \tag{5.109}$$

The initial conditions are usually set to $w_0 = 0$ at time $t_0 = 0$. From the initial conditions it follows that the transition probability

$$p(w,t_0|w_0,t_0) = \delta(w - w_0). \tag{5.110}$$

In order to derive the probability distribution $p(w,t|w_0,t_0)$, we investigate the characteristic function, Eq. (2.7), that is given by

$$\phi(s,t) \equiv \langle \exp(isw) \rangle = \int dw \; p(w,t|w_0,t_0) \exp(isw). \tag{5.111}$$

It has the time derivative

$$\begin{aligned} \frac{\partial \phi(s,t)}{\partial t} &= \frac{\partial}{\partial t} \int dw \; p(w,t|w_0,t_0) \exp(isw) \\ &= \int dw \; \exp(isw) \frac{\partial p(w,t|w_0,t_0)}{\partial t}. \end{aligned} \tag{5.112}$$

We can now insert the Fokker-Planck equation and get

$$\frac{\partial \phi(s,t)}{\partial t} = \int dw \; \exp(isw) D \frac{\partial^2 p(w,t|w_0,t_0)}{\partial w^2}. \tag{5.113}$$

D is constant and can be placed in front of the integral,

$$\frac{\partial \phi(s,t)}{\partial t} = D \int dw \; \exp(isw) \frac{\partial^2}{\partial w^2} p(w,t|w_0,t_0). \tag{5.114}$$

With partial integration, this gives

$$\frac{\partial \phi(s,t)}{\partial t} = D \left[\exp(isw) \frac{\partial}{\partial w} p(w,t|w_0,t_0) \right]_{-\infty}^{\infty} - D \int dw \; \left(\frac{\partial}{\partial w} \exp(isw) \right) \frac{\partial}{\partial w} p(w,t|w_0,t_0). \tag{5.115}$$

We assume that the transition probability density and its derivative are bounded, i.e. they fall of rapidly enough for large values $|w|$. The equation then simplifies to

$$\begin{aligned}\frac{\partial\phi(s,t)}{\partial t} &= -D\int dw\ \left(\frac{\partial}{\partial w}\exp(isw)\right)\frac{\partial}{\partial w}p(w,t|w_0,t_0)\\ &= -isD\int dw\ \exp(isw)\frac{\partial}{\partial w}p(w,t|w_0,t_0).\end{aligned} \tag{5.116}$$

We perform integration by parts a second time,

$$\frac{\partial\phi(s,t)}{\partial t} = -isD\left[\exp(isw)p(w,t|w_0,t_0)\right]_{-\infty}^{\infty} + isD\int dw\ \left(\frac{\partial}{\partial w}\exp(isw)\right)p(w,t|w_0,t_0). \tag{5.117}$$

The boundary terms vanish again, such that the expression results in

$$\begin{aligned}\frac{\partial\phi(s,t)}{\partial t} &= isD\int dw\ \left(\frac{\partial}{\partial w}\exp(isw)\right)p(w,t|w_0,t_0)\\ &= -s^2D\int dw\ \exp(isw)p(w,t|w_0,t_0)\\ &= -s^2D\ \phi(s,t).\end{aligned} \tag{5.118}$$

Inserting the characteristic function, Eq. (5.111), this simplifies to

$$\frac{\partial\phi(s,t)}{\partial t} = -s^2D\ \phi(s,t). \tag{5.119}$$

This partial differential equation is solved by

$$\phi(s,t) = \exp\left[-Ds^2(t-t_0)\right]\phi(s,t_0). \tag{5.120}$$

To see this, we compute its partial derivative with respect to time,

$$\begin{aligned}\frac{\partial\phi(s,t)}{\partial t} &= \frac{\partial}{\partial t}\exp\left[-Ds^2(t-t_0)\right]\phi(s,t_0)\\ &= -Ds^2\exp\left[-Ds^2(t-t_0)\right]\phi(s,t_0)\\ &= -Ds^2\phi(s,t).\end{aligned} \tag{5.121}$$

From the initial conditions it follows with Eq. (5.110) that

$$\begin{aligned}\phi(s,t_0) &= \int dw\ p(w,t_0|w_0,t_0)\exp(isw)\\ &= \int dw\ \delta(w-w_0)\exp(isw)\\ &= \exp(isw_0).\end{aligned} \tag{5.122}$$

So, the full characteristic function is given by

$$\phi(s,t) = \int dw\ p(w,t|w_0,t_0)\exp(isw) = \exp\left[isw_0 - Ds^2(t-t_0)\right]. \tag{5.123}$$

Its Fourier inverse is given by

$$\mathcal{F}^{-1}[\phi(s,t)] \equiv \frac{1}{2\pi}\int ds\ \phi(s,t)\exp(-isw). \tag{5.124}$$

Inserting our result for the characteristic function, we have

$$\mathcal{F}^{-1}[\phi(s,t)] = \frac{1}{2\pi}\int ds \int dw'\ p(w',t|w_0,t_0)\exp[isw']\exp[-isw]. \tag{5.125}$$

The probability density $p(w',t|w_0,t_0)$ is independent of the variable s. Therefor the expression can be rearranged to

$$\mathcal{F}^{-1}[\phi(s,t)] = \int dw'\ p(w',t|w_0,t_0)\ \frac{1}{2\pi}\int ds\ \exp[is(w'-w)]. \tag{5.126}$$

We identify a representation of the δ-Distribution, in particular

$$\frac{1}{2\pi}\int ds\ \exp[is(w'-w)] \equiv \delta(w'-w). \tag{5.127}$$

Substituting this representation into our earlier expression gives

$$\mathcal{F}^{-1}[\phi(s,t)] = \int dw'\ p(w',t|w_0,t_0)\ \delta(w'-w) = p(w,t|w_0,t_0). \tag{5.128}$$

Inserting Eq. (5.124), we arrive at an expression for the probability distribution in terms of its characteristic function,

$$p(w,t|w_0,t_0) = \frac{1}{2\pi}\int ds\ \phi(s,t)\exp(-isw). \tag{5.129}$$

Substituting our result for the characteristic function, Eq. (5.123), gives

$$\begin{aligned} p(w,t|w_0,t_0) &= \frac{1}{2\pi}\int ds \exp\left[isw_0 - Ds^2(t-t_0)\right]\exp[-isw] \\ &= \frac{1}{2\pi}\int ds \exp\left[isw_0 - Ds^2(t-t_0) - isw\right] \\ &= \frac{1}{2\pi}\int ds \exp\left[is(w_0-w) - Ds^2(t-t_0)\right]. \end{aligned} \tag{5.130}$$

After completing the square, the term in brackets is given by

$$T = -D(t-t_0)\left(s - \frac{i(w_0-w)}{2D(t-t_0)}\right)^2 - \frac{(w_0-w)^2}{4D(t-t_0)}. \tag{5.131}$$

To see this, we compute

$$\begin{aligned} T &= -D(t-t_0)\left[s - \frac{i(w_0-w)}{2D(t-t_0)}\right]^2 - \frac{(w_0-w)^2}{4D(t-t_0)} \\ &= -D(t-t_0)\left[s^2 - s\frac{i(w_0-w)}{D(t-t_0)} - \frac{(w_0-w)^2}{4D^2(t-t_0)^2}\right] - \frac{(w_0-w)^2}{4D(t-t_0)} \\ &= -Ds^2(t-t_0) + is(w_0-w). \end{aligned} \tag{5.132}$$

The full expression is thus given by

$$\begin{aligned} p(w,t|w_0,t_0) &= \frac{1}{2\pi}\int ds \exp\left[-D(t-t_0)\left(s-\frac{i(w_0-w)}{2D(t-t_0)}\right)^2 - \frac{(w_0-w)^2}{4D(t-t_0)}\right] \\ &= \frac{1}{2\pi}\exp\left[-\frac{(w_0-w)^2}{4D(t-t_0)}\right]\int ds \exp\left[-D(t-t_0)\left(s-\frac{i(w_0-w)}{2D(t-t_0)}\right)^2\right]. \end{aligned} \tag{5.133}$$

We now define $u = \sqrt{D(t-t_0)}\left(s - \frac{i(w_0-w)}{2D(t-t_0)}\right)$. Then the derivative is $\frac{du}{ds} = \sqrt{D(t-t_0)}$. Performing a substitution with u, Eq. (5.133) simplifies to

$$p(w,t|w_0,t_0) = \frac{1}{2\pi}\frac{1}{\sqrt{D(t-t_0)}}\exp\left[-\frac{(w_0-w)^2}{4D(t-t_0)}\right]\int du \exp\left[-u^2\right]. \tag{5.134}$$

The standard Gauss integral is solved by

$$\int du \exp\left[-u^2\right] = \sqrt{\pi}. \tag{5.135}$$

Inserting this, the expression is given by

$$\begin{aligned} p(w,t|w_0,t_0) &= \frac{1}{2\pi}\frac{\sqrt{\pi}}{\sqrt{D(t-t_0)}}\exp\left[-\frac{(w_0-w)^2}{4D(t-t_0)}\right]\int du \exp\left[-u^2\right] \\ &= \frac{1}{\sqrt{2\pi\cdot 2D(t-t_0)}}\exp\left[-\frac{(w_0-w)^2}{4D(t-t_0)}\right]\cdot\int du \exp\left[-u^2\right] \end{aligned} \tag{5.136}$$

We recognize this result as a Gaussian probability distribution, see Eq. (2.11), and we can directly read off the average

$$\langle W(t)\rangle = w_0 = 0 \tag{5.137}$$

and the variance

$$Var(W(t)) \equiv \langle W(t)^2\rangle - \langle W(t)\rangle^2 = \langle W(t)^2\rangle = 2D(t-t_0) = 2Dt \tag{5.138}$$

of the Wiener process. Here we assumed initial conditions $w_0 = 0$ at time $t_0 = 0$. Since the average is $\langle W(t)\rangle = 0$, the variance is identical to the second moment,

$$\langle W(t)^2\rangle = 2Dt. \tag{5.139}$$

In the future we will mostly consider a **standard Wiener process** with diffusion coefficient $B_{ij} = 1$, such that variance and second moment evaluate to

$$\langle W(t)^2\rangle = t. \tag{5.140}$$

We briefly mention some further properties of the Wiener process, that are formally derived in [26]. The paths of the Wiener process are continuous, irregular and non-differentiable. Additionally, non overlapping Wiener increments $\Delta W_i \equiv W(t_i) - W(t_{i-1})$ are statistically independent. Since $W(t)$ is a stochastic process, we should also consider the autocovariance

matrix $\mathbf{C}'$, see Eq. (2.9). Let's assume times $0 < s \leq t$. Then the coefficients of the autocovariance are

$$\begin{aligned} C'_{s,t} &\equiv \langle (W(s) - \langle W(s) \rangle)(W(t) - \langle W(t) \rangle) \rangle \\ &= \langle W(t)W(s) \rangle - \langle W(t) \rangle \langle W(s) \rangle. \end{aligned} \tag{5.141}$$

The averages $\langle W(t) \rangle$ and $\langle W(s) \rangle$ are 0, see Eq. 5.137, such that

$$C'_{s,t} = \langle W(t)W(s) \rangle. \tag{5.142}$$

We now rewrite this expression,

$$\begin{aligned} C'_{s,t} &= \langle W(t)W(s) \rangle \\ &= \langle [W(t) - W(s) + W(s)]W(s) \rangle \\ &= \langle [W(t) - W(s)]W(s) + W(s)^2 \rangle \\ &= \langle [W(t) - W(s)]W(s) \rangle + \langle W(s)^2 \rangle. \end{aligned} \tag{5.143}$$

From Eq. (5.137) we have $\langle W(s)^2 \rangle = s$. Therefor the autocovariance can be expressed by

$$C'_{s,t} = \langle [W(t) - W(s)]W(s) \rangle + s. \tag{5.144}$$

The initial conditions are $t_0 = 0$ and $W(t_0) = w_0 = 0$. We can therefore subtract a term $W(t_0) = 0$ in the expression,

$$\begin{aligned} C'_{s,t} &= \langle [W(t) - W(s)] \, [W(s) - 0] \rangle + s \\ &= \langle [W(t) - W(s)] \, [W(s) - W(t_0)] \rangle + s. \end{aligned} \tag{5.145}$$

The increments $\Delta W_1 = W(t) - W(s)$ and $\Delta W_2 = W(s) - W(t_0)$ are not overlapping, since $0 = t_0 < s \leq t$, and are thus statistically independent. It follows for the autocovariance that

$$\begin{aligned} C'_{s,t} &= \langle W(t) - W(s) \rangle \langle W(s) - W(t_0) \rangle + s \\ &= \langle W(t) - W(s) \rangle \langle W(s) \rangle + s \\ &= \langle W(t) \rangle \langle W(s) \rangle - \langle W(s) \rangle \langle W(s) \rangle + s \\ &= s, \end{aligned} \tag{5.146}$$

where in the last step we computed the averages, Eq. (5.137). Since we assumed $s < t$, it follows that

$$\forall \; t, t' > 0 : C'_{t,t'} = \langle W(t)W(t') \rangle = \min(t, t'). \tag{5.147}$$

We can now also state the first and second moment of the Wiener increment, defined by $\Delta W_t = W_{t_{i+1}} - W_{t_i}$. The first moment of the increment is

$$\langle \Delta W_t \rangle = \langle W_{t_{i+1}} - W_{t_i} \rangle = \langle W_{t_{i+1}} \rangle - \langle W_{t_i} \rangle = 0 \tag{5.148}$$

and the second moment is

$$\begin{aligned} \langle \Delta W_t^2 \rangle &= \langle (W_{t_{i+1}} - W_{t_i})(W_{t_{i+1}} - W_{t_i}) \rangle \\ &= \langle W_{t_{i+1}}^2 \rangle - 2 \langle W_{t_i} W_{t_{i+1}} \rangle + \langle W_{t_i}^2 \rangle \\ &= t_{i+1} - 2 \min(t, t_{i+1}) + t_i \\ &= t_{i+1} - 2t_i + t_i \\ &= t_{i+1} - t_i. \end{aligned} \tag{5.149}$$

Building on the one dimensional Wiener process, we define a multivariate standard Wiener process $\boldsymbol{W}(t) = [W_1(t), ..., W_n(t)]$ via the multivariable Fokker-Planck equation

$$\frac{\partial}{\partial t} p(\boldsymbol{w}, t|\boldsymbol{w}_0, t_0) = \frac{1}{2} \sum_i \frac{\partial^2}{\partial w_i^2} p(\boldsymbol{w}, t|\boldsymbol{w}_0, t_0) \tag{5.150}$$

with solution

$$p(\boldsymbol{w}, t|\boldsymbol{w}_0, t_0) = [2\pi \cdot (t - t_0)]^{-n/2} \exp\left[-\frac{(\boldsymbol{w}_0 - \boldsymbol{w})^2}{2(t - t_0)}\right]. \tag{5.151}$$

Then, the average is given by

$$\langle \boldsymbol{W}(t) \rangle = \boldsymbol{w}_0 = \boldsymbol{0} \tag{5.152}$$

and the covariance coefficient for variables i, j is

$$\begin{aligned} C_{ij} &= \langle (W_i(t) - \langle W_i(t) \rangle)(W_j(t) - \langle W_j(t) \rangle) \rangle \\ &= \langle W_i(t) W_j(t) \rangle - \langle W_i(t) \rangle \langle W_j(t) \rangle \\ &= \langle W_i(t) W_j(t) \rangle \end{aligned} \tag{5.153}$$

From the independence of the processes, follows for

$$i \neq j \implies \langle W_i(t) W_j(t) \rangle = \langle W_i(t) \rangle \langle W_j(t) \rangle = w_{0,i} w_{0,j} = 0. \tag{5.154}$$

For $i = j$ the coefficient reduces to the second moment

$$\left\langle W(t)^2 \right\rangle = t - t_0 = t. \tag{5.155}$$

So, in general

$$C_{ij} = \delta_{ij} \left\langle W(t)^2 \right\rangle = \delta_{ij} t. \tag{5.156}$$

Let's investigate the moments of a difference $\boldsymbol{W}(t) - \boldsymbol{W}(s)$. The first moment is

$$\langle \boldsymbol{W}(t) - \boldsymbol{W}(s) \rangle = \langle \boldsymbol{W}(t) \rangle - \langle \boldsymbol{W}(s) \rangle = \boldsymbol{0}. \tag{5.157}$$

In combination with the autocovariance, Eq. (5.147), the second moment has elements

$$\begin{aligned} C_{ij,ts} &= \langle (W_i(t) - W_i(s))(W_j(t) - W_j(s)) \rangle \\ &= \langle W_i(t) W_j(t) \rangle - \langle W_i(t) W_j(s) \rangle - \langle W_i(s) W_j(t) \rangle + \langle W_i(s) W_j(s) \rangle \\ &= \delta_{ij}(t - 2\min(t, s) + s) \\ &= \delta_{ij}(t - 2\min(t, s) + s) \\ &= \delta_{ij}(\max(t, s) - \min(t, s)). \end{aligned} \tag{5.158}$$

We now want to investigate the drift and diffusion terms of the multivariable Fokker-Planck equation. The drift vector, see Eq. (5.12), has coefficients

$$\begin{aligned} A_i(\boldsymbol{w}, t) &= \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int d\boldsymbol{w}' (w_i' - w_i) p(\boldsymbol{w}', t + \Delta t|\boldsymbol{w}, t) \\ &= \lim_{\Delta t \to 0} \frac{\langle W_i(t + \Delta t) - W_i(t) \rangle}{\Delta t} \\ &= \lim_{\Delta t \to 0} \frac{\langle W_i(t + \Delta t) \rangle - \langle W_i(t) \rangle}{\Delta t} \\ &= 0, \end{aligned} \tag{5.159}$$

where we used Eq. (5.137) for computing the averages.
The diffusion matrix has coefficients

$$\begin{aligned} B_{ij}(\boldsymbol{w},t) &= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\int d\boldsymbol{w}'(w_i'-w_i)(w_j'-w_j)p(\boldsymbol{w}',t+\Delta t|\boldsymbol{w},t) \\ &= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\Big[\langle (W_i(t+\Delta t)-W_i(t))(W_j(t+\Delta t)-W_j(t))\rangle\Big] \\ &= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\Big[\langle W_i(t+\Delta t)W_j(t+\Delta t)\rangle - \langle W_i(t+\Delta t)W_j(t)\rangle \\ &\qquad - \langle W_i(t)W_j(t+\Delta t)\rangle + \langle W_i(t)W_j(t)\rangle\Big]. \end{aligned} \tag{5.160}$$

We first assume $i \neq j$. Then, using the results for the covariance matrix, we have

$$\begin{aligned} B_{ij}(\boldsymbol{w},t) = \lim_{\Delta t\to 0}\frac{1}{\Delta t}\Big[&\langle W_i(t+\Delta t)\rangle\langle W_j(t+\Delta t)\rangle - \langle W_i(t+\Delta t)\rangle\langle W_j(t)\rangle \\ &- \langle W_i(t)\rangle\langle W_j(t+\Delta t)\rangle + \langle W_i(t)\rangle\langle W_j(t)\rangle\Big]. \end{aligned} \tag{5.161}$$

It follows that $B_{ij}(\boldsymbol{w},t)=0$. Now we assume $i=j$. Building on the covariance matrix as well as the autocovariance matrix, the expression is given by

$$\begin{aligned} Bij(\boldsymbol{w},t) &= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\Big[\langle W(t+\Delta t)^2\rangle - \langle W(t+\Delta t)W(t)\rangle - \langle W(t)W(t+\Delta t)\rangle + \langle W(t)^2\rangle\Big] \\ &= \lim_{\Delta t\to 0}\frac{t+\Delta t-\min(t+\Delta t,t)-\min(t+\Delta t,t)+t}{\Delta t} \\ &= \lim_{\Delta t\to 0}\frac{t+\Delta t-t-t+t}{\Delta t} \\ &= \lim_{\Delta t\to 0}\frac{\Delta t}{\Delta t}. \end{aligned} \tag{5.162}$$

And it follows that $Bij=1$. So, the probability distribution reproduces the drift vector and diffusion matrix, expected from the definition of the one dimensional Wiener process. What happens if the Wiener process is scaled by a constant a? Then the average stays zero, but the covariance is

$$\begin{aligned} C_{ij} &= \langle (aW_i(t)-\langle aW_i(t)\rangle)(aW_j(t)-\langle aW_j(t)\rangle)\rangle \\ &= a^2\langle (W_i(t)-\langle W_i(t)\rangle)(W_j(t)-\langle W_j(t)\rangle)\rangle \\ &= a^2 t. \end{aligned} \tag{5.163}$$

5.2.5 Connection between Langevin noise and Wiener process

Consider again the Langevin noise term $\boldsymbol{\xi}(t)$. For simplicity, we focus only on one component $\xi(t)$ and set for the correlation strength $2D\gamma^2=1$. We follow the steps in Ref. [26] and define the integral

$$u(t) = \int_0^t dt'\xi(t'). \tag{5.164}$$

Because the average of the noise $\langle \xi(t') \rangle = 0$, see Eq. (5.10), the average of $u(t)$ is

$$\langle u(t) \rangle = \left\langle \int_0^t dt' \xi(t') \right\rangle = \int_0^t dt' \left\langle \xi(t') \right\rangle = 0. \tag{5.165}$$

The second moment of the noise is given by $\langle \xi(s)\xi(\tau) \rangle = \delta(s - \tau)$, Eq. (5.11). So, the second moment of $u(t)$ must be

$$\begin{aligned} \langle u(t)^2 \rangle &= \left\langle \int_0^t ds \xi(s) \int_0^t d\tau \xi(\tau) \right\rangle \\ &= \int_0^t ds \int_0^t d\tau \left\langle \xi(s)\xi(\tau) \right\rangle \\ &= \int_0^t ds \int_0^t d\tau \ \delta(s - \tau) \\ &= \int_0^t ds = t \end{aligned} \tag{5.166}$$

and the variance of $u(t)$ is

$$\langle u(t)^2 \rangle - \langle u(t) \rangle^2 = \langle u(t)^2 \rangle = t. \tag{5.167}$$

We assume realizations $u = \int_0^t dt' \xi(t')$ at time t and $u' = \int_0^{t+\Delta t} dt' \xi(t')$ at time $t + \Delta t$. Then, we can derive the drift term, Eq. (5.12),

$$\begin{aligned} A(x,t) &= \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int du' \left(u' - u\right) p(u', t + \Delta t | u, t) \\ &= \lim_{\Delta t \to 0} \frac{\langle u' - u \rangle}{\Delta t} \\ &= \lim_{\Delta t \to 0} \frac{\langle \int_0^{t+\Delta t} dt' \ \xi(t') - \int_0^t dt' \ \xi(t') \rangle}{\Delta t} \\ &= \lim_{\Delta t \to 0} \frac{\langle \int_t^{t+\Delta t} dt' \ \xi(t') \rangle}{\Delta t} \\ &= \lim_{\Delta t \to 0} \frac{\int_t^{t+\Delta t} dt' \ \langle \xi(t') \rangle}{\Delta t} \end{aligned} \tag{5.168}$$

and with $\langle \xi(t') \rangle = 0$ it follows that $A(x,t) = 0$. The diffusion term, Eq. (5.13), is given by

$$\begin{aligned}
B(x,t) &= \lim_{\Delta t \to 0} \frac{1}{\Delta t} \int du'(u'-u)^2 p(u', t+\Delta t | u, t) \\
&= \lim_{\Delta t \to 0} \frac{\langle (u'-u)^2 \rangle}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{\left\langle \left(\int_0^{t+\Delta t} dt' \; \xi(t') - \int_0^t dt' \; \xi(t') \right) \left(\int_0^{t+\Delta t} ds \; \xi(s) - \int_0^t ds \; \xi(s) \right) \right\rangle}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{\left\langle \left(\int_t^{t+\Delta t} dt' \; \xi(t') \right) \left(\int_t^{t+\Delta t} ds \; \xi(s) \right) \right\rangle}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{\left\langle \int_t^{t+\Delta t} \int_t^{t+\Delta t} dt' ds \; \xi(t') \xi(s) \right\rangle}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{\int_t^{t+\Delta t} \int_t^{t+\Delta t} dt' ds \; \langle \xi(t') \xi(s) \rangle}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{\int_t^{t+\Delta t} \int_t^{t+\Delta t} dt' ds \; \delta(t'-s)}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{\int_t^{t+\Delta t} dt'}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{t + \Delta t - t}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{\Delta t}{\Delta t}.
\end{aligned} \tag{5.169}$$

And we again have $B(x,t) = 1$. So, $u(t)$ is actually the Wiener process $W(t)$ and one could informally write

$$W(t) = \int_0^t dt' \xi(t') \implies \xi(t) = \frac{\partial W(t)}{\partial t}. \tag{5.170}$$

But the Wiener process is not differentiable! One should therefore define the integral via increments

$$W(t) = \int_0^t dW := \lim_{n \to \infty} \sum_{k=0}^{n-1} \Delta W_k = \lim_{n \to \infty} \sum_{k=0}^{n-1} \int_{t_k}^{t_{k+1}} d\tau \; \xi(\tau) \tag{5.171}$$

with $t_k = k \Delta t$ for some time increment Δt.
Let's now assume the multicomponent Langevin noise term $\boldsymbol{\xi}(t)$ and set $2D\gamma^2 = a$ It has covariance

$$\begin{aligned}
C_{ij}^{(\xi)} &= \langle (\xi_i(t) - \langle \xi_i(t) \rangle)(\xi_j(t) - \langle \xi_j(t) \rangle) \rangle \\
&= \langle \xi_i(t) \xi_j(t) \rangle - \langle \xi_i(t) \rangle \langle \xi_j(t) \rangle \\
&= \langle \xi_i(t) \xi_j(t) \rangle \\
&= a \delta_{ij}
\end{aligned} \tag{5.172}$$

and autocovariance

$$\begin{aligned} C_{ts}^{(\xi)} &= \langle(\boldsymbol{\xi}(t) - \langle\boldsymbol{\xi}(t)\rangle)(\boldsymbol{\xi}(s) - \langle\boldsymbol{\xi}(s)\rangle)\rangle \\ &= \langle\boldsymbol{\xi}(t)\boldsymbol{\xi}(s)\rangle - \langle\boldsymbol{\xi}(t)\rangle\langle\boldsymbol{\xi}(s)\rangle \\ &= \langle\boldsymbol{\xi}(t)\boldsymbol{\xi}(s)\rangle \\ &= a\delta(t-s). \end{aligned} \tag{5.173}$$

Combined, we have

$$\langle(\xi_i(t) - \langle\xi_i(t)\rangle)(\xi_j(s) - \langle\xi_j(s)\rangle)\rangle = a\delta_{ij}\delta(t-s). \tag{5.174}$$

Then, we define the multivariable realization $\boldsymbol{u} = \int_0^t dt'\boldsymbol{\xi}(t')$ at time t as well as the realization $\boldsymbol{u}' = \int_0^{t+\Delta t} dt'\boldsymbol{\xi}(t')$ at time t'. We derive the diffusion matrix by

$$\begin{aligned} B_{ij}(\boldsymbol{x},t) &= \lim_{\Delta t\to 0}\frac{1}{\Delta t}\int d\boldsymbol{x}'(u_i'-u_i)(u_j'-u_j)p(\boldsymbol{u}',t+\Delta t|\boldsymbol{u},t) \\ &= \lim_{\Delta t\to 0}\frac{\langle(u_i'-u_i)(u_j'-u_j)\rangle}{\Delta t} \\ &= \lim_{\Delta t\to 0}\frac{\left\langle\left(\int_0^{t+\Delta t}dt'\ \xi_i(t') - \int_0^t dt'\ \xi_i(t')\right)\left(\int_0^{t+\Delta t}ds\ \xi_j(t') - \int_0^t ds\ \xi_j(t')\right)\right\rangle}{\Delta t} \\ &= \lim_{\Delta t\to 0}\frac{\left\langle\left(\int_t^{t+\Delta t}dt'\ \xi_i(t')\right)\left(\int_t^{t+\Delta t}ds\ \xi_j(t')\right)\right\rangle}{\Delta t} \\ &= \lim_{\Delta t\to 0}\frac{\left\langle\int_t^{t+\Delta t}\int_t^{t+\Delta t}dt'ds\ \xi_i(t')\xi_j(s)\right\rangle}{\Delta t} \\ &= \lim_{\Delta t\to 0}\frac{\int_t^{t+\Delta t}\int_t^{t+\Delta t}dt'ds\ \langle\xi_i(t')\xi_j(s)\rangle}{\Delta t} \\ &= a\lim_{\Delta t\to 0}\frac{\int_t^{t+\Delta t}\int_t^{t+\Delta t}dt'ds\ \delta_{ij}\delta(t'-s)}{\Delta t} \\ &= a\delta_{ij}\lim_{\Delta t\to 0}\frac{\int_t^{t+\Delta t}dt'}{\Delta t} \\ &= a\delta_{ij}\lim_{\Delta t\to 0}\frac{\Delta t}{\Delta t} \\ &= a\delta_{ij}. \end{aligned} \tag{5.175}$$

Now, one could informally write

$$\sqrt{a}\boldsymbol{W}(t) = \int_0^t dt'\boldsymbol{\xi}(t'). \tag{5.176}$$

We will see a reappearance of the Wiener process in the next chapter. Following the steps in Refs. [50] and [51]. We define the general Langevin equation by

$$\frac{dx}{dt} = a(x,t) + b(x,t)\xi(t), \tag{5.177}$$

where $\xi(t)$ is Gaussian white noise. This means its first moment is $\langle \xi(t) \rangle = 0$ and its second moment is $\langle \xi(t_1)\xi(t_2) \rangle = 2D\delta(t_1 - t_2)$ with correlation strength D. We further assume that odd correlation functions of $\xi(t)$ vanish and that otherwise we have

$$\langle \xi(t_1)\xi(t_2)...\xi(t_{2n}) \rangle = (2D)^n \left[\sum_{\frac{(2n!)}{(2^n n!)}} \delta(t_{j_1} - t_{j_2})...\delta(t_{j_{2n-1}} - t_{j_{2n}}) \right]. \tag{5.178}$$

The term $a(x,t)$ is referred to as drift term and $b(x,t)\xi(t)$ as diffusion term. The diffusion term introduces the stochastic nature of this stochastic differential equation. We restate the Langevin equation in integral form,

$$x(t + \Delta t) - x(t) = \int_t^{t+\Delta t} dt' \; a[x(t'), t'] + \int_t^{t+\Delta t} dt' \; b[x(t'), t']\xi(t'). \tag{5.179}$$

We set note that We restate the Langevin equation in integral form,

$$[x(t'), t'] = a[x(t) + (x(t') - x(t)), t'] \tag{5.180}$$

and analogously,

$$b[x(t'), t'] = b[x(t) + (x(t') - x(t)), t'] \tag{5.181}$$

With this, we can perform a Taylor expansion around $x(t)$ for the drift and diffusion coefficients,

$$a[x(t'), t'] = a[x(t), t'] + [x(t') - x(t)] \frac{\partial a}{\partial x}\bigg|_{x(t)} + \cdots, \tag{5.182}$$

$$b[x(t + \Delta t), t + \Delta t] = b[x(t), t'] + [x(t') - x(t)] \frac{\partial b}{\partial x}\bigg|_{x(t)} + \cdots. \tag{5.183}$$

The integral does then evaluate to

$$\begin{aligned} x(t + \Delta t) - x(t) = & \int_t^{t+\Delta t} dt' \; a[x(t), t'] \\ & + \int_t^{t+\Delta t} dt' \; [x(t') - x(t)] \frac{\partial a}{\partial x}\bigg|_{x(t')} + \cdots \\ & + \int_t^{t+\Delta t} dt' \; b[x(t), t']\xi(t') \\ & + \int_t^{t+\Delta t} dt' \; [x(t') - x(t)] \frac{\partial b}{\partial x}\bigg|_{x(t')} \xi(t') + \cdots \end{aligned} \tag{5.184}$$

We notice that the differences $x(t') - x(t)$ can again be expressed as integrals of the form

$$x(t') - x(t) = \int_t^{t'} d\tilde{t} \; a[x(\tilde{t}), \tilde{t}] + \int_t^{t'} d\tilde{t} \; b[x(\tilde{t}), \tilde{t}]\xi(\tilde{t}). \tag{5.185}$$

Inserting this into the previous expression gives

$$\begin{aligned} x(t+\Delta t) - x(t) = &\int_t^{t+\Delta t} dt'\ a[x(t),t'] \\ &+ \int_t^{t+\Delta t} dt'\ \left.\frac{\partial a}{\partial x}\right|_{x(t')} \int_t^{t'} d\tilde{t}\ a[x(\tilde{t}),\tilde{t}] \\ &+ \int_t^{t+\Delta t} dt'\ \left.\frac{\partial a}{\partial x}\right|_{x(t')} \int_t^{t'} d\tilde{t}\ b[x(\tilde{t}),\tilde{t}]\xi(\tilde{t}) + \cdots \\ &+ \int_t^{t+\Delta t} dt'\ b[x(t),t']\xi(t') \\ &+ \int_t^{t+\Delta t} dt'\ \left.\frac{\partial b}{\partial x}\right|_{x(t')} \xi(t') \int_t^{t'} d\tilde{t}\ a[x(\tilde{t}),\tilde{t}] \\ &+ \int_t^{t+\Delta t} dt'\ \left.\frac{\partial b}{\partial x}\right|_{x(t')} \xi(t') \int_t^{t'} d\tilde{t}\ b[x(\tilde{t}),\tilde{t}]\xi(\tilde{t}) + \cdots . \end{aligned} \tag{5.186}$$

Now, taking the average,

$$\langle x(t+\Delta t) - x(t)\rangle \tag{5.187}$$

and invoking the correlation functions of the noise $\xi(t)$, gives

$$\begin{aligned} \langle x(t+\Delta t) - x(t)\rangle = &\int_t^{t+\Delta t} dt'\ a[x(t),t'] \\ &+ 2D \int_t^{t+\Delta t} dt'\ \left.\frac{\partial b}{\partial x}\right|_{x(t')} \xi(t') \int_t^{t'} d\tilde{t}\ b[x(\tilde{t}),\tilde{t}]\xi(\tilde{t})\delta(\tilde{t}-t'). \end{aligned} \tag{5.188}$$

Here, terms of order $\mathcal{O}(\Delta t^2)$ were neglected. This includes terms with at least four occurrences of $\xi(t)$. Additionally, terms where the noise $\xi(t)$ occurs an odd number of times, vanished. Choosing an appropriate delta distribution, one derives

$$A(x,t) = \lim_{\Delta t\to 0} \frac{1}{\Delta t}\langle x(t+\Delta t) - x(t)\rangle = a(x,t) + Db(x,t)\frac{\partial b(x,t)}{\partial x}. \tag{5.189}$$

In an analogous way one derives the diffusion coefficient

$$B(x,t) = \lim_{\Delta t\to 0} \frac{1}{\Delta t}\langle [x(t+\Delta t) - x(t)][x(t+\Delta t) - x(t)]\rangle = 2Db^2(x,t). \tag{5.190}$$

5.2.6 Stochastic differential equations

We don't want to establish the whole framework of stochastic calculus in this work. But we give a very brief introduction to the Itô process. Following Ref. [52], a stochastic process $X(t)$ is an **Itô process**, if it is of the form

$$X_t = X_0 + \int_0^t a(X_s)ds + \int_0^t b(X_s)dW_s. \tag{5.191}$$

Here, X_0 is an initial value, $\int_0^t a(X_s)ds$ is the drift term, $\int_0^t b(X_s)dW_s$ the diffusion term and W is a Wiener process. In differential form this can be stated as

$$dX_t = X_0 + a(X_t)dt + b(X_t)dW_t. \tag{5.192}$$

A discrete approximation is provided by the **Euler-Maruyama approximation**. Let $t_0 = \tau_0 < \tau_1 < ... < \tau_n < ... < \tau_N = T$ be a discretization of the time interval $[t_0, T]$ and consider another continuous time stochastic process $Y = (t)$. It is called an Euler approximation of $X(t)$, if it satisfies the iterative scheme

$$\begin{aligned} Y_{n+1} &= Y_n + a(\tau_n, Y_n)(\tau_{n+1} - \tau_n) + b(\tau_n, Y_n)(W_{\tau_{n+1}} - W_{\tau_n}) \\ &= Y_n + a(\tau_n, Y_n)\Delta_n + b(\tau_n, Y_n)\Delta_W \end{aligned} \tag{5.193}$$

for $n = 0, 1, ..., N-1$ and initial value $Y_0 = X_0$. Here $Y_n := Y(\tau_n)$, $\Delta_n = \tau_{n+1} - \tau_n$ is the n-th time increment and $\Delta_W = W_{\tau_{n+1}} - W_{\tau_n}$ is the n-th random increment.

Chapter 6

Theoretical derivations

In this work experimental results as well as numerical simulations supported the hypothesis that SGD favors flat minima. Equipped with the necessary theoretical concepts from stochastic dynamics we are now able to explore theoretical explanations for these observations.

6.1 Continuous approximation of stochastic gradient descent

The updating rules of GD and SGD are discrete. To investigate them with a broad mathematical framework, it is helpful to establish continuous time approximations of the updating rules.

6.1.1 Gradient flow

Equation (4.4) presented SGD as GD with an additional noise term. Therefor we first investigate the GD method. As discussed before, GD is characterized by the **parameter updating rule**, Eq. (3.18),

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t). \tag{6.1}$$

For a continuous approximation of GD we follow the steps in Refs. [53] and [54]. We first introduce a continuous time measure τ and define the **gradient flow** $\boldsymbol{\Theta}(\tau)$ as the solution to the ordinary differential equation

$$\dot{\boldsymbol{\Theta}}(\tau) = \frac{d\boldsymbol{\Theta}(\tau)}{d\tau} = -\nabla_{\boldsymbol{\Theta}} L\left(\boldsymbol{\Theta}(\tau)\right) \tag{6.2}$$

with initial condition $\boldsymbol{\Theta}(0) = \boldsymbol{\theta}_0$. For small learning rate η, a Taylor expansion of the gradient flow $\boldsymbol{\Theta}(\tau + \eta)$ around point τ, is given by

$$\begin{aligned}
\boldsymbol{\Theta}(\tau + \eta) &= \boldsymbol{\Theta}(\tau) + \eta \frac{d\boldsymbol{\Theta}(\tau)}{d\tau} + \frac{1}{2}\eta^2 \frac{d^2\boldsymbol{\Theta}(\tau)}{d^2\tau} + \dots \\
&= \boldsymbol{\Theta}(\tau) + \eta \frac{d\boldsymbol{\Theta}(\tau)}{d\tau} + \mathcal{O}(\eta^2) \\
&= \boldsymbol{\Theta}(\tau) + \eta \dot{\boldsymbol{\Theta}}(\tau) + \mathcal{O}(\eta^2) \\
&\approx \boldsymbol{\Theta}(\tau) + \eta \dot{\boldsymbol{\Theta}}(\tau),
\end{aligned} \tag{6.3}$$

where in the last line, we neglected terms of order η^2 or higher. We assume that the continuous time variable τ is defined on the real interval $[0, T]$ with $T = N\eta \in \mathbb{R}$ and $N \in \mathbb{N}$. In the discrete Euler approximation, see Ref. [52], this interval has discretization $\{\tau_t\}_{t=0}^{N} \subseteq [0, T]$ with $0 = \tau_0 < \tau_1 < \tau_2 < \cdots < \tau_N = T$ and the step size is given by

$\Delta\tau_t = \tau_{t+1} - \tau_t = (t+1)\eta - t\eta = \eta$. Applying this discretization, we identify $\boldsymbol{\Theta}(\tau_t)$ with $\boldsymbol{\theta}_t$ and $\boldsymbol{\Theta}(\tau_t + \eta)$ with $\boldsymbol{\theta}_{t+1}$. Together with $\eta\dot{\boldsymbol{\Theta}}(\tau_t) = -\eta\nabla_{\boldsymbol{\Theta}} L\left(\boldsymbol{\Theta}(\tau_t)\right)$, this establishes the GD algorithm

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) \tag{6.4}$$

as discrete approximation of the continuous gradient flow $\boldsymbol{\Theta}(\tau)$. Analogously, gradient flow can be viewed as continuous approximation of GD.

6.1.2 Stochastic modified equations

We now want to derive a continuous time approximation of the discrete SGD updating rule. SGD is an optimization technique invoking the **parameter updating rule**

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\nabla_{\boldsymbol{\theta}} L^{\mu}(\boldsymbol{\theta}_t) \overset{4.4}{=} \boldsymbol{\theta}_t - \eta\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) + \sqrt{\eta}\boldsymbol{\xi}(\boldsymbol{\theta}_t). \tag{6.5}$$

In the random shift model, see section 4.1, there is an ensemble of minibatch loss landscapes $\{L^{\mu}\}_{\mu}$ where the $\{\mu_i\}_{i\in\mathbb{N}}$ are independent uniform random variables with values in $\{1, 2, ...n\}$, [55]. The discrete SGD updating rule is the GD updating rule with an additional noise term $\sqrt{\eta}\boldsymbol{\xi} = \eta[\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) - \nabla_{\boldsymbol{\theta}} L^{\mu}(\boldsymbol{\theta}_t)]$, which depends on the parameters $\boldsymbol{\theta}_t$. Invoking Eq. (6.2), we therefore expect the continuous approximation of SGD to be of the form

$$d\boldsymbol{\Theta}(\tau) = -\nabla L(\boldsymbol{\Theta}(\tau))d\tau + f(\boldsymbol{\Theta}(\tau))d\tau,$$

where the stochastic term $f(\boldsymbol{\Theta}(\tau))d\tau$ is a function of $\boldsymbol{\Theta}(\tau)$, that has to be specified. Approximating discrete stochastic gradient algorithms by stochastic differential equations is referred to as the method of stochastic modified equations by Ref. [56]. Variants of this method have been applied and investigated in several publications, including Refs. [57], [58], [54], [59], [60], [61], [55], [62], [63], [64] and [56]. In Ref. [56] it is shown that for a weak approximation of Eq. (4.4) one should set

$$f(\boldsymbol{\Theta}(\tau))d\tau := \eta\sqrt{\mathbf{C}(\boldsymbol{\Theta}(\tau))}d\boldsymbol{W}(\tau).$$

Here, $\mathbf{C}$ is the covariance matrix, Eq. (4.10), and $\boldsymbol{W}(\tau)$ is a multidimensional Wiener process. This leads to the stochastic differential equation

$$d\boldsymbol{\Theta}(\tau) = -\nabla L(\boldsymbol{\Theta}(\tau))d\tau + \eta\sqrt{\mathbf{C}(\boldsymbol{\Theta}(\tau))}d\boldsymbol{W}(\tau). \tag{6.6}$$

Reference [56] motivates this by noting that Eq. (6.6) is an Itô stochastic differential equation, see Eq. (5.191),

$$d\boldsymbol{\Theta}(\tau) = \boldsymbol{a}d\tau + \boldsymbol{b}\boldsymbol{W}(\tau) \tag{6.7}$$

with drift term $\boldsymbol{a}(\tau, \boldsymbol{\Theta}(\tau)) = -\nabla L(\boldsymbol{\Theta}(\tau))$ and diffusion term $\boldsymbol{b}(\tau, \boldsymbol{\Theta}(\tau)) = \sqrt{\eta\mathbf{C}(\boldsymbol{\Theta}(\tau))}$. It can be approximated by Euler-Maruyama discretization, see Eq. (5.193) and Ref. [52]. Again, we first introduce the discretization $\{\tau_t\}_{t=0}^{N} \subseteq [0, T]$ with $0 = \tau_0 < \tau_1 < \tau_2 < \cdots < \tau_N = T$ and step size $\Delta\tau_t = \eta$. We first set $\boldsymbol{b} = \mathbf{0}$. Then, the remaining ordinary differential equation resembles the total differential of the continuous approximation of gradient descent, Eq. 6.2, with discrete counterpart

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t). \tag{6.8}$$

Next, we observe the stochastic part, introduced by $d\boldsymbol{W}(\tau)$ for $\boldsymbol{b} \neq \boldsymbol{0}$. The Wiener increment is given by $\Delta \boldsymbol{W}_t = \boldsymbol{W}_{\tau_{t+1}} - \boldsymbol{W}_{\tau_t}$. From Eq. (5.157) and Eq. (5.158) we know that the first moment of the Wiener increment is

$$\langle \Delta \boldsymbol{W}_t \rangle = \boldsymbol{0} \tag{6.9}$$

and the second moment has elements

$$\langle \Delta W_{ij,t}^2 \rangle = \delta_{ij}(\tau_{t+1} - \tau_t) = \delta_{ij}\Delta\tau = \delta_{ij}\eta. \tag{6.10}$$

The complete Euler-Maruyama discretization is then,

$$\begin{aligned} \boldsymbol{\theta}_{t+1} &= \boldsymbol{\theta}_t - \nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t)\Delta\tau_t + \eta\sqrt{\mathbf{C}(\boldsymbol{\theta}_t)}\Delta\boldsymbol{W}_t \\ &= \boldsymbol{\theta}_t - \eta\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}_t) + \eta\sqrt{\mathbf{C}(\boldsymbol{\theta}_t)}(\boldsymbol{W}_{(t+1)\eta} - \boldsymbol{W}_{t\eta}). \end{aligned} \tag{6.11}$$

It turns out the first and second moments of the update steps $\Delta\boldsymbol{\theta}_t$ are equal for the SGD updating rule Eq. (3.19) and the Euler approximation Eq. (6.11). So they are similar in a distributional sense. This is the heuristic justification for viewing Eq. (6.6) as continuous time approximation of Eq. (3.19). Further mathematical rigorous results are derived by Ref. [56].
The continuous approximation has the Fokker-Planck equation

$$\begin{aligned} \frac{\partial p(\boldsymbol{\Theta}, \tau|\boldsymbol{\Theta}_0, \tau_0)}{\partial \tau} &= -\sum_i \frac{\partial}{\partial \Theta_i}\left[A_i(\boldsymbol{\Theta}, t)p(\boldsymbol{\Theta}, t|\boldsymbol{\Theta}_0, t_0)\right] \\ &\quad + \frac{1}{2}\sum_{ij}\frac{\partial^2}{\partial\Theta_i\partial\Theta_j}[B_{ij}(\boldsymbol{\Theta}, t)p(\boldsymbol{\Theta}, t|\boldsymbol{\Theta}_0, t_0)] \\ &= -\sum_i \frac{\partial}{\partial \Theta_i}\left[-\frac{\partial}{\partial\Theta_i}L(\boldsymbol{\Theta}(\tau))(\boldsymbol{\Theta}, t)p(\boldsymbol{\Theta}, t|\boldsymbol{\Theta}_0, t_0)\right] \\ &\quad + \frac{1}{2}\sum_{ij}\frac{\partial^2}{\partial\Theta_i\partial x_j}[C_{ij}(\boldsymbol{\Theta}, t)p(\boldsymbol{\Theta}, t|\boldsymbol{\Theta}_0, t_0)]. \end{aligned} \tag{6.12}$$

The drift vector $\boldsymbol{A}$ has elements $A_i(\boldsymbol{\Theta}, t) = -\frac{\partial}{\partial\Theta_i}L(\boldsymbol{\Theta}(\tau))$ and the diffusion matrix $\boldsymbol{B}$ has elements $B_{ij}(\boldsymbol{\Theta}, t) = C_{ij}(\boldsymbol{\Theta}, t)$. Additionally, we can view Eq. (6.11) as the integral of a general Langevin equation

$$\dot{\boldsymbol{\theta}} = -\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta}) + \boldsymbol{\xi}(\boldsymbol{\theta}).$$

6.2 Effective loss function

In this section we want to construct an effective potential, the effective loss function, that is underlying the training dynamics. We again follow the steps in Ref. [21] and continue our analysis by specifying the loss function as in the two-dimensional model, i.e.

$$L(x, y) = \frac{1}{2}\lambda_0\varepsilon x^2 + \frac{1}{2}\lambda(x)y^2 \tag{6.13}$$

with flat variable x and sharp variable y. Similar to Ref. [21], we redefine the diffusion matrix by

$$\mathbf{B} \equiv \eta\mathbf{C}/2 = \eta\sigma\mathbf{H}^2 = \Delta_S\mathbf{H}^2. \tag{6.14}$$

The relation $\mathbf{C} = 2\sigma\mathbf{H}^2$ was derived in the random shift model, Eq. (4.30), and the effective noise $\Delta_S \equiv \eta\sigma$, Eq. (4.30) was defined in the numerical analysis. The Hessian $\mathbf{H}$ was derived in the loss model, Eq. (4.60). Then, the diffusion matrix $\mathbf{B}(x, y)$ is given by

$$\begin{aligned}\mathbf{B} &= \Delta_S \begin{bmatrix} \lambda_0\varepsilon + \frac{1}{2}\lambda''(x)y^2 & \lambda'(x)y \\ \lambda'(x)y & \lambda(x). \end{bmatrix} \begin{bmatrix} \lambda_0\varepsilon + \frac{1}{2}\lambda''(x)y^2 & \lambda'(x)y \\ \lambda'(x)y & \lambda(x). \end{bmatrix} \\ &= \Delta_S \begin{bmatrix} \left(\lambda_0\varepsilon + \frac{1}{2}\lambda''(x)y^2\right)^2 + \lambda'(x)^2y^2 & \lambda'(x)y\left(\lambda_0\varepsilon + \frac{1}{2}\lambda''(x)y^2\right) + \lambda(x)\lambda'(x)y \\ \lambda'(x)y\left(\lambda_0\varepsilon + \frac{1}{2}\lambda''(x)y^2\right) + \lambda(x)\lambda'(x)y & \lambda'(x)^2y^2 + \lambda(x)^2 \end{bmatrix}.\end{aligned} \tag{6.15}$$

Analogously, the Drift vector $\boldsymbol{A}$ is defined by

$$\boldsymbol{A}(x, y) = \nabla_{\boldsymbol{\theta}} L = \begin{pmatrix} \frac{\partial}{\partial x} L(x, y) \\ \frac{\partial}{\partial y} L(x, y) \end{pmatrix} = \begin{pmatrix} \lambda_0\varepsilon x + \frac{1}{2}\lambda'(x)y^2 \\ \lambda(x)y \end{pmatrix}. \tag{6.16}$$

From the Langevin equation, Eq. (6.1.2), it follows for the two-dimensional case that the time evolution of the flat variable is

$$\dot{x} = A_1(x, y) + \xi_x(x, y) = \lambda_0\varepsilon x + \frac{1}{2}\lambda'(x)y^2 + \xi_x(x, y). \tag{6.17}$$

Analogously, the time evolution of the sharp variable is

$$\dot{y} = A_2(x, y) + \xi_y(x, y) = \lambda(x)y + \xi_y(x, y). \tag{6.18}$$

The corresponding Fokker-Planck equation is

$$\begin{aligned}\frac{\partial p(x, y, t)}{\partial t} &= \frac{\partial}{\partial x}\left[A_1(x, y)p(x, y, t)\right] + \frac{\partial}{\partial y}\left[A_2(x, y)p(x, y, t)\right] \\ &\quad + \frac{\partial^2}{\partial x \partial x}[B_{11}(x, y)p(x, y)] + \frac{\partial^2}{\partial y \partial x}[B_{12}(x, y)p(x, y)] \\ &\quad + \frac{\partial^2}{\partial x \partial y}[B_{21}(x, y)p(x, y)] + \frac{\partial^2}{\partial y \partial y}[B_{22}(x, y)p(x, y)].\end{aligned} \tag{6.19}$$

6.2.1 Adiabatic elimination

We are interested in steady state solutions to the Fokker-Planck equation, Eq. (6.19). For this, we first need to introduce the notion of time separation. Reference [21] defines the timescale

$$\tau_x = \left[\frac{\partial}{\partial x} A_1(x, y)\right]^{-1} = \left[\varepsilon\lambda_0 + \frac{1}{2}\lambda''(x)y^2\right]^{-1} \tag{6.20}$$

for the flat variable x and the timescale

$$\tau_y = \left[\frac{\partial}{\partial x} A_2(x, y)\right]^{-1} = \lambda(x)^{-1} \tag{6.21}$$

for the sharp variable y. The assumption, backed by our empirical results, is that there is a time separation between the phase where the sharp variable y has the largest influence and the phase where the flat variable x has the largest influence on the time evolution of the loss function. The large eigenvalue associated with the sharp variable y imply that the

early evolution of the loss function is mainly driven by sharp variable and quickly reaches a local minimum. The rest of the training dynamics are then mostly influenced by the slow time evolution of the x variable. We therefor assume that $\tau_y \ll \tau_x$. We additionally assume a small noise level Δ_S and small $\varepsilon \ll 1$. Further, we require that the cross terms B_{12} and B_{21} of the diffusion matrix vanish, such that the diffusion matrix is given by

$$\mathbf{B} = \Delta_S \begin{bmatrix} \left(\lambda_0\varepsilon + \frac{1}{2}\lambda''(x)y^2\right)^2 + \lambda'(x)^2y^2 & 0 \\ 0 & \lambda'(x)^2y^2 + \lambda(x)^2 \end{bmatrix} \tag{6.22}$$

and the Fokker-Planck equation is given by

$$\begin{aligned} \frac{\partial p(x,y,t)}{\partial t} &= \frac{\partial}{\partial x}\left[A_1(x,y)p(x,y,t)\right] + \frac{\partial}{\partial y}\left[A_2(x,y)p(x,y,t)\right] \\ &\quad + \frac{\partial^2}{\partial x \partial x}[B_{11}(x,y)p(x,y,t)] + \frac{\partial^2}{\partial y \partial y}[B_{22}(x,y)p(x,y,t)] \\ &= L_x p(x,y,t) + L_y p(x,y,t). \end{aligned} \tag{6.23}$$

Here, we introduced the Fokker-Planck operators

$$L_x(x,y) = \frac{\partial}{\partial x}A_1(x,y) + \frac{\partial^2}{\partial x \partial x}B_{11}(x,y), \tag{6.24}$$

$$L_y(x,y) = \frac{\partial}{\partial y}A_2(x,y) + \frac{\partial^2}{\partial y \partial y}B_{22}(x,y). \tag{6.25}$$

We are interested in the steady state solution, that is defined by

$$\frac{\partial p_{ss}(x,y,t)}{\partial t} = 0. \tag{6.26}$$

There is no explicit time dependence and we can write $p_{ss}(x,y)$. This expression can always be stated in terms of the one variable probability distribution $p_{ss}(x)$ and the conditional probability distribution $p_{ss}(x|y)$,

$$p_{ss}(x,y) = p_{ss}(x|y)p_{ss}(x). \tag{6.27}$$

We now need to derive $p_{ss}(x|y)$ and $p_{ss}(x)$. Under the assumption of time separation, we can perform adiabatic elimination of the fast variable y, following the steps in [65] and [46]. We first restate the operator $L_y(x,y)$ by factoring out a large constant $\alpha > \tau_y^{-1}$, such that

$$L_y(x,y) = \alpha \mathcal{L}_y(x,y). \tag{6.28}$$

We then fix the variable x, rendering x a parameter. For fixed x, $\mathcal{L}_y(x,y)$ does only depend on y. We denote this by writing $\mathcal{L}_y(y;x)$. We analogously define $L_x(x;y)$ for fixed y. Since $L_x(x;y)$ and $\mathcal{L}_y(y;x)$ each only depend on one variable, we can work with our results from section (5.2), especially from the subsection about the method of eigenvalue expansion. We derived that there are eigenvalue equations

$$\mathcal{L}_y(y;x)\varphi_n(y;x) = -\mu_n(x)\varphi_n(y;x) \tag{6.29}$$

for the operator $\mathcal{L}(y;x)$ with $n \geq 0$. The stationary equation is

$$\mathcal{L}(x|y)\varphi_0(y;x) = -\mu_0(x)\varphi_0(y;x) = 0 \tag{6.30}$$

with eigenvalue $\mu_0(x) = 0$. Additionally, we introduced the operator $L = e^{\phi(y)/2} L_{FP} e^{-\phi(y)/2}$, see Eq. (5.95), with eigenvalue equation $L\psi_n(y;x) = -\mu_n(x)\psi_n(y;x)$ and we established the relation $\varphi_n = \psi_n \sqrt{\frac{\varphi_0}{N}}$, see Eq. (5.97). For ease of computation we set the normalization constant to $N = 1$, such that $\varphi_n(y;x) = \psi_n(y;x)\sqrt{\varphi_0(y;x)}$. We also introduce the eigenvalue equation of the adjoint operator,

$$\mathcal{L}^\dagger(y;x)\overline{\varphi_n(y;x)} = -\mu_n(x)\overline{\varphi_n(y;x)} \tag{6.31}$$

and the eigenvalue relation

$$\overline{\varphi_n(y;x)} = \psi_n(y;x)/\sqrt{\varphi_0(y;x)} = \varphi_n(y;x)/\varphi_0(y;x). \tag{6.32}$$

We can now derive that

$$\overline{\varphi_0(y;x)} = \varphi_0(y;x)/\varphi_0(y;x) = 1. \tag{6.33}$$

Additionally, we have the orthonormality relation

$$\int_{-\infty}^{\infty} dy\ \overline{\psi_n(y;x)}\psi_m(y;x) = \int_{-\infty}^{\infty} dy\ \overline{\varphi_n(y;x)}\varphi_m(y;x) = \delta_{nm}, \tag{6.34}$$

and the completeness relation,

$$\delta(y - y') = \sum_n \overline{\psi_n(y;x)}\psi_n(y'|x) = \sum_n \overline{\varphi_n(y;x)}\varphi_n(y'|x). \tag{6.35}$$

Since the set of eigenfunctions is complete, we can now expand the full probability density $p(x, y, t)$ of both variables by

$$p(x, y, t) = \sum_{m=0}^{\infty} c_m(x, t)\varphi_m(y;x) \tag{6.36}$$

with coefficients $c_m(x, t)$. We can express the y-independent probability density $p(x, t)$ by the integral

$$p(x, t) = \int dy\ p(x, y, t). \tag{6.37}$$

Inserting the expansion of $p(x, y, t)$ into this expression, gives

$$p(x, t) = \int dy \sum_{m=0}^{\infty} c_m(x, t)\varphi_m(y;x). \tag{6.38}$$

We can always multiply with $1 = \overline{\varphi_0(y;x)}$, such that

$$\begin{aligned} p(x, t) &= \int dy \sum_{m=0}^{\infty} c_m(x, t)\varphi_m(y;x) \cdot 1 \\ &= \int dy \sum_{m=0}^{\infty} c_m(x, t)\varphi_m(y;x)\overline{\varphi_0(y;x)}. \end{aligned} \tag{6.39}$$

Then, assuming suitable conditions, such that limes and integral are interchangeable, it follows that

$$p(x, t) = \sum_{m=0}^{\infty} c_m(x, t) \int dy\ \varphi_m(y;x)\overline{\varphi_0(y;x)}. \tag{6.40}$$

Now, invoking the orthonormality relation gives

$$p(x,t) = \sum_{m=0}^{\infty} c_m(x,t)\delta_{m0} = c_0(x,t). \tag{6.41}$$

Inserting the expansion of $p(x,y,t)$ into the Fokker Planck equation, Eq. (6.23), leads to

$$\frac{\partial p(x,y,t)}{\partial t} = \frac{\partial}{\partial t}\sum_{m=0}^{\infty} \varphi_m(y;x)c_m(x,t). \tag{6.42}$$

In operator representation we have

$$\frac{\partial p(x,y,t)}{\partial t} = L_x \sum_{m=0}^{\infty} \varphi_m(y;x)c_m(x,t) + L_y \sum_{m=0}^{\infty} \varphi_m(y;x)c_m(x,t). \tag{6.43}$$

With the substitute operator $\alpha\mathcal{L}_y$, this gives

$$\frac{\partial p(x,y,t)}{\partial t} = L_x \sum_{m=0}^{\infty} \varphi_m(y;x)c_m(x,t) + \alpha\mathcal{L}_y \sum_{m=0}^{\infty} \varphi_m(y;x)c_m(x,t). \tag{6.44}$$

By letting the operator act on the eigenfunctions, we find that

$$\frac{\partial p(x,y,t)}{\partial t} = L_x \sum_{m=0}^{\infty} \varphi_m(y;x)c_m(x,t) - \alpha \sum_{m=0}^{\infty} \mu_m(x)\varphi_m(y;x)c_m(x,t). \tag{6.45}$$

We multiply this equation from the left by $\overline{\varphi_n(y;x)}$, which doesn't explicitly depend on the time variable t, such that

$$\begin{aligned}\frac{\partial}{\partial t}\sum_{m=0}^{\infty} \overline{\varphi_n(y;x)}\varphi_m(y;x)c_m(x,t) = &\sum_{m=0}^{\infty} \overline{\varphi_n(y;x)}L_x\varphi_m(y;x)c_m(x,t) \\ &- \alpha \sum_{m=0}^{\infty} \mu_m(x)\overline{\varphi_n(y;x)}\varphi_m(y;x)c_m(x,t)\end{aligned} \tag{6.46}$$

and integrate the resulting expressions over the variable y,

$$\begin{aligned}\int dy \frac{\partial}{\partial t}\sum_{m=0}^{\infty} c_m(x,t)\overline{\varphi_n(y;x)}\varphi_m(y;x) = &\int dy \sum_{m=0}^{\infty} \overline{\varphi_n(y;x)}L_x\ c_m(x,t)\varphi_m(y;x) \\ &- \alpha \int dy \sum_{m=0}^{\infty} \overline{\varphi_n(y;x)}c_m(x,t)\mu_m(x)\varphi_m(y;x).\end{aligned} \tag{6.47}$$

We again assume that we can interchange the limits and the integrals. Then the expression can be rearranged to

$$\begin{aligned}\frac{\partial}{\partial t}\sum_{m=0}^{\infty} c_m(x,t) \int dy\ \varphi_m(y;x)\overline{\varphi_n(y;x)} = &\sum_{m=0}^{\infty} \int dy\ \overline{\varphi_n(y;x)}L_x\varphi_m(y;x)c_m(x,t) \\ &- \alpha \sum_{m=0}^{\infty} c_m(x,t)\mu_m(x) \int dy\ \varphi_m(y;x)\overline{\varphi_n(y;x)}.\end{aligned} \tag{6.48}$$

By invoking the orthonormality relation, we find that

$$\frac{\partial}{\partial t}\sum_{m=0}^{\infty} c_m(x,t)\delta_{nm} = \sum_{m=0}^{\infty}\int dy\ \overline{\varphi_n(y;x)}L_x\varphi_m(y;x)c_m(x,t) - \alpha\sum_{m=0}^{\infty} c_m(x,t)\mu_m(x)\delta_{nm}. \tag{6.49}$$

Applying the Kronecker-deltas gives

$$\frac{\partial}{\partial t}c_n(x,t) = \sum_{m=0}^{\infty} L_{n,m}c_m(x,t) - \alpha\mu_n(x)c_n(x,t), \tag{6.50}$$

where we introduced the operators

$$L_{n,m} = \int dy\ \overline{\varphi_n(y,x)}L_x\varphi_m(y;x). \tag{6.51}$$

We rearrange terms to

$$\frac{\partial}{\partial t}c_n(x,t) + \alpha\mu_n(x)c_n(x,t) = \sum_{m=0}^{\infty} L_{n,m}c_m(x,t), \tag{6.52}$$

and factorize the coefficient $c_n(x,t)$,

$$\left[\frac{\partial}{\partial t} + \alpha\mu_n(x)\right]c_n(x,t) = \sum_{m=0}^{\infty} L_{n,m}c_m(x,t). \tag{6.53}$$

From this the stationary solution follows by inserting the eigenvalue $\mu_0 = 0$,

$$\left[\frac{\partial}{\partial t} + \alpha\mu_0(x)\right]c_0(x,t) = \dot{c_0}(x,t) = \sum_{m=0}^{\infty} L_{0,m}c_m(x,t). \tag{6.54}$$

Now we introduce an approximation. Since we are interested in large time scales compared to $(\alpha\mu_1)^{-1}$, we neglect the $\dot{c_n}(x,t)$ terms for $n \geq 1$, such that

$$\left[\frac{\partial}{\partial t} + \alpha\mu_n(x)\right]c_n(x,t) \approx \alpha\mu_n(x)c_n(x,t) = \sum_{m=0}^{\infty} L_{n,m}c_m(x,t). \tag{6.55}$$

We divide by $\alpha\mu_n(x)$ and find that

$$c_n(x,t) = (\alpha\mu_n(x))^{-1}\sum_{m=0}^{\infty} L_{n,m}c_m(x,t). \tag{6.56}$$

In order for this operation to be well-defined, we require $\mu_n(x) \neq 0$. This implies the choice $n > 0$. Since the expression is recursive, all $c_m(x,t)$ with $m > 0$ are of order α^{-1}. We thus have

$$c_n(x,t) = (\alpha\mu_n(x))^{-1}L_{n,0}c_0 + (\alpha\mu_n(x))^{-1}\mathcal{O}(\alpha^{-1}) = (\alpha\mu_n(x))^{-1}L_{n,0}c_0 + \mathcal{O}(\alpha^{-2}). \tag{6.57}$$

Neglecting terms of order $\mathcal{O}(\alpha^{-2})$, one obtains

$$c_n(x,t) = (\alpha\mu_n(x))^{-1} L_{n,0} c_0 \tag{6.58}$$

for $n > 0$. We rewrite Eq. (6.54) for the steady state,

$$\dot{c_0}(x,t) = \sum_{m=0}^{\infty} L_{0,m} c_m = L_{0,0} c_0 + \sum_{m=1}^{\infty} L_{0,m} c_m. \tag{6.59}$$

Inserting Eq. (6.58) into this expression gives

$$\begin{aligned} \dot{c_0}(x,t) &= L_{0,0} c_0 + \sum_{m=1}^{\infty} L_{0,m} (\alpha\mu_m(x))^{-1} L_{m,0} c_0 \\ &= L_{0,0} c_0 + \alpha^{-1} \sum_{m=1}^{\infty} L_{0,m} (\mu_m(x))^{-1} L_{m,0} c_0. \end{aligned} \tag{6.60}$$

So, ignoring terms of order (α^{-1}), we have

$$\dot{c_0}(x,t) = L_{0,0} c_0. \tag{6.61}$$

From Eq. (6.41) we know that the $c_0(x,t) = p(x,t)$. We combine this with our derived expression for $\dot{c_0}(x,t)$ and find

$$\frac{\partial p(x,t)}{\partial t} = L_{0,0} p(x,t). \tag{6.62}$$

We now want to understand the form of the operator $L_{0,0}$. From Eq. (6.51) we have the definition that

$$L_{0,0} = \int dy \; \overline{\varphi_0(y;x)} L_x \varphi_0(y;x). \tag{6.63}$$

With $\overline{\varphi_0(y;x)} = 1$ this expression simplifies to

$$L_{0,0} = \int dy \; L_x \varphi_0(y;x). \tag{6.64}$$

We know insert the definition of the Fokker-Planck operator, see Eq. (6.24), and arrive at

$$L_{0,0} = \int dy \; \frac{\partial}{\partial x} A_1(x,y) \varphi_0(y;x) + \int dy \; \frac{\partial^2}{\partial x \partial x} B_{11}(x,y) \varphi_0(y;x). \tag{6.65}$$

We restate this expression by

$$L_{0,0} = -\frac{\partial}{\partial x} \tilde{A}_1(x) + \frac{\partial^2}{\partial x^2} \tilde{D}_{11}(x), \tag{6.66}$$

where we introduced a modified drift term

$$\tilde{A}_1(x) = \langle A_1(x,y) \rangle_y \equiv \int A_1(x,y) \varphi_0(y,x) dy \tag{6.67}$$

and a modified diffusion term

$$\tilde{D}_{11}(x) = \langle B_{11}(x,y) \rangle_y \equiv \int B_{11}(x,y) \varphi_0(y,x) dy. \tag{6.68}$$

We additionally introduced the notation

$$\langle \cdot \rangle_y \equiv \int \cdot \varphi_0(y,x) dy. \tag{6.69}$$

The time evolution for the probability density $p(x,t)$ is thus given by the Fokker-Planck equation

$$\begin{aligned} \frac{\partial p(x,t)}{\partial t} &= -\frac{\partial}{\partial x}\tilde{A}_1(x)p(x,t) + \frac{\partial^2}{\partial x^2}\tilde{B}_{11}(x)p(x,t) \\ &= \frac{\partial}{\partial x}\left[-\tilde{A}_1(x)p(x,t) + \frac{\partial}{\partial x}\tilde{B}_{11}(x)p(x,t)\right] \end{aligned} \tag{6.70}$$

In particular for the steady state distribution, we have

$$\frac{\partial p_{ss}(x,t)}{\partial t} = \frac{\partial}{\partial x}\left[-\tilde{A}_1(x)p_{ss}(x,t) + \frac{\partial}{\partial x}\tilde{B}_{11}(x)p_{ss}(x,t)\right] = 0. \tag{6.71}$$

Inserting the coefficients of our loss model, we derive

$$\frac{\partial p_{ss}(x,t)}{\partial t} = \frac{\partial}{\partial x}\left[\varepsilon\lambda_0 x + \frac{1}{2}\lambda'(x)\langle y^2\rangle_y + \frac{d}{dx}\langle B_{11}(x,y)\rangle_y\right] = 0. \tag{6.72}$$

Now to the conditional probability distribution $p_{ss}(y|x)$. Since x is a fixed parameter, the steady state distribution $\varphi_0(y;x)$ should be viewed as a conditional probability. We follow [21] and identify $\varphi_0(y;x) = p_{ss}(y|x)$. The Fokker-Planck equation for the steady state is then given by

$$\begin{aligned} \frac{\partial}{\partial t}p_{ss}(y|x) &= L_x p_{ss}(y|x) + \alpha\mathcal{L}_y p_{ss}(y|x) \\ &= L_x p_{ss}(y|x) + \alpha\mu_0(x)p_{ss}(y|x). \end{aligned} \tag{6.73}$$

Since the eigenvalue is $\mu_0(x) = 0$, this expression simplifies to

$$\frac{\partial}{\partial t}p_{ss}(y|x). = L_x p_{ss}(y|x) \tag{6.74}$$

By inserting the characteristic of the steady state that $\frac{\partial}{\partial t}p_{ss}(y|x) = 0$, we conclude that $L_x p_{ss}(y|x) = 0$. Additionally, we note that $\alpha\mathcal{L}_y = L_y$. Overall, we can write

$$\frac{\partial}{\partial t}p_{ss}(y|x) = L_y p_{ss}(y|x) = 0. \tag{6.75}$$

Inserting the definition of the Fokker-Planck operator, Eq. (6.24), we find

$$\frac{\partial}{\partial t}p_{ss}(y|x) = \frac{\partial}{\partial y}\left[A_2(x,y) + \frac{\partial}{\partial y}B_{22}(x,y)\right]p_{ss}(y|x) = 0. \tag{6.76}$$

We now insert the drift and diffusion coefficients from our model, such that

$$\frac{\partial}{\partial t}p_{ss}(y|x) = \frac{\partial}{\partial y}\left[\lambda(x)y + \frac{\partial}{\partial y}\left(\lambda'(x)^2y^2 + \lambda(x)^2\right)\right]p_{ss}(y|x) = 0. \tag{6.77}$$

6.2.2 Derivation of the effective loss function

We make an approximation for the diffusion matrix by fixing its entries at the limit $x \to 0$. In our loss model, taking the limit $x \to 0$ produces the following term for the eigenvalue expansion, see Eq. (4.54),

$$\lambda(x) = \sum_{n=0}^{\infty} \lambda_n x^{2n} = \lambda_0 + \sum_{n=1}^{\infty} \lambda_n x^{2n} \overset{(x\to 0)}{\to} \lambda_0. \tag{6.78}$$

The first derivative is then given by

$$\lambda'(x) = \sum_{n=0}^{\infty} 2n\lambda_n x^{2n-1} \overset{(x\to 0)}{\to} 0, \tag{6.79}$$

and the second derivative is given by

$$\lambda''(x) = \sum_{n=0}^{\infty} 2n(2n-1)\lambda_n x^{2n-2} = 2\lambda_1 + \sum_{n=2}^{\infty} 2n(2n-1)\lambda_n x^{2n-2} \overset{(x\to 0)}{\to} 2\lambda_1. \tag{6.80}$$

For the diffusion matrix $\mathbf{B}(x=0,y)$ one obtains

$$\begin{aligned}\mathbf{B} &= \Delta_S \begin{bmatrix} \left(\lambda_0\varepsilon + \frac{1}{2}\lambda''(x)y^2\right)^2 + \lambda'(x)^2y^2 & \lambda'(x)y\left(\lambda_0\varepsilon + \frac{1}{2}\lambda''(x)y^2\right) + \lambda(x)\lambda'(x)y \\ \lambda'(x)y\left(\lambda_0\varepsilon + \frac{1}{2}\lambda''(x)y^2\right) + \lambda(x)\lambda'(x)y & \lambda'(x)^2y^2 + \lambda(x)^2 \end{bmatrix} \\ &\overset{(x\to 0)}{\to} \Delta_S \begin{bmatrix} \left(\varepsilon\lambda_0 + \lambda_1 y^2\right)^2 & 0 \\ 0 & \lambda_0^2 \end{bmatrix}\end{aligned} \tag{6.81}$$

with elements $B_{11} = \Delta_S\left(\varepsilon\lambda_0 + \lambda_1 y^2\right)^2$ and $B_{22} = \Delta_S\lambda_0^2$. Specifically the coefficient $B_{22} = \lambda_0^2$ is constant under this approximation. Let us now investigate the steady state distribution $p_{ss}(y|x)$. Its time evolution is given by the Fokker-Planck equation

$$\frac{\partial}{\partial t}p_{ss}(y|x) = \frac{\partial}{\partial y}\left[\lambda(x)y + \frac{\partial}{\partial y}B_{22}\right]p_{ss}(y|x). \tag{6.82}$$

We let the differential operators act on the probability distribution and find

$$\frac{\partial}{\partial t}p_{ss}(y|x) = \frac{\partial}{\partial y}\left[\lambda(x)y\ p_{ss}(y|x)\right] + \frac{\partial^2}{\partial y^2}\left[B_{22}p_{ss}(y|x)\right]. \tag{6.83}$$

Since B_{22} is constant and $\lambda(x)$ is independent of the variable y, they can both be placed in front of the operators, such that

$$\frac{\partial}{\partial t}p_{ss}(y|x) = \lambda(x)\frac{\partial}{\partial y}\left[y\ p_{ss}(y|x)\right] + B_{22}\frac{\partial^2}{\partial y^2}p_{ss}(y|x). \tag{6.84}$$

From the product rule it follows that

$$\frac{\partial}{\partial t}p_{ss}(y|x) = \lambda(x)p_{ss}(y|x) + \lambda(x)y\frac{\partial}{\partial y}p_{ss}(y|x) + B_{22}\frac{\partial^2}{\partial y^2}p_{ss}(y|x). \tag{6.85}$$

This is a partial differential equation of second degree. Reference [21] propose

$$p_{ss}(y|x) = \sqrt{\frac{\lambda(x)}{2\pi B_{22}}} \exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right] \tag{6.86}$$

as a solution. To see that this is indeed a solution, we first compute its first derivative with respect to the variable y. We find

$$\frac{\partial}{\partial y}p_{ss}(y|x) = \frac{\partial}{\partial y}\left(\sqrt{\frac{\lambda(x)}{2\pi B_{22}}} \exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right]\right). \tag{6.87}$$

The term $\sqrt{\frac{\lambda(x)}{2\pi B_{22}}}$ is constant, such that the expression can be rearranged into

$$\frac{\partial}{\partial y}p_{ss}(y|x) = \sqrt{\frac{\lambda(x)}{2\pi B_{22}}}\frac{\partial}{\partial y} \exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right]. \tag{6.88}$$

Differentiating the exponential function gives, by the chain rule,

$$\frac{\partial}{\partial y}p_{ss}(y|x) = \sqrt{\frac{\lambda(x)}{2\pi B_{22}}} \exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right]\frac{\partial}{\partial y}\left(-\frac{\lambda(x)}{2B_{22}}y^2\right). \tag{6.89}$$

And finally we arrive at the expression

$$\frac{\partial}{\partial y}p_{ss}(y|x) = -\frac{\lambda(x)y}{B_{22}}\sqrt{\frac{\lambda(x)}{2\pi B_{22}}} \exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right]. \tag{6.90}$$

Comparing this result to the proposed solution, Eq. (6.86), we find that

$$\frac{\partial}{\partial y}p_{ss}(y|x) = -\frac{\lambda(x)y}{B_{22}}p_{ss}. \tag{6.91}$$

Next, we compute the second derivative

$$\frac{\partial^2}{\partial y^2}p_{ss}(y|x) = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial y}p_{ss}(y|x)\right) = \frac{\partial}{\partial y}\left(-\frac{\lambda(x)y}{B_{22}}\sqrt{\frac{\lambda(x)}{2\pi B_{22}}} \exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right]\right). \tag{6.92}$$

With the product rule this gives

$$\begin{aligned}\frac{\partial^2}{\partial y^2}p_{ss}(y|x) = &-\sqrt{\frac{\lambda(x)}{2\pi B_{22}}}\frac{\lambda(x)y}{B_{22}}\frac{\partial}{\partial y}\left(\exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right]\right)\\ &-\sqrt{\frac{\lambda(x)}{2\pi B_{22}}} \exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right]\frac{\partial}{\partial y}\left(\frac{\lambda(x)y}{B_{22}}\right).\end{aligned} \tag{6.93}$$

The first term behaves analogously to the computations above and we find

$$\left(\frac{\lambda(x)y}{B_{22}}\right)^2\sqrt{\frac{\lambda(x)}{2\pi B_{22}}} \exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right]. \tag{6.94}$$

Here we recognize that

$$\left(\frac{\lambda(x)y}{B_{22}}\right)^2\sqrt{\frac{\lambda(x)}{2\pi B_{22}}}\exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right]=-\frac{\lambda(x)y}{B_{22}}\frac{\partial}{\partial y}p_{ss}(y|x). \tag{6.95}$$

For the second term we simply need the result $\frac{\partial}{\partial y}\left(\frac{\lambda(x)y}{B_{22}}\right)=\frac{\lambda(x)}{B_{22}}$. The second term is then given by

$$-\frac{\lambda(x)}{B_{22}}\sqrt{\frac{\lambda(x)}{2\pi B_{22}}}\exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right]. \tag{6.96}$$

Comparing this expression with the first differential, we observe that

$$-\frac{\lambda(x)}{B_{22}}\sqrt{\frac{\lambda(x)}{2\pi B_{22}}}\exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right]=\frac{1}{y}\frac{\partial}{\partial y}p_{ss}(y|x). \tag{6.97}$$

Combining both terms we now have the result for the second derivative, that is

$$\frac{\partial^2}{\partial y^2}p_{ss}(y|x)=-\frac{\lambda(x)y}{B_{22}}\frac{\partial}{\partial y}p_{ss}(y|x)+\frac{1}{y}\frac{\partial}{\partial y}p_{ss}(y|x). \tag{6.98}$$

Rearranging terms, we have

$$0=B_{22}\frac{\partial^2}{\partial y^2}p_{ss}(y|x)+\lambda(x)y\frac{\partial}{\partial y}p_{ss}(y|x).-B_{22}\frac{1}{y}\frac{\partial}{\partial y}p_{ss}(y|x) \tag{6.99}$$

We now insert the result for the first derivative, Eq. (6.91), into the expression and arrive at

$$\begin{aligned}0&=B_{22}\frac{\partial^2}{\partial y^2}p_{ss}(y|x)+\lambda_0 y\frac{\partial}{\partial y}p_{ss}(y|x)+B_{22}\frac{1}{y}\frac{\lambda(x)y}{B_{22}}p_{ss}(y|x)\\&=B_{22}\frac{\partial^2}{\partial y^2}p_{ss}(y|x)+\lambda(x)y\frac{\partial}{\partial y}p_{ss}(y|x)+\lambda(x)p_{ss}(y|x)\end{aligned} \tag{6.100}$$

in agreement with Eq. (6.85) and $\frac{\partial}{\partial t}p_{ss}(y|x)=0$. Given this solution for the conditional probability $p_{ss}(y|x)$, we can now derive the modified diffusion coefficient $\tilde{B}_{11}=\langle B_{11}\rangle_y$, see Eq. (6.68). We first state

$$\langle B_{11}\rangle_y=\int dy\ B_{11}(x,y)p_{ss}(y|x). \tag{6.101}$$

We insert the diffusion coefficient $B_{11}(x,y)=\Delta_S\left(\varepsilon\lambda_0+\lambda_1y^2\right)^2$ and obtain

$$\langle B_{11}\rangle_y=\int dy\ \Delta_S\left(\varepsilon\lambda_0+\lambda_1y^2\right)^2\sqrt{\frac{\lambda(x)}{2\pi B_{22}}}p_{ss}(y|x). \tag{6.102}$$

In a next step we insert our result for the conditional probability $p_{ss}(y|x)$, such that

$$\langle B_{11}\rangle_y=\int dy\ \Delta_S\left(\varepsilon\lambda_0+\lambda_1y^2\right)^2\sqrt{\frac{\lambda(x)}{2\pi B_{22}}}\exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right]. \tag{6.103}$$

We now complete the multiplication $\left(\varepsilon\lambda_0 + \lambda_1 y^2\right)^2 = \left(\varepsilon^2\lambda_0^2 + 2\varepsilon\lambda_0\lambda_1 y^2 + \lambda_1^2 y^4\right)$ and arrive at the expression

$$\langle B_{11}\rangle_y = \int dy\ \Delta_S \left(\varepsilon^2\lambda_0^2 + 2\varepsilon\lambda_0\lambda_1 y^2 + \lambda_1^2 y^4\right) \sqrt{\frac{\lambda(x)}{2\pi B_{22}}} \exp\left[-\frac{\lambda(x)}{2B_{22}} y^2\right]. \tag{6.104}$$

Further, we introduce the term $a \equiv \frac{\lambda(x)}{2B_{22}}$ that is independent of y. Then, the expression simplifies to

$$\langle B_{11}\rangle_y = \int dy\ \Delta_S \left(\varepsilon^2\lambda_0^2 + 2\varepsilon\lambda_0\lambda_1 y^2 + \lambda_1^2 y^4\right) \sqrt{\frac{a}{\pi}} e^{-ay^2}. \tag{6.105}$$

Expanding the bracket and invoking the linearity of the integral gives three terms

$$\begin{aligned}\langle B_{11}\rangle_y = \Delta_S\varepsilon^2\lambda_0^2\sqrt{\frac{a}{\pi}} \int dy\ e^{-ay^2} \\ + 2\Delta_S\varepsilon\lambda_0\lambda_1\sqrt{\frac{a}{\pi}} \int dy\ y^2\ e^{-ay^2} \\ + \Delta_S\lambda_1^2\sqrt{\frac{a}{\pi}} \int dy\ y^4 e^{-ay^2}.\end{aligned} \tag{6.106}$$

We will go through each term separately. We define $u = \sqrt{a}y$ with derivative $\frac{du}{dy} = \sqrt{a}$. In the first term we substitute u, such that

$$\begin{aligned}\Delta_S\varepsilon^2\lambda_0^2\sqrt{\frac{a}{\pi}} \int dy\ e^{-ay^2} &= \Delta_S\varepsilon^2\lambda_0^2\sqrt{\frac{a}{\pi}} \int du \frac{1}{\sqrt{a}}\ e^{-u^2} \\ &= \Delta_S\varepsilon^2\lambda_0^2\frac{1}{\sqrt{\pi}} \int du\ e^{-u^2}.\end{aligned} \tag{6.107}$$

With the standard Gauss integral $\int du\ e^{-u^2} = \sqrt{\pi}$, we find

$$\begin{aligned}\Delta_S\varepsilon^2\lambda_0^2\sqrt{\frac{a}{\pi}} \int dy\ e^{-ay^2} &= \Delta_S\varepsilon^2\lambda_0^2\frac{\sqrt{\pi}}{\sqrt{\pi}} \\ &= \Delta_S\varepsilon^2\lambda_0^2 \\ &= \varepsilon^2 B_{22},\end{aligned} \tag{6.108}$$

where in the last step we substituted $B_{22} = \Delta_S\lambda_0^2$. Now the second term. Because of the equality $-\frac{\partial}{\partial a}e^{-ay^2} = y^2 e^{-ay^2}$, the second term can be expressed by

$$2\Delta_S\varepsilon\lambda_0\lambda_1\sqrt{\frac{a}{\pi}} \int dy\ y^2\ e^{-ay^2} = -2\Delta_S\varepsilon\lambda_0\lambda_1 \int dy\ \frac{\partial}{\partial a} e^{-ay^2}. \tag{6.109}$$

The integral is in regard to the variable y, thus we can place the differential operator in front of the integral, such that

$$2\Delta_S\varepsilon\lambda_0\lambda_1\sqrt{\frac{a}{\pi}} \int dy\ y^2\ e^{-ay^2} = -2\Delta_S\varepsilon\lambda_0\lambda_1\sqrt{\frac{a}{\pi}}\frac{\partial}{\partial a} \int dy\ e^{-ay^2}. \tag{6.110}$$

We solve the Gauss integral analogously to the first term and thus have

$$2\Delta_S\varepsilon\lambda_0\lambda_1\sqrt{\frac{a}{\pi}} \int dy\ y^2\ e^{-ay^2} = -2\Delta_S\varepsilon\lambda_0\lambda_1\sqrt{\frac{a}{\pi}}\frac{\partial}{\partial a}\sqrt{\frac{\pi}{a}}. \tag{6.111}$$

Computing the derivative then gives

$$2\Delta_S\varepsilon\lambda_0\lambda_1\sqrt{\frac{a}{\pi}}\int dy\ y^2\ e^{-ay^2} = 2\Delta_S\varepsilon\lambda_0\lambda_1\sqrt{\frac{a}{\pi}}\sqrt{\frac{\pi}{a}}\cdot\frac{1}{a}\cdot\frac{1}{2}$$
$$= \Delta_S\varepsilon\lambda_0\lambda_1\cdot\frac{1}{a}. \tag{6.112}$$

The constant a was defined by $a \equiv \frac{\lambda(x)}{2B_{22}}$. Inserting this into the expression for the second term, we find

$$\Delta_S\varepsilon\lambda_0\lambda_1\sqrt{\frac{a}{\pi}}\int dy\ y^2\ e^{-ay^2} = 2\Delta_S\varepsilon\lambda_0\lambda_1\frac{B_{22}}{\lambda(x)}$$
$$= 2\Delta_S\varepsilon\lambda_0\lambda_1 B_{22}\lambda(x)^{-1}. \tag{6.113}$$

For the last term we make the rearrangement $y^4 = y^3\cdot y$, such that

$$\Delta_S\lambda_1^2\sqrt{\frac{a}{\pi}}\int dy\ y^4 e^{-ay^2} = \Delta_S\lambda_1^2\sqrt{\frac{a}{\pi}}\int dy\ y^3 y e^{-ay^2}. \tag{6.114}$$

Now we can perform partial integration and derive

$$\Delta_S\lambda_1^2\sqrt{\frac{a}{\pi}}\int dy\ y^4 e^{-ay^2} = \Delta_S\lambda_1^2\sqrt{\frac{a}{\pi}}\left(\left[y^3\left(\frac{-e^{-ay^2}}{2a}\right)\right]_{y=-\infty}^{y=\infty} - \int dy\ 3y^2\left(\frac{-e^{-ay^2}}{2a}\right)\right). \tag{6.115}$$

The boundary term vanishes and the expression simplifies to

$$\Delta_S\lambda_1^2\sqrt{\frac{a}{\pi}}\int dy\ y^4 e^{-ay^2} = -3\Delta_S\lambda_1^2\sqrt{\frac{a}{\pi}}\int dy\ y^2\left(\frac{-e^{-ay^2}}{2a}\right)$$
$$= \frac{3}{2a}\Delta_S\lambda_1^2\sqrt{\frac{a}{\pi}}\int dy\ y^2 e^{-ay^2}. \tag{6.116}$$

We can now proceed analogous to the second term. We first rewrite the expression as

$$\Delta_S\lambda_1^2\sqrt{\frac{a}{\pi}}\int dy\ y^4 e^{-ay^2} = -\frac{3}{2a}\Delta_S\lambda_1^2\sqrt{\frac{a}{\pi}}\frac{\partial}{\partial a}\int dy\ e^{-ay^2}. \tag{6.117}$$

The integral then evaluates to

$$\Delta_S\lambda_1^2\sqrt{\frac{a}{\pi}}\int dy\ y^4 e^{-ay^2} = -\frac{3}{2a}\Delta_S\lambda_1^2\sqrt{\frac{a}{\pi}}\frac{\partial}{\partial a}\sqrt{\frac{\pi}{a}} \tag{6.118}$$

and taking the derivative, we find

$$\Delta_S\lambda_1^2\sqrt{\frac{a}{\pi}}\int dy\ y^4 e^{-ay^2} = \frac{3}{2a}\Delta_S\lambda_1^2\sqrt{\frac{a}{\pi}}\sqrt{\frac{\pi}{a}}\cdot\frac{1}{a}\cdot\frac{1}{2}$$
$$= \frac{3}{4a^2}\Delta_S\lambda_1^2. \tag{6.119}$$

We now insert $a = \frac{\lambda(x)}{2B_{22}}$ and have the result

$$\begin{aligned}\Delta_S \lambda_1^2 \sqrt{\frac{a}{\pi}} \int dy\ y^4 e^{-ay^2} &= \frac{3}{4} \Delta_S \lambda_1^2 \frac{4B_{22}^2}{\lambda(x)^2} \\ &= 3\Delta_S \lambda_1^2 B_{22}^2 \lambda(x)^{-2}. \end{aligned} \tag{6.120}$$

Combining all terms, the modified diffusion coefficient is given by

$$\langle B_{11} \rangle_y = \varepsilon^2 B_{22} + 2\Delta_S \varepsilon \lambda_0 \lambda_1 B_{22} \lambda(x)^{-1} + 3\Delta_S \lambda_1^2 B_{22}^2 \lambda(x)^{-2}. \tag{6.121}$$

We now define the anisotropy

$$\kappa \equiv \frac{B_{22}}{\langle B_{11} \rangle_y}. \tag{6.122}$$

We insert the result for the diffusion coefficient $\langle B_{11} \rangle_y$ and find

$$\kappa = \frac{B_{22}}{\varepsilon^2 B_{22} + 2\Delta_S \varepsilon \lambda_0 \lambda_1 B_{22} \lambda(x)^{-1} + 3\Delta_S \lambda_1^2 B_{22}^2 \lambda(x)^{-2}}. \tag{6.123}$$

Factorizing B_{22}, this simplifies to

$$\kappa = \frac{1}{\varepsilon^2 + 2\Delta_S \varepsilon \lambda_0 \lambda_1 \lambda(x)^{-1} + 3\Delta_S \lambda_1^2 B_{22} \lambda(x)^{-2}}. \tag{6.124}$$

We now insert the diffusion coefficient $B_{22} = \Delta_S \lambda_0^2$ and derive

$$\kappa = \frac{1}{\varepsilon^2 + 2\Delta_S \varepsilon \lambda_0 \lambda_1 \lambda(x)^{-1} + 3\Delta_S^2 \lambda_0^2 \lambda_1^2 \lambda(x)^{-2}}. \tag{6.125}$$

We again make an approximation, proposed by Ref. [21], and fix the anisotropy in the limit $x \to 0$. Then all occurrences of $\lambda(x)$ are replaced by λ_0, such that

$$\kappa = \frac{1}{\varepsilon^2 + 2\Delta_S \varepsilon \lambda_1 + 3\Delta_S^2 \lambda_1^2}. \tag{6.126}$$

Next, we want to calculate the second moment of the variable y, that is

$$\langle y^2 \rangle_y = \int dy\ y^2 p_{ss}(y|x). \tag{6.127}$$

Inserting the solution for the steady state $p_{ss}(y|x)$, we find

$$\langle y^2 \rangle_y = \int dy\ y^2 \sqrt{\frac{\lambda(x)}{2\pi B_{22}}} \exp\left[-\frac{\lambda(x)}{2B_{22}} y^2 \right]. \tag{6.128}$$

Here we again make the substitution $a = \frac{\lambda(x)}{2B_{22}}$, such that the expression is given by

$$\langle y^2 \rangle_y = \sqrt{\frac{a}{\pi}} \int dy\ y^2 e^{-ay^2}. \tag{6.129}$$

This is solved analogously to the occasion before. We first rewrite this as

$$\langle y^2 \rangle_y = -\sqrt{\frac{a}{\pi}} \frac{\partial}{\partial a} \int dy\ e^{-ay^2}. \tag{6.130}$$

This expression then evaluates to

$$\begin{aligned}\langle y^2\rangle_y &= -\sqrt{\frac{a}{\pi}}\frac{\partial}{\partial a}\sqrt{\frac{\pi}{a}} \\ &= \frac{1}{2a}.\end{aligned} \tag{6.131}$$

Now we re-substituting $a = \frac{\lambda(x)}{2B_{22}}$ and obtain

$$\langle y^2\rangle_y = \frac{B_{22}}{\lambda(x)}. \tag{6.132}$$

The drift term $A_1(x, y)$ is given by

$$A_1(x, y) = \lambda_0\varepsilon x + \frac{1}{2}\lambda'(x)y^2 \tag{6.133}$$

and the modified drift term $\tilde{A}_1(x)$ is given by

$$\tilde{A}_1(x) = \langle A_1(x, y)\rangle_y \equiv \int A_1(x, y)p_{ss}(y|x)dy. \tag{6.134}$$

We insert the drift term $\tilde{A}_1(x)$ into the modified one and find

$$\begin{aligned}\langle A_1(x, y)\rangle_y &= \int \left[\lambda_0\varepsilon x + \frac{1}{2}\lambda'(x)y^2\right] p_{ss}(y|x)dy \\ &= \int dy\ \lambda_0\varepsilon x p_{ss}(y|x) + \int dy\ \frac{1}{2}\lambda'(x)y^2 p_{ss}(y|x).\end{aligned} \tag{6.135}$$

We investigate each term. In the first term we insert the solution of $p_{ss}(y|x)$, such that

$$\lambda_0\varepsilon x \int dy\ p_{ss}(y|x) = \lambda_0\varepsilon x \int dy\ \sqrt{\frac{\lambda(x)}{2\pi B_{22}}}\exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right]. \tag{6.136}$$

Again, we make the substitution $a = \frac{\lambda(x)}{2B_{22}}$. We find

$$\lambda_0\varepsilon x \int dy\ p_{ss}(y|x) = \lambda_0\varepsilon x\sqrt{\frac{a}{\pi}}\int dy\ e^{-ay^2}. \tag{6.137}$$

The solution method is analogous to the previous cases. As a result we have

$$\begin{aligned}\lambda_0\varepsilon x \int dy\ p_{ss}(y|x) &= \lambda_0\varepsilon x\sqrt{\frac{a}{\pi}}\sqrt{\frac{\pi}{a}} \\ &= \lambda_0\varepsilon x.\end{aligned} \tag{6.138}$$

We rearrange the second term, such that

$$\begin{aligned}\int dy\ \frac{1}{2}\lambda'(x)y^2 p_{ss}(y|x) &= \frac{1}{2}\lambda'(x)\int dy\ y^2 p_{ss}(y|x) \\ &= \frac{1}{2}\lambda'(x)\langle y^2\rangle_y.\end{aligned} \tag{6.139}$$

Inserting our result for $\langle y^2 \rangle_y$, we find

$$\int dy \; \frac{1}{2}\lambda'(x) y^2 p_{ss}(y|x) = \frac{1}{2}\lambda'(x)\frac{B_{22}}{\lambda(x)}. \tag{6.140}$$

The modified drift coefficient is then given by

$$\langle A_1(x,y) \rangle_y = \lambda_0 \varepsilon x + \frac{1}{2}\lambda'(x)\frac{B_{22}}{\lambda(x)}. \tag{6.141}$$

We remind that the modified diffusion coefficient is $\langle B_{11} \rangle_y = \frac{B_{22}}{\kappa}$. And now we can state the Fokker-Planck equation for the time evolution of $p(x)_{ss}$. It is given by

$$\frac{\partial}{\partial t} p_{ss}(x) = \frac{d}{dx}\left[\langle A_1(x,y) \rangle_y + \frac{d}{dx}\langle B_{11}(x,y) \rangle_y \right] p_{ss}(x). \tag{6.142}$$

Inserting the coefficients, we derive

$$\begin{aligned} \frac{\partial}{\partial t} p_{ss}(x) &= \frac{d}{dx}\left[\varepsilon\lambda_0 x + \frac{1}{2}\lambda'(x)\frac{B_{22}}{\lambda(x)} + \frac{d}{dx}\frac{B_{22}}{\kappa} \right] p_{ss}(x) \\ &= \varepsilon\lambda_0 \frac{d}{dx} x p_{ss}(x) + \frac{B_{22}}{2}\frac{d}{dx}\frac{\lambda'(x)}{\lambda(x)} p_{ss}(x) + \frac{B_{22}}{\kappa}\frac{d^2}{dx^2} p_{ss}(x). \end{aligned} \tag{6.143}$$

We compute the derivatives for each term separately. First, we have

$$\begin{aligned} \varepsilon\lambda_0 \frac{d}{dx} x p_{ss}(x) &= \varepsilon\lambda_0 p_{ss}(x) + \varepsilon\lambda_0 x \frac{d}{dx} p_{ss}(x) \\ &= \varepsilon\lambda_0 \left[p_{ss}(x) + x\frac{d}{dx} p_{ss}(x) \right]. \end{aligned} \tag{6.144}$$

For the second term, we have

$$\begin{aligned} \frac{B_{22}}{2}\frac{d}{dx}\frac{\lambda'(x)}{\lambda(x)} p_{ss}(x) &= \frac{B_{22}}{2}\left[p_{ss}(x)\frac{d}{dx}\frac{\lambda'(x)}{\lambda(x)} + \frac{\lambda'(x)}{\lambda(x)}\frac{d}{dx} p_{ss}(x) \right] \\ &= \frac{B_{22}}{2}\left[p_{ss}(x)\frac{\lambda''(x)}{\lambda(x)} - p_{ss}(x)\frac{\lambda'(x)^2}{\lambda(x)^2} + \frac{\lambda'(x)}{\lambda(x)}\frac{d}{dx} p_{ss}(x) \right] \end{aligned} \tag{6.145}$$

and the last term stays

$$\frac{\partial}{\partial t} p_{ss}(x) = \frac{B_{22}}{\kappa}\frac{d^2}{dx^2} p_{ss}(x). \tag{6.146}$$

Overall, we find

$$\begin{aligned} \frac{\partial}{\partial t} p_{ss}(x) = \varepsilon\lambda_0 &\left[p_{ss}(x) + x\frac{d}{dx} p_{ss}(x) \right] \\ &+ \frac{B_{22}}{2}\left[p_{ss}(x)\frac{\lambda''(x)}{\lambda(x)} - p_{ss}(x)\frac{\lambda'(x)^2}{\lambda(x)^2} + \frac{\lambda'(x)}{\lambda(x)}\frac{d}{dx} p_{ss}(x) \right] \\ &+ \frac{B_{22}}{\kappa}\frac{d^2}{dx^2} p_{ss}(x). \end{aligned} \tag{6.147}$$

Rearranging terms and noting that $\frac{\partial}{\partial t} p_{ss}(x) = 0$, we obtain

$$\begin{aligned} 0 = &\left[\varepsilon\lambda_0 + \frac{B_{22}}{2}\frac{\lambda''(x)}{\lambda(x)} + \frac{B_{22}}{2}\frac{\lambda'(x)^2}{\lambda(x)^2}\right] p_{ss}(x) \\ &+ \left[\varepsilon\lambda_0 x + \frac{\lambda'(x)}{\lambda(x)}\right]\frac{d}{dx}p_{ss}(x) \\ &+ \frac{B_{22}}{\kappa}\frac{d^2}{dx^2}p_{ss}(x). \end{aligned} \tag{6.148}$$

This ordinary differential equation of second degree is solved by

$$p_{ss}(x) = \frac{1}{Z_{ss}} \exp\left[-\frac{\kappa\varepsilon\lambda_0}{2B_{22}}x^2\right] \lambda(x)^{-\frac{\kappa}{2}} \tag{6.149}$$

with normalization factor

$$Z_{ss} = \int dx\ p_{ss}(x). \tag{6.150}$$

This solution was proposed by [21]. To see that it actually is a solution, we first compute the first derivative

$$\begin{aligned} \frac{d}{dx}p_{ss}(x) &= \frac{d}{dx}\left(\frac{1}{Z_{ss}} \exp\left[-\frac{\kappa\varepsilon\lambda_0}{2B_{22}}x^2\right] \lambda(x)^{-\frac{\kappa}{2}}\right) \\ &= \frac{1}{Z_{ss}}\lambda(x)^{-\frac{\kappa}{2}}\frac{d}{dx}\exp\left[-\frac{\kappa\varepsilon\lambda_0}{2B_{22}}x^2\right] + \frac{1}{Z_{ss}}\exp\left[-\frac{\kappa\varepsilon\lambda_0}{2B_{22}}x^2\right]\frac{d}{dx}\lambda(x)^{-\frac{\kappa}{2}}. \end{aligned} \tag{6.151}$$

For the first term we have

$$\begin{aligned} \frac{1}{Z_{ss}}\lambda(x)^{-\frac{\kappa}{2}}\frac{d}{dx}\exp\left[-\frac{\kappa\varepsilon\lambda_0}{2B_{22}}x^2\right] &= -\frac{1}{Z_{ss}}\lambda(x)^{-\frac{\kappa}{2}}\exp\left[-\frac{\kappa\varepsilon\lambda_0}{2B_{22}}x^2\right]\frac{d}{dx}\frac{\kappa\varepsilon\lambda_0}{2B_{22}}x^2 \\ &= -\frac{1}{Z_{ss}}\frac{\kappa\varepsilon\lambda_0 x}{B_{22}}\lambda(x)^{-\frac{\kappa}{2}}\exp\left[-\frac{\kappa\varepsilon\lambda_0}{2B_{22}}x^2\right]. \end{aligned} \tag{6.152}$$

We recognize that this term is related to the proposed solution, by

$$-\frac{1}{Z_{ss}}\frac{\kappa\varepsilon\lambda_0 x}{B_{22}}\lambda(x)^{-\frac{\kappa}{2}}\exp\left[-\frac{\kappa\varepsilon\lambda_0}{2B_{22}}x^2\right] = -\frac{\kappa\varepsilon\lambda_0 x}{B_{22}}p_{ss}(x). \tag{6.153}$$

The second term evaluates to

$$\frac{1}{Z_{ss}}\exp\left[-\frac{\kappa\varepsilon\lambda_0}{2B_{22}}x^2\right]\frac{d}{dx}\lambda(x)^{-\frac{\kappa}{2}} = -\frac{1}{Z_{ss}}\frac{\kappa}{2}\lambda(x)^{-\frac{\kappa}{2}}\lambda(x)^{-1}\lambda'(x)\exp\left[-\frac{\kappa\varepsilon\lambda_0}{2B_{22}}x^2\right]. \tag{6.154}$$

Again, we can again express this result in terms of $p_{ss}(x)$. We find

$$-\frac{1}{Z_{ss}}\frac{\kappa}{2}\lambda(x)^{-\frac{\kappa}{2}}\lambda(x)^{-1}\lambda'(x)\exp\left[-\frac{\kappa\varepsilon\lambda_0}{2B_{22}}x^2\right] = -\frac{\kappa}{2}\lambda(x)^{-1}\lambda'(x)p_{ss}(x). \tag{6.155}$$

Merging both results, the first derivative is given by

$$\begin{aligned} \frac{d}{dx}p_{ss}(x) &= -\frac{\kappa\varepsilon\lambda_0 x}{B_{22}}p_{ss}(x) - \frac{\kappa}{2}\lambda(x)^{-1}\lambda'(x)p_{ss}(x) \\ &= -\kappa\left(\frac{\varepsilon\lambda_0 x}{B_{22}} + \frac{1}{2}\frac{\lambda'(x)}{\lambda(x)}\right)p_{ss}(x). \end{aligned} \tag{6.156}$$

Taking the second derivative, we obtain

$$\begin{aligned}\frac{d^2}{dx^2}p_{ss}(x) &= -\kappa\frac{d}{dx}\left(\frac{\varepsilon\lambda_0 x}{B_{22}} - \frac{1}{2}\frac{\lambda'(x)}{\lambda(x)}\right)p_{ss}(x) \\ &= -\kappa\frac{\varepsilon\lambda_0}{B_{22}}\frac{d}{dx}x\ p_{ss}(x) - \kappa\frac{1}{2}\frac{d}{dx}\frac{\lambda'(x)}{\lambda(x)}p_{ss}(x). \end{aligned} \tag{6.157}$$

We multiply both sides by $\frac{B_{22}}{\kappa}$ and find

$$\frac{B_{22}}{\kappa}\frac{d^2}{dx^2}p_{ss}(x) = -\varepsilon\lambda_0\frac{d}{dx}x\ p_{ss}(x) - \frac{B_{22}}{2}\frac{d}{dx}\frac{\lambda'(x)}{\lambda(x)}p_{ss}(x). \tag{6.158}$$

This can now be rearranged into the form

$$\varepsilon\lambda_0\frac{d}{dx}x\ p_{ss}(x) + \frac{B_{22}}{2}\frac{d}{dx}\frac{\lambda'(x)}{\lambda(x)}p_{ss}(x) + \frac{B_{22}}{\kappa}\frac{d^2}{dx^2}p_{ss}(x) = 0. \tag{6.159}$$

This result is in perfect agreement with the Fokker-Planck equation given by Eq. (6.144). Finally, we arrive at an expression for the two variable steady state distribution,

$$P_{ss}(x,y) = P_{ss}(x)P_{ss}(y|x). \tag{6.160}$$

We insert the results, Eq. (6.149) and Eq. (6.2.2), and find that the full expression is given by

$$P_{ss}(x,y) = \frac{1}{Z_{ss}}\exp\left[-\frac{\kappa\varepsilon\lambda_0}{2B_{22}}x^2\right]\lambda(x)^{-\frac{\kappa}{2}}\cdot\sqrt{\frac{\lambda(x)}{2\pi B_{22}}}\exp\left[-\frac{\lambda(x)}{2B_{22}}y^2\right]. \tag{6.161}$$

We combine the exponential terms, such that

$$P_{ss}(x,y) = \frac{1}{Z_{ss}}\lambda(x)^{-\frac{\kappa}{2}}\cdot\sqrt{\frac{\lambda(x)}{2\pi B_{22}}}\exp\left[-\frac{\lambda(x)+\kappa\varepsilon\lambda_0}{2B_{22}}y^2\right]. \tag{6.162}$$

We further note that $\sqrt{\lambda(x)} = \lambda(x)^{-1/2}$ and combine it with $\lambda(x)^{-\frac{\kappa}{2}}$. This gives

$$P_{ss}(x,y) = \frac{1}{Z_{ss}}\frac{1}{\sqrt{2\pi B_{22}}}\exp\left[-\frac{\lambda(x)+\kappa\varepsilon\lambda_0}{2B_{22}}y^2\right]\lambda(x)^{-\frac{\kappa-1}{2}}. \tag{6.163}$$

We abbreviate the expression by introducing the constant $Z'_{ss} \equiv Z_{ss}\sqrt{2\pi B_{22}}$. The final expression is then

$$P_{ss}(x,y) = \frac{1}{Z'_{ss}}\exp\left[-\frac{\kappa\varepsilon\lambda_0 x^2 + \lambda(x)y^2}{2B_{22}}\right]\lambda(x)^{-\frac{\kappa-1}{2}}. \tag{6.164}$$

We can now define the effective loss by

$$L_{eff}(x,y) \equiv -B_{22}\ln[p_{ss}(x,y)] \tag{6.165}$$

with effective temperature B_{22}, see Ref. [21]. We insert our approximative result for the steady state distribution $p_{ss}(x,y)$ and find

$$L_{eff}(x,y) = -B_{22}\ln\left[\frac{1}{Z'_{ss}}\exp\left(-\frac{\kappa\varepsilon\lambda_0 x^2 + \lambda(x)y^2}{2B_{22}}\right)\lambda(x)^{-\frac{\kappa-1}{2}}\right]. \tag{6.166}$$

The product within the logarithm can be equally stated as a sum of the logarithms of the factors. This then gives

$$L_{eff}(x,y) = -B_{22}\left(\ln\left[\frac{1}{Z'_{ss}}\right] + \ln\left[\exp\left(-\frac{\kappa\varepsilon\lambda_0 x^2 + \lambda(x)y^2}{2B_{22}}\right)\right] + \ln\left[\lambda(x)^{-\frac{\kappa-1}{2}}\right]\right). \tag{6.167}$$

Taking the logarithm of the exponential function, we obtain

$$L_{eff}(x,y) = -B_{22}\left(\ln\left[\frac{1}{Z'_{ss}}\right] + -\frac{\kappa\varepsilon\lambda_0 x^2 + \lambda(x)y^2}{2B_{22}} + \ln\left[\lambda(x)^{-\frac{\kappa-1}{2}}\right]\right). \tag{6.168}$$

Additionally, we have

$$\ln\left[\lambda(x)^{-\frac{\kappa-1}{2}}\right] = (\kappa-1)\ln\left[\lambda(x)^{-1/2}\right] \tag{6.169}$$

and

$$\ln\left[\frac{1}{Z'_{ss}}\right] = -\ln\left[Z'_{ss}\right]. \tag{6.170}$$

With this we transform the expression into

$$L_{eff}(x,y) = -B_{22}\left(-\ln\left[Z'_{ss}\right] - \frac{\kappa\varepsilon\lambda_0 x^2 + \lambda(x)y^2}{2B_{22}} + (\kappa-1)\ln\left[\lambda(x)^{-1/2}\right]\right). \tag{6.171}$$

We multiply out the bracket and find

$$L_{eff}(x,y) = B_{22}\ln\left[Z'_{ss}\right] + \frac{1}{2}\left(\kappa\varepsilon\lambda_0 x^2 + \lambda(x)y^2\right) - B_{22}\left(\kappa-1\right)\ln\left[\lambda(x)^{-1/2}\right]. \tag{6.172}$$

We now insert the flatness $F(x)$ that is defined by $1/\sqrt{\lambda(x)}$, see Ref. [21], and note that $(\kappa-1)$=-$(1-\kappa)$. The expression for the effective loss is thus given by

$$L_{eff}(x,y) = B_{22}\ln\left[Z'_{ss}\right] + \frac{1}{2}\left(\kappa\varepsilon\lambda_0 x^2 + \lambda(x)y^2\right) + B_{22}\left(1-\kappa\right)\ln\left[F(x)\right]. \tag{6.173}$$

We further neglect the constant term, such that the full expression is given by

$$L_{eff}(x,y) \approx \frac{1}{2}\left(\kappa\varepsilon\lambda_0 x^2 + \lambda(x)y^2\right) + B_{22}\left(1-\kappa\right)\ln\left[F(x)\right]. \tag{6.174}$$

We now want to investigate in which way the effective loss differs from the loss function given by $L = \lambda_0\varepsilon x^2 + \frac{1}{2}\lambda(x)y^2$. We write $L_{eff}(x,y) = L(x,y) + L_{SGD}(x)$ and derive the difference by computing

$$L_{SGD}(x) = L_{eff}(x,y) - L(x,y). \tag{6.175}$$

Inserting both terms, we find

$$L_{SGD}(x) = \frac{1}{2}\left(\kappa\varepsilon\lambda_0 x^2 + \lambda(x)y^2\right) - B_{22}(\kappa-1)\ln[F(x)] - \frac{1}{2}\lambda_0\varepsilon x^2 - \frac{1}{2}\lambda(x)y^2. \tag{6.176}$$

Now the terms with $\frac{1}{2}\lambda(x)y^2$ cancel, such that

$$L_{SGD}(x) = \frac{1}{2}\kappa\varepsilon\lambda_0 x^2 - B_{22}(\kappa - 1)\ln[F(x)] - \frac{1}{2}\lambda_0\varepsilon x^2. \tag{6.177}$$

We factorize the term $\varepsilon\lambda_0 x^2$ and obtain

$$L_{SGD}(x) = (\kappa - 1)\frac{1}{2}\varepsilon\lambda_0 x^2 - B_{22}(\kappa - 1)\ln[F(x)]. \tag{6.178}$$

Rewriting $(\kappa - 1) = -(1 - \kappa)$ transforms the expression into

$$L_{SGD}(x) = (\kappa - 1)\frac{1}{2}\varepsilon\lambda_0 x^2 + B_{22}(1 - \kappa)\ln[F(x)]. \tag{6.179}$$

In a last step we notice that for the loss function at $y = 0$ we have $L(x, y = 0) = \frac{1}{2}\varepsilon\lambda_0 x^2$ and arrive at the expression

$$L_{SGD}(x) = (\kappa - 1)\left[L(x, y = 0) + B_{22}\ln[F(x)]\right]. \tag{6.180}$$

This leads to our concluding result. The effective loss $L_{eff}(x, y)$ can be expressed by the sum of the loss $L(x, y)$ and an additional flatness dependent loss term $L_{SGD}(x)$. Overall, we find

$$\begin{aligned} L_{eff}(x, y) &= L(x, y) + L_{SGD}(x) \\ &= \lambda_0\varepsilon x^2 + \frac{1}{2}\lambda(x)y^2 + (\kappa - 1)\frac{1}{2}\varepsilon\lambda_0 x^2 + B_{22}(1 - \kappa)\ln[F(x)]. \end{aligned} \tag{6.181}$$

For a completely degenerate loss landscape, $\varepsilon = 0$, this simplifies to

$$\begin{aligned} L_{eff}(x, y) &= L(x, y) + L_{SGD}(x) \\ &= \frac{1}{2}\lambda(x)y^2 + B_{22}(1 - \kappa)\ln[F(x)]. \end{aligned} \tag{6.182}$$

We assume $\kappa > 1$. The function $\ln[F(x)]$ is monotonically increasing with the flatness $F(x)$. The values of $F(x)$ near a minimum should be in the interval $]0, 1]$, such that $\ln[F(x)] \leq 0$. It follows that in this interval the absolute value of $|\ln[F(x)]|$ is monotonically decreasing. Now, with anisotropy $\kappa > 1$, the SGD loss term $L_{SGD}(x) = B_{22}(1 - \kappa)\ln[F(x)]$ is positive. So, a larger flatness decreases the loss term L_{SGD} and in this way the effective loss L_{eff}. It follows that anisotropic noise, as observed for SGD, has a higher probability of converging towards flatter solutions than less anisotropic noise, e.g. GD. Thus, within the model, SGD favors flat loss minima.

Chapter 7

Conclusion

Following the work by ref. [21] we investigated the hypothesis that SGD favors flat loss minima. We started by conducting experiments and that formed the basis of our empirical assumptions. Especially important is the assumption of a valley of connected loss minima as well as the degeneracy of the Hessian spectrum. We constructed a model of SGD and performed numerical experiments for two interesting cases of loss landscapes. Based on this model and methods from stochastic dynamics we were able to derive an effective loss function for SGD that is influenced by a flatness dependent term. We recognized that for large anisotropy the effective loss decreases when the flatness increases. Minimizing the effective loss can therefor drive the system towards flatter loss minima than more isotropic algorithms. In this work we only derived the results for an approximation with constant anisotropy κ and diffusion coefficient B_{22}. Reference [21] additionally investigate a landscape dependent anisotropy κ and again demonstrate the emergence of a flatness dependent loss term. They also further discuss the validity of the assumptions relevant for the method of adiabatic elimination. It was however not the aim of this work to reproduce every result in the work by [21], but rather to connect insights from their work with a much broader introductory background. This ranged from presenting fundamentals of machine learning to developing the stochastic dynamics framework for diffusion processes. It's also important to point out that assumptions like the minimal valley might very well not be applicable in the vast majority of real and large machine learning settings. However, while our experiments were performed on a relatively small toy model, recent research, see ref. [66], suggests that the minimal valley view of the loss landscape does indeed hold for certain types of large language models based on the transformer architecture, see Ref. [14]. We further want to note that the link between flatness and generalization still awaits a sufficient theoretical explanation, although there is promising research, e.g. by [67]. Overall we find that stochastic dynamics forms a rich framework for the theoretical investigation of training dynamics.

Bibliography

[1] A. M. Turing, "Computing machinery and intelligence", *Mind; a quarterly review of psychology and philosophy*, vol. 59, no. 236, pp. 433–460, 1950, ISSN: 00264423, 14602113. JSTOR: 2251299.

[2] Lovelace, Ada and Menabrea, L.F., "Sketch of The Analytical Engine. Invented by Charles Babbage", *Bibliothèque Universelle de Genève*, vol. 82, 1842.

[3] J. McCarthy, M. L. Minsky, N. Rochester, and C. E. Shannon, "A Proposal for the Dartmouth Summer Research Project on Artificial Intelligence, August 31, 1955", *AI Magazine*, vol. 27, no. 4, pp. 12–12, Dec. 2006, ISSN: 2371-9621. DOI: 10.1609/aimag.v27i4.1904.

[4] W. S. McCulloch and W. Pitts, "A logical calculus of the ideas immanent in nervous activity", *The Bulletin of Mathematical Biophysics*, vol. 5, no. 4, pp. 115–133, Dec. 1943, ISSN: 0007-4985, 1522-9602. DOI: 10.1007/BF02478259.

[5] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

[6] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks", in *Advances in Neural Information Processing Systems*, vol. 25, Curran Associates, Inc., 2012.

[7] D. E. Rumelhart, G. E. Hinton, and R. J. Williams, "Learning representations by back-propagating errors", *Nature*, vol. 323, no. 6088, pp. 533–536, Oct. 1986, ISSN: 1476-4687. DOI: 10.1038/323533a0.

[8] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning", *Nature*, vol. 521, no. 7553, pp. 436–444, May 2015, ISSN: 1476-4687. DOI: 10.1038/nature14539.

[9] V. Mnih et al., *Playing Atari with Deep Reinforcement Learning*, Dec. 2013. DOI: 10.48550/arXiv.1312.5602. arXiv: 1312.5602 [cs.LG].

[10] J. Jumper et al., "Highly accurate protein structure prediction with AlphaFold", *Nature*, vol. 596, no. 7873, pp. 583–589, Aug. 2021, ISSN: 0028-0836, 1476-4687. DOI: 10.1038/s41586-021-03819-2.

[11] C. Matek, S. Krappe, C. Münzenmayer, T. Haferlach, and C. Marr, "Highly accurate differentiation of bone marrow cell morphologies using deep neural networks on a large image data set", *Blood*, vol. 138, no. 20, pp. 1917–1927, Nov. 2021, ISSN: 0006-4971. DOI: 10.1182/blood.2020010568.

[12] C. Matek, S. Schwarz, K. Spiekermann, and C. Marr, "Human-level recognition of blast cells in acute myeloid leukaemia with convolutional neural networks", *Nature Machine Intelligence*, vol. 1, no. 11, pp. 538–544, Nov. 2019, ISSN: 2522-5839. DOI: 10.1038/s42256-019-0101-9.

[13] OpenAI et al., *GPT-4 Technical Report*, Mar. 2024. DOI: 10.48550/arXiv.2303.08774. arXiv: 2303.08774 [cs.CL].

[14] A. Vaswani et al., "Attention is all you need", in *Advances in Neural Information Processing Systems*, I. Guyon et al., Eds., vol. 30, Curran Associates, Inc., 2017.

[15] Y. Bengio et al., "Managing extreme AI risks amid rapid progress", *Science*, vol. 384, no. 6698, pp. 842–845, May 2024, ISSN: 0036-8075, 1095-9203. DOI: 10.1126/science.adn0117. Accessed: Aug. 28, 2026.

[16] R. Bommasani et al., *On the Opportunities and Risks of Foundation Models*, Jul. 2022. DOI: 10.48550/arXiv.2108.07258. arXiv: 2108.07258 [cs.LG].

[17] A. Turner, L. Smith, R. Shah, A. Critch, and P. Tadepalli, "Optimal policies tend to seek power", in *Advances in Neural Information Processing Systems*, M. Ranzato, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, Eds., vol. 34, Curran Associates, Inc., 2021, pp. 23 063–23 074.

[18] A. Critch and S. Russell, *TASRA: A Taxonomy and Analysis of Societal-Scale Risks from AI*, Jun. 2023. DOI: 10.48550/arXiv.2306.06924. arXiv: 2306.06924 [cs.AI].

[19] A. M. Turing, "Intelligent Machinery, A Heretical Theory*", *Philosophia Mathematica*, vol. 4, no. 3, pp. 256–260, Sep. 1996, ISSN: 1744-6406, 0031-8019. DOI: 10.1093/philmat/4.3.256.

[20] F. Bach, *Learning Theory from First Principles* (Adaptive Computation and Machine Learning). Cambridge, Massachusetts London, England: The MIT Press, 2024, ISBN: 978-0-262-04944-3.

[21] N. Yang, C. Tang, and Y. Tu, "Stochastic Gradient Descent Introduces an Effective Landscape-Dependent Regularization Favoring Flat Solutions", *Physical Review Letters*, vol. 130, no. 23, p. 237 101, Jun. 2023. DOI: 10.1103/PhysRevLett.130.237101. arXiv: 2206.01246.

[22] S. Hochreiter and J. Schmidhuber, "Flat Minima", *Neural Computation*, vol. 9, no. 1, pp. 1–42, Jan. 1997, ISSN: 0899-7667, 1530-888X. DOI: 10.1162/neco.1997.9.1.1.

[23] J.-F. Le Gall, *Measure Theory, Probability, and Stochastic Processes* (Graduate Texts in Mathematics 295). Cham, Switzerland: Springer, 2022, ISBN: 978-3-031-14205-5 978-3-031-14204-8.

[24] T. Tao, *An Introduction to Measure Theory* (Graduate Studies in Mathematics 126). Providence, Rhode Island: American Mathematical Society, 2011, ISBN: 978-1-4704-6640-4 978-1-4704-1187-9.

[25] P. Brémaud, *Probability Theory and Stochastic Processes* (Universitext). Cham: Springer, 2020, ISBN: 978-3-030-40183-2.

[26] C. W. Gardiner and C. W. Gardiner, *Stochastic Methods: A Handbook for the Natural and Social Sciences* (Springer Series in Synergetics), 4th ed. Berlin: Springer, 2009, ISBN: 978-3-540-70712-7.

[27] N. G. van Kampen, *Stochastic Processes in Physics and Chemistry* (North-Holland Personal Library), 3rd ed. Amsterdam, The Netherlands Oxford, UK: Elsevier, 2007, ISBN: 978-0-444-52965-7.

[28] P. Grohs and G. Kutyniok, *Mathematical Aspects of Deep Learning*. Cambridge, UK [etc.]: Cambridge university press, 2023, ISBN: 978-1-316-51678-2.

[29] Y. LeCun, C. Cortes, C. Burges, et al., "MNIST handwritten digit database", 2010.

[30] A. LeNail, "NN-SVG: Publication-Ready Neural Network Architecture Schematics", *Journal of Open Source Software*, vol. 4, no. 33, p. 747, Jan. 2019, ISSN: 2475-9066. DOI: 10.21105/joss.00747.

[31] R. D. Nelson, *The Penguin Dictionary of Mathematics*, 4th ed. London: Penguin, 2008, ISBN: 978-1-84972-600-9.

[32] Z. Yao, A. Gholami, K. Keutzer, and M. Mahoney, *PyHessian: Neural Networks Through the Lens of the Hessian*, Mar. 2020. DOI: 10.48550/arXiv.1912.07145. arXiv: 1912.07145 [cs.LG]. Accessed: Sep. 2, 2026.

[33] V. Nair and G. E. Hinton, "Rectified linear units improve restricted boltzmann machines", in *Proceedings of the 27th International Conference on International Conference on Machine Learning*, ser. ICML'10, Madison, WI, USA: Omnipress, 2010, pp. 807–814, ISBN: 978-1-60558-907-7.

[34] Y. Feng and Y. Tu, "The inverse variance–flatness relation in stochastic gradient descent is critical for finding flat minima", *Proceedings of the National Academy of Sciences*, vol. 118, no. 9, e2015617118, Mar. 2021. DOI: 10.1073/pnas.2015617118.

[35] M. Wei and D. J. Schwab, *How noise affects the Hessian spectrum in overparameterized neural networks*, Oct. 2019. DOI: 10.48550/arXiv.1910.00195. arXiv: 1910.00195 [cs].

[36] A. Altland and J. von Delft, *Mathematics for Physicists: Introductory Concepts and Methods*. Cambridge New York: Cambridge university press, 2019, ISBN: 978-1-108-47122-0.

[37] K. Arvai, *Kneed (Version v0.8.5) [Computer software] Zenodo*, Mar. 2026. DOI: 10.5281/zenodo.8127224. Accessed: Aug. 28, 2026.

[38] V. Satopa, J. Albrecht, D. Irwin, and B. Raghavan, "Finding a "kneedle" in a haystack: Detecting knee points in system behavior", in *31st International Conference on Distributed Computing Systems Workshops*, 2011, pp. 166–171.

[39] R. Brown, "XXVII. *A brief account of microscopical observations made in the months of June, July and August* 1827, *on the particles contained in the pollen of plants; and on the general existence of active molecules in organic and inorganic bodies*", *The Philosophical Magazine*, vol. 4, no. 21, pp. 161–173, Sep. 1828, ISSN: 1941-5850, 1941-5869. DOI: 10.1080/14786442808674769.

[40] A. Einstein, "Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen", *Annalen der Physik*, vol. 322, no. 8, pp. 549–560, 1905. DOI: 10.1002/andp.19053220806. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/andp.19053220806.

[41] M. von Smoluchowski, "Zur kinetischen theorie der brownschen molekularbewegung und der suspensionen", *Annalen der Physik*, vol. 326, no. 14, pp. 756–780, 1906. DOI: 10.1002/andp.19063261405. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/andp.19063261405.

[42] D. Lemons and A. Gythiel, "Paul langevin's 1908 paper "on the theory of brownian motion"", *American Journal of Physics - AMER J PHYS*, vol. 65, pp. 1079–1081, 1997.

[43] D. Tong, *Fluid Mechanics: Volume 4 Lectures on Theoretical Physics*. Cambridge: Cambridge University Press, 2025, ISBN: 978-1-009-59468-4.

[44] R. P. Feynman, "The Feynman lectures on physics. Volume 1: Mainly mechanics, radiation, and heat", in The new millennium edition, paperback first published, New York: Basic Books, 2011, ISBN: 978-0-465-02493-3.

[45] Frey, Erwin, *PhysicsOfLifeLMU: Thermodynamics, Statistical Mechanics, Nonequilibrium Physics and My Teaching Philosophy*, https://www.youtube.com/watch?v=-pEPKnuN1iY, May 2025. Accessed: Aug. 29, 2026.

[46] H. Risken, *The Fokker-Planck Equation: Methods of Solution and Applications* (Springer Series in Synergetics 18), Second edition. Berlin Heidelberg: Springer, 1996, ISBN: 978-3-642-61544-3.

[47] D. S. Zhang, G. W. Wei, D. J. Kouri, and D. K. Hoffman, "Distributed approximating functional approach to the Fokker–Planck equation: Eigenfunction expansion", *The Journal of Chemical Physics*, vol. 106, no. 12, pp. 5216–5224, Mar. 1997, ISSN: 0021-9606, 1089-7690. DOI: 10.1063/1.473520.

[48] R. Shankar, *Principles of Quantum Mechanics*, Second edition, corrected at 19th printing. New York, NY: Springer, 2014, ISBN: 978-0-306-44790-7.

[49] D. Tong, *Quantum Mechanics* (Lectures on Theoretical Physics volume 3). Cambridge: Cambridge University press, 2025, ISBN: 978-1-009-59485-1.

[50] J. L. Garcia-Palacios, *Introduction to the theory of stochastic processes and Brownian motion problems*, Jan. 2007. DOI: 10.48550/arXiv.cond-mat/0701242. arXiv: cond-mat/0701242.

[51] S. F. Kwok, *Langevin and Fokker-Planck Equations and Their Generalizations: Descriptions and Solutions*. Singapore Hackensack: World scientific publishing, 2018, ISBN: 978-981-322-840-5.

[52] P. E. Kloeden and E. Platen, *Numerical Solution of Stochastic Differential Equations* (Applications of Mathematics, Stochastic Modelling and Applied Probability 23), Corrected third printing. Berlin, Heidelberg: Springer, 1999, ISBN: 978-3-662-12616-5.

[53] D. Scieur, V. Roulet, F. Bach, and A. dAspremont, "Integration methods and optimization algorithms", in *Advances in Neural Information Processing Systems*, I. Guyon et al., Eds., vol. 30, Curran Associates, Inc., 2017.

[54] B. Shi, W. Su, and M. I. Jordan, "On Learning Rates and Schrödinger Operators", *Journal of Machine Learning Research*, vol. 24, no. 379, pp. 1–53, 2023, ISSN: 1533-7928.

[55] A. Lanconelli and C. S. A. Lauria, "A note on diffusion limits for stochastic gradient descent", *Journal of Approximation Theory*, vol. 309, p. 106 160, Aug. 2025, ISSN: 0021-9045. DOI: 10.1016/j.jat.2025.106160.

[56] Q. Li, C. Tai, and W. E, "Stochastic Modified Equations and Dynamics of Stochastic Gradient Algorithms I: Mathematical Foundations", *Journal of Machine Learning Research*, vol. 20, no. 40, pp. 1–47, 2019, ISSN: 1533-7928.

[57] K. Jin, J. Latz, C. Liu, and C.-B. Schönlieb, *A Continuous-time Stochastic Gradient Descent Method for Continuous Data*, Dec. 2021. DOI: 10.48550/arXiv.2112.03754. arXiv: 2112.03754 [cs.LG].

[58] M. Rosca, Y. Wu, C. Qin, and B. Dherin, *On a continuous time model of gradient descent dynamics and instability in deep learning*, Sep. 2023. DOI: 10.48550/arXiv.2302.01952. arXiv: 2302.01952 [stat.ML].

[59] K. Liu, L. Ziyin, and M. Ueda, "Noise and Fluctuation of Finite Learning Rate Stochastic Gradient Descent", in *Proceedings of the 38th International Conference on Machine Learning*, PMLR, Jul. 2021, pp. 7045–7056.

[60] J. Latz, "Analysis of stochastic gradient descent in continuous time", *Statistics and Computing*, vol. 31, no. 4, p. 39, Jul. 2021, ISSN: 0960-3174, 1573-1375. DOI: 10.1007/s11222-021-10016-8.

[61] S. Ankirchner and S. Perko, "A Comparison of Continuous-Time Approximations to Stochastic Gradient Descent", *Journal of Machine Learning Research*, vol. 25, no. 13, pp. 1–55, 2024, ISSN: 1533-7928.

[62] W. Hu, C. J. Li, L. Li, and J.-G. Liu, *On the diffusion approximation of nonconvex stochastic gradient descent*, Mar. 2018. DOI: 10.48550/arXiv.1705.07562. arXiv: 1705.07562 [stat].

[63] Z. Li, S. Malladi, and S. Arora, *On the Validity of Modeling SGD with Stochastic Differential Equations (SDEs)*, Jun. 2021. DOI: 10.48550/arXiv.2102.12470. arXiv: 2102.12470 [cs.LG].

[64] Q. Li, C. Tai, and W. E, "Stochastic Modified Equations and Adaptive Stochastic Gradient Algorithms", in *Proceedings of the 34th International Conference on Machine Learning*, PMLR, Jul. 2017, pp. 2101–2110.

[65] K. Kaneko, "Adiabatic Elimination by the Eigenfunction Expansion Method", *Progress of Theoretical Physics*, vol. 66, no. 1, pp. 129–142, Jul. 1981, ISSN: 0033-068X. DOI: 10.1143/PTP.66.129.

[66] Z. Liu, Y. Liu, J. Gore, and M. Tegmark, *Neural Thermodynamic Laws for Large Language Model Training*, May 2025. DOI: 10.48550/arXiv.2505.10559. arXiv: 2505.10559 [cs.LG].

[67] M. Haddouche, P. Viallard, U. Simsekli, and B. Guedj, "A PAC-Bayesian Link Between Generalisation and Flat Minima", in *Proceedings of The 36th International Conference on Algorithmic Learning Theory*, PMLR, Feb. 2025, pp. 481–511.

Declaration on AI usage

In this work AI tools were used in the form of large language models, in particular the models by Anthropic. The use of AI tools was limited to evaluation and optimization of self-implemented code, including the code for the machine learning experiments as well as the numerical experiments. Additionally, the AI tools assisted in the extension of self-implemented code for the creation of plots as well as the LaTeX document. Regarding the figures, the AI use was purely instrumental in implementing the plotting code. The decision of what data to represent in which way within the figures was mine. Importantly, the use of AI tools does not extend to the written text, interpretations of experimental data or physical and mathematical derivations.

Selbständigkeitserklärung

Hiermit erkläre ich, die vorliegende Arbeit selbständig verfasst zu haben und keine anderen als die in der Arbeit angegebenen Quellen und Hilfsmittel benutzt zu haben.

Ort, Datum der Abgabe Unterschrift